

Cylinder loop weights and planar nesting

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Abstract

We determine the growth exponent, at every fixed positive loop fugacity, of the critical honeycomb partition function for disjoint polygons separating two prescribed markers on a balanced cylinder. At fugacity two the cylinder exponent is $1/6$; the corresponding planar nesting exponent and middle-strip nesting exponent are $1/12$.

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1 Introduction

The critical activity fixes the weight of each honeycomb polygon, but the sum over disjoint polygons surrounding a point requires control on every spatial scale. We determine the growth of this nesting partition function by first calculating a partition function for polygons separating two prescribed points on a cylinder. Two independent pressure estimates then relate the cylinder to planar nests. The same analysis also gives bounds for large polygon counts and for the length of critical self-avoiding paths in a strip.

1.1 The model and its exponents

We realize the honeycomb lattice as the dual of the equilateral triangular tiling with side length one and a horizontal edge direction. A triangular band has height $\sqrt{3}/2$. A polygon is an unoriented simple cycle of the dual graph. Its length $|P|$ is its number of vertices, and its critical weight is $\rho^{|P|}$, where

$$\rho = \frac{1}{\sqrt{2 + \sqrt{2}}}.$$

Nienhuis predicted this activity through the dilute $O(n)$ model [18]; Duminil-Copin and Smirnov proved that its reciprocal is the honeycomb connective constant [5]. Their parafermionic observable supplies the local boundary cancellation used here. We also use the finite strip transfer construction and the critical strip-crossing estimate of [19].

Identify the plane with $\mathbb{C}$, with the horizontal lattice direction real and a face center at zero. For even $N = 2m$, quotient by the translation $iN\sqrt{3}/2$ and mark the images of 0 and $me^{2\pi i/3}$. Their vertical separation is m bands. This is the balanced cylinder geometry used throughout: a lattice cut follows m edges in direction $e^{2\pi i/3}$ and then m in direction $e^{\pi i/3}$, closing after one period. Compactifying the cylinder's two infinite ends gives a sphere. A polygon *separates the markers* if they lie in distinct components of its complement on that sphere. Such a polygon may wind around the cylinder, or may be contractible and enclose exactly one marker in its disk.

Give every separating polygon an additional factor χ , called its loop fugacity, and define

$$Z_N(\chi) = \sum_{\mathcal{P}} \prod_{P \in \mathcal{P}} \chi \rho^{|P|}.$$

The sum runs over finite families of pairwise vertex-disjoint separating polygons, including the empty family with weight one. The transfer construction in Section 2 proves convergence of these masses. Every separating polygon meets the finite arc of the prescribed cut between the markers, so the number of disjoint polygons is bounded at fixed N . Thus Z_N is a polynomial with nonnegative coefficients and $Z_N(0) = 1$.

Theorem 1.1 (Cylinder fugacity exponent). *For every fixed $\chi > 0$, along even N ,*

$$Z_N(\chi) = N^{\tau(\chi)+o(1)}, \quad \tau(\chi) = \begin{cases} \frac{1}{6} - \frac{2}{3\pi^2} (\arccos(\chi/2))^2, & 0 < \chi \leq 2, \\ \frac{1}{6} + \frac{2}{3\pi^2} (\operatorname{arcosh}(\chi/2))^2, & \chi \geq 2. \end{cases}$$

For the planar counterpart, let $P_\chi(r)$ be the sum of $\prod_{P \in \mathcal{P}} \chi \rho^{|P|}$ over disjoint nests around one fixed face center, with every polygon of diameter at most r ; again include the empty nest. Theorem 11.2 proves the bounded-factor comparison

$$P_\chi(c_0 N)^2 \leq Z_N(\chi) \leq C_{\chi, c_0} P_\chi(c_0 N)^2$$

for every fixed $\chi \geq 0$ and sufficiently small fixed $c_0 > 0$. Consequently, for every fixed $\chi > 0$,

$$P_\chi(r) = r^{\tau(\chi)/2+o(1)}.$$

The square has a geometric origin. Removing essential polygons and large contractible polygons costs a bounded factor; the remaining polygons form two independent small planar nests, one near each marker. Figure 3 shows the two planar neighborhoods. The two pressure calculations below provide the estimates needed for this removal.

In particular, $Z_N(2) = N^{1/6+o(1)}$ and $P_2(r) = r^{1/12+o(1)}$. Fugacity two arises in the boundary winding identity because an enclosing polygon has two orientations. If S_k is the same fugacity-two nesting mass around a middle face of a strip of $2k$ bands, with no diameter restriction, then Corollary 11.3 gives

$$S_k = k^{1/12+o(1)}.$$

1.2 Uniform counts and path length

The fixed-fugacity theorem concerns one parameter at a time. To control polygon counts that grow with N , we prove a separate estimate uniform over an unbounded fugacity range. Define the critical coefficient mass $w_N(a)$ by

$$Z_N(\chi) = \sum_{a \geq 0} w_N(a) \chi^a.$$

Thus $w_N(a)$ sums $\prod_P \rho^{|P|}$ over families with exactly a separating polygons. Section 8 proves

$$Z_N(2 \cosh v) \leq \exp(C \log N (1 + v^2)) \quad (v \geq 0),$$

with C independent of v , and hence

$$w_N(a) \leq \exp\left(C \log N - c \frac{a^2}{\log N}\right) \quad (a \geq 0).$$

These are unnormalized coefficient masses. The uniform fugacity bound uses the coarse real-sector estimate before any counts are truncated; the fixed-parameter asymptotic alone would not give it.

The nesting law also controls a path-length observable. A *port* is the midpoint of a triangular-tile edge. In a strip of h bands, consider self-avoiding paths from a fixed bottom boundary port to either boundary, with no other boundary visit. They include their initial and final half-edges, and $n(\gamma)$ counts the visited dual vertices. Proposition 11.5 proves

$$\sum_{\gamma} \rho^{n(\gamma)} n(\gamma) \leq h^{13/12+o(1)}.$$

Under the normalized critical weights, their expected length conditional on reaching the top boundary of the bottom $\lfloor h/4 \rfloor$ bands is $h^{4/3+o(1)}$ (Proposition 11.6). The difference between these exponents comes from the conditioning event, whose unnormalized mass is comparable to $h^{-1/4}$ by the strip-crossing theorem of [19, Theorem 1.1]. A positive identity relating visits to an interior edge to nests around its two adjacent faces supplies the required length comparison.

1.3 Proof strategy

The proof has an algebraic and analytic route to the cylinder exponent, followed by a geometric route to nesting. We describe the two routes separately because the pressure estimates do not follow from the cylinder asymptotic.

A finite representation with its normalization. Section 2 constructs the cylinder vacuum from a polynomial open-strip state. A vacuum state records the pairings of occupied ports made by paths on one side of a cut, with no external source. Contracting the two half-cylinder states with the separation rule gives Z_N . Degree bounds and reductions at special parameters determine this contraction exactly.

Writing $\chi = 2 \cos \eta$, a finite contour deformation expresses its numerator as a sum of particle integrals with factor $e^{i\eta M}$. The integer M fixes the two groups' elementary particle totals as $m - M$ and $m + M$. Each group contains main particles and residue-generated far single particles and fused pairs, the latter counted twice in the elementary degree. Only terms of elementary degree N enter the identity.

The logarithmic contribution of the particle integrals. Section 3 centers each main group at the mean density $m\rho(u)$, where ρ is a fixed probability density with exponential tails. Walls at distance proportional to $\log N$ leave only $O(1)$ mean mass outside; this is the spatial scale of the particle calculation. The real quadratic energy of the remaining fluctuations controls their imbalance, their far-particle counts, and excursions beyond that interval. A Legendre orthogonal-polynomial ensemble gives a positive reference law for the two large interior groups. Subtracting its logarithmic interaction removes the diagonal singularity, so the remaining energy is meaningful on empirical particle measures.

The coefficient of $\log N$ must be kept throughout this comparison. The relative fluctuation determinant contributes $7/8$, the two reference partition masses contribute $-1/2$, and the scalar vacuum denominator contributes $-5/24$. Their sum is

$$\frac{7}{8} - \frac{1}{2} - \frac{5}{24} = \frac{1}{6}.$$

To obtain the quadratic term in η , the phase $e^{i\eta M}$ must be retained during integration. Section 4 proves the reference integration bounds. Section 5 realizes the needed complex displacement by moving actual particles and computing its Jacobian. Because the interior particle numbers also change with M , Section 6 constructs a holomorphic approximation to the integer-dimensional integrals before moving the imbalance sum. The combined particle and imbalance-contour deformations convert the phase into a quadratic contribution to the real action. These steps yield, locally uniformly for $|\operatorname{Re} \eta| < \pi$,

$$\frac{\log |Z_N(2 \cos \eta)|}{\log N} \leq \frac{1}{6} - \frac{2}{3\pi^2} \operatorname{Re}(\eta^2) + o(1).$$

Section 7 obtains the matching lower exponent from this harmonic upper envelope. The anchor $\eta = \pi/2$ has the exact value $Z_N(0) = 1$; subharmonicity propagates near equality, and coefficient positivity transfers it to each fixed positive fugacity. Thus no lower estimate of a single complex sector is needed.

From the cylinder to planar polygons and strip paths. Section 9 varies the weight of essential loops in an annular transfer. Its first two pressure derivatives at zero fugacity control both their total mass and their first horizontal-span moment. These bounds make the cost of removing separating essential polygons uniform in N . Section 10 instead varies the weight of all loops in an open strip. Its first pressure derivative at zero fugacity is affine in N up to an error $O(N^{-1})$. Counting each shape's vertical placements and comparing two strip widths then gives

$$\sum_{[P]: \operatorname{diam} P > r} \rho^{|P|} \leq C(1+r)^{-2}.$$

Here $[P]$ is an unrooted, unoriented polygon modulo triangular-lattice translations. This estimate controls large contractible polygons. Section 11 uses both removal bounds to separate the two planar nests, then proves the length–nest identity and the raw and conditioned strip-length estimates.

1.4 Earlier work

The electric exponent has a direct Coulomb-gas predecessor. Gamsa and Cardy study the change in weight of loops separating marked points [7, Section 3, equation (6)]. In the dilute zero-loop specialization, their scaling dimension for $\chi = 2 \cos \eta$, $0 < \chi \leq 2$, is $x = \eta^2/(3\pi^2) - 1/12$, giving the predicted two-point power $-2x = \tau(\chi)$. Their argument assumes the Gaussian-field scaling description. Cardy's annulus calculation likewise permits an independent weight for winding loops and gives the corresponding radial power [1]. The present argument establishes the finite lattice normalization and the estimates required for the cylinder and planar observables above; it also proves the stated positive-fugacity branch for $\chi > 2$.

Periodic dilute-loop transfer matrices with seams that change wrapping loop weights were analyzed by Zhou and Batchelor [24]. Their spectral calculation provides related electric-parameter formulas. The observable here is a contraction with two specified markers, and includes contractible polygons surrounding one marker. The finite contraction and its nonvanishing scalar normalization

must therefore be identified in this geometry. Ikhlef and Cardy related local parafermionic equations to the Yang–Baxter weights of dilute-loop models [14]. Glazman and Manolescu proved invariance of the boundary two-point function for columnwise rhombic half-planes with angles in $[\pi/3, 2\pi/3]$ [10]. We use their local-weight convention, specialized to the regular honeycomb lattice in the final statements.

Polynomial exchange equations and special-rapidity reductions have earlier uses in integrable loop constructions. Di Francesco and Zinn-Justin use them to prove a multiparameter sum rule for the dense $O(1)$ model [2]. Garbali and Nienhuis develop polynomial states and fusion recurrences for the dilute $O(1)$ open strip, and obtain determinant expressions for its normalization [8, 9]. They also note that the polynomial-state method applies at zero loop weight [8, Section 4]. Here the open-strip polynomial state is supplied by [19]; we construct its periodic limit, the normalized vacuum quotient, and the two-marker contraction with their physical interpretation.

The analytic reference measure is a classical orthogonal-polynomial ensemble. Andréief’s identity gives its Gram normalization and tilted factorial determinants [22]. The circle comparison is the finite upper bound associated with the strong Szegő theorem [21]; its Schur-polynomial proof is related to the unitary trace calculations of Diaconis–Shahshahani and Diaconis–Evans [4, 3]. We include these arguments in the required normalization. To retain the logarithmic exponent, the present application also needs uniform source bounds for growing physical tests, a particle realization of the complex quadratic change of variables, and continuation of the integer dimensions. These are the roles of Sections 4–6.

2 Cylinder partition functions and contour sectors

Our first task is to express the positive cylinder observable as a finite sum to which asymptotic analysis can be applied. We prove two exact representations: a sum over three labels at each site, and an integral over finitely many contour sectors. The normalization is part of each identity. We then estimate that normalization on the balanced physical cylinder. The construction begins with the strip estimate needed to control paths in a fixed-width periodic slab. The positive length–nest comparison will be proved with its applications in Section 11.

We use the honeycomb lattice dual to the equilateral triangular lattice of side length one. A port is the midpoint of a tile edge. A path between ports visits each honeycomb vertex at most once, includes its endpoint half-edges, and has weight $\rho^{n(\gamma)}$, where $n(\gamma)$ counts visited vertices. A polygon of length r has r vertices. Put

$$\lambda = \frac{\pi}{8}, \quad d = \cos(2\lambda) = \frac{1}{\sqrt{2}}, \quad c = \cos(3\lambda), \quad \rho = \frac{1}{\sqrt{2 + \sqrt{2}}}.$$

For the strip of H triangular bands, let $\mathcal{A}_H$ be the mass of paths from one fixed bottom port to the other bottom ports and let $\mathcal{B}_H$ be their mass to the top boundary. We use the following results from [19, Theorem 1.1 and Lemmas 2.1, 2.3]:

$$c\mathcal{A}_H + \mathcal{B}_H = 1, \quad 0 < \mathcal{B}_H \asymp H^{-1/4}, \quad \mathcal{B}_H \text{ is nonincreasing in } H. \quad (1)$$

The same source proves exponential decay of a one-defect transfer at each fixed height, and boundedness of the corresponding vacuum half-state. Here a defect means one strand attached to a specified source and continuing to the current cut. We will name each use of these finite-height facts.

2.1 Strip input and the critical walk bound

The critical activity is the reciprocal of the connective constant, as proved by Duminil-Copin and Smirnov [5]. For later use we give the short deduction from the strip estimates; it also supplies a subexponential bound without a limit interchange.

Proposition 2.1 (Critical weight). *If c_n is the number of n -edge self-avoiding walks from a specified honeycomb vertex and $\mu = \lim_{n \rightarrow \infty} c_n^{1/n}$, then $\rho = \mu^{-1}$. Moreover, $c_n \rho^n = \exp(o(n))$ as an upper bound.*

Proof. We use the bridge-decomposition method of Hammersley and Welsh [13]; see also Madras and Slade [16] and Duminil-Copin and Hammond [6, Proposition 2.1]. We give the decomposition with the present honeycomb endpoint conventions, using the positive strip bound to control each weak bridge.

Submultiplicativity gives existence of μ . Split a walk at a vertex of minimum height, choosing the first such vertex in a fixed traversal order when necessary. Its two tails start at that vertex and stay above its height. Translate this vertex to one of two fixed representatives, according to its triangular-band type. Recording the two tails and the split time recovers the original walk, because its original root is fixed.

Decompose each nonempty tail by stopping at its last global maximum, then at the last minimum of the remainder, and continuing alternately. Every piece is a weak bridge: all its vertices lie between the heights of its endpoints. There are no horizontal edges. Its positive spans, measured in thirds of a triangular band, are integers, are nonincreasing, and are strictly decreasing after a possible equality of the first two. Their sum is at most $2n$. Consequently there are $O(\sqrt{n})$ pieces and at most $\exp(O(\sqrt{n} \log(n+1)))$ possible span lists, even after recording the split time and the two initial types.

For a fixed initial vertex and a specified directed span, the total $\rho^{\#\text{edges}}$ -weight of weak bridges is bounded by an absolute constant. To obtain this from the strip bound $\mathcal{B}_H \leq 1$, choose horizontal band boundaries immediately beyond the two endpoint heights. At a lower endpoint of lower-center type append the vertical half-edge to the lower boundary. At the other type first append the fixed descending edge, then the vertical half-edge. Make the reflected construction at the upper endpoint. All added vertices are outside the old height range, so the extension preserves self-avoidance and has bounded multiplicity. The starting boundary port and strip height are determined by the starting vertex and span. The extension adds at most three vertex-weight factors relative to the old edge count. The same argument applies to downward bridges by reflection.

For each span list, ignore avoidance between successive pieces and sum their weights successively. This gives a constant to the power $O(\sqrt{n})$, and hence

$$c_n \rho^n \leq \exp(O(\sqrt{n} \log(n+1))).$$

Taking n th roots yields $\rho \leq \mu^{-1}$. If the inequality were strict, choose r with $\rho < r < \mu^{-1}$. The unrestricted walk series at r converges, so its tail at ρ decays exponentially in length. A bridge crossing N bands has length at least a positive constant times N , up to endpoint constants. Its mass would therefore decay exponentially in N , contradicting $\mathcal{B}_N \asymp N^{-1/4}$ from the strip calculation. $\square$

2.2 The polynomial cylinder vacuum

A cylinder is obtained by making the band direction periodic. Cutting it along a column leaves a cyclic list of ports; the state on one side of that cut records which ports are joined. We first construct this state algebraically, then prove that at physical parameters it is the absolutely convergent sum

of half-cylinder configurations. Thus the algebraic normalization and the positive physical state will be identified before they are contracted.

The local operators and the open-strip input. For complex u , write $C = \cos \lambda$, $a(u) = \cos(2u - 3\lambda)$ and

$$\mathcal{D}(u) = \sin(2\lambda + u) \sin(3\lambda + u) = \frac{C + a(u)}{2}.$$

The six tile coefficients, in the normalization of [10, equation (1)], are

$$\begin{aligned} A(u) &= \frac{d \sin(3\lambda - u)}{\mathcal{D}(u)}, & U(u) &= \frac{d \sin u}{\mathcal{D}(u)}, & B(u) &= \frac{\sin u \sin(3\lambda - u)}{\mathcal{D}(u)}, \\ E(u) &= \frac{\sin(3\lambda - u) \sin(2\lambda - u)}{\mathcal{D}(u)}, & F(u) &= \frac{\sin(u - \lambda) \sin u}{\mathcal{D}(u)}. \end{aligned} \quad (2)$$

The sixth coefficient, that of the empty tile, is one. The two-slot operator $R(u)$ acts on noncrossing partial pairings. Order its inputs as (south, west) and its outputs as (east, north). Thus the input cap joins south to west and the output cup joins east to north. The coefficient A belongs to either south–east or west–north alone; B belongs to either south–north or west–east alone; U belongs to the cap alone or cup alone. The two simultaneous A connections have coefficient E , and the simultaneous cap and cup have coefficient F . The empty diagram has coefficient one. This is the usual two-slot notation in which A labels a single vertical connection and B a single diagonal connection. Figure 1 shows the port order and the physical column construction used below. Composition glues occupied slots, annihilates occupancy mismatches, and assigns weight zero to every closed cycle. Transpose is reflection exchanging input and output slots. The map S_2 sends a vacancy to a vacancy plus ρ times a cup and sends an occupied strand to either of two output slots with coefficient ρ . The map S_3 , with no input slots, is the sum of the vacant diagram and a cup.

The physical geometry fixes how these coefficients count honeycomb vertices. A row of parameter u_i is a rhombus with horizontal sides of length one and upward side vector $e^{iu_i/t}$, where $t = 3/8$; a left-to-right column uses $R(3\lambda - u_i)$. At $u_i = \lambda$ or 2λ the upward side makes angle $\pi/3$ or $2\pi/3$ with the horizontal. Splitting the rhombus along its short diagonal gives two equilateral triangles. Each occupied triangle contributes one factor ρ , while an unvisited triangle contributes one. This follows either from the local construction in [19, Section 2], or by substituting in the weights:

v	$A(v)$	$U(v)$	$B(v)$	$E(v)$	$F(v)$
λ	ρ	ρ^2	ρ^2	ρ^2	0
2λ	ρ^2	ρ	ρ^2	0	ρ^2 .

Thus these column diagrams have exactly the vertex weights of the honeycomb paths and polygons, including the endpoint half-edge convention.

The open-strip results we need are recorded together, so that no cylinder statement is implicit in the import. Their complete proofs are the local diagram identities, scalar reduction identities, and vacuum degree bound in [19, Lemmas 2.2, 3.1, 3.2].

Proposition 2.2 (Open-strip polynomial input). *Put $D(a, b) = a^2 + b^2 + 2dab - d^2$ and define*

$$p_N(a_1, \dots, a_N) = \prod_{i < j} \frac{D(a_i, a_j)}{a_i - a_j} \text{Pf} \left[\frac{(a_i - a_j)(a_i + a_j + \rho)}{D(a_i, a_j)} \right].$$

In odd size the matrix is augmented by a column of ones and its negative transpose; $p_0 = p_1 = 1$. This is a symmetric polynomial of per-variable degree at most $N - 1$, with leading coefficient p_{N-1} . There is a polynomial vacuum vector W_N with empty component p_N and the following properties, where $P_N = W_N/p_N$ is interpreted as a rational vector.

1. The local operators satisfy

$$\begin{aligned} R_2(v)R_1(u+v)R_2(u) &= R_1(u)R_2(u+v)R_1(v), \\ R(u)R(-u) &= I, \quad R(j\lambda) = S_j S_j^\dagger, \\ R_2(u)R_1(u+j\lambda)(I \otimes S_j) &= (S_j \otimes I)\hat{R}_j(u) \quad (j = 2, 3), \end{aligned}$$

where $\hat{R}_2(u) = R(u + \lambda)$ and $\hat{R}_3(u)$ is the one-slot identity. Subscripts indicate the two consecutive slots on which R acts. The vector P_N satisfies adjacent braid covariance:

$$P_N(\dots, u_i, u_{i+1}, \dots) = R_i(u_{i+1} - u_i)P_N(\dots, u_{i+1}, u_i, \dots).$$

At $u_{i+1} = u_i + j\lambda$ it equals $S_j P_{\text{red}}$; for $j = 2$ the pair is replaced by $u_i + \lambda$, and for $j = 3$ it is deleted.

2. For $f_j = \cos(w + j\lambda)$ the scalar reductions are

$$\begin{aligned} p(Y, f_0, f_6) &= p(Y)(f_0 + f_6 + \rho) \prod_{z \in Y} (z - f_{-6})(z - f_{12}), \\ p(Y, f_{-2}, f_2) &= p(Y, f_0)(f_{-2} + f_2 + \rho) \prod_{z \in Y} (z + f_0). \end{aligned}$$

They also hold with all shifts negated.

3. In the variables e^{iu_k} , W_N is Laurent polynomial with exponents between $-(2N - 2)$ and $2N - 2$. Shifting u_k by π changes a component's sign exactly when slot k is occupied. In its last variable $y = u_N$, the vacant part is a polynomial in $b = a(y)$ of degree at most $N - 1$ with leading coefficient W_{N-1} ; the occupied part is $\sin y$ times a polynomial in b of degree at most $N - 2$.
4. These last-variable polynomials are the unique interpolants at $y_i = u_i + 3\lambda$, $i < N$. Write $b_i = a(y_i)$, $d_i = a(u_i - 3\lambda)$, $e_i = a(u_i + 6\lambda)$, and let $\mathcal{R}_i$ move the inserted row y_i from immediately after row i to the last position by successive braid boxes. Their vector data are

$$Y_i = (a(u_i) + b_i + \rho) \prod_{\substack{j < N \\ j \neq i}} (a(u_j) - d_i)(a(u_j) - e_i) \mathcal{R}_i S_3 W_{N-2}.$$

Thus, with $\ell_i(b) = \prod_{j < N, j \neq i} (b - b_j)/(b_i - b_j)$,

$$\begin{aligned} W_N^{\text{vac}}(b) &= W_{N-1} \prod_{i < N} (b - b_i) + \sum_{i < N} (Y_i)_{\text{vac}} \ell_i(b), \\ W_N^{\text{occ}}(y) &= \sin y \sum_{i < N} \frac{(Y_i)_{\text{occ}}}{\sin y_i} \ell_i(a(y)). \end{aligned}$$

These identities hold at generic arguments and then by polynomial continuation. In each W_{N-2} the interpolated and selected rows are omitted.

We now use these exact finite identities to construct the periodic state. No infinite-cylinder summation is included in the proposition.

Make the row index periodic, with $1, \dots, N$ in cyclic order on a cut. A state is a noncrossing partial pairing of these ports with disk connectivity. Equivalently, compactify the open end of a half-cylinder to a point and allow no strand to end there. Its basis is the same as the linear-order vacuum basis. Until the separating-loop contraction is defined below, every closed cycle has weight zero, including a cycle that winds around the cylinder. Put

$$\beta = 2\lambda = \frac{\pi}{4}, \quad q = e^{i\beta}, \quad x_i = e^{2iu_i}, \quad D_0(x, y) = x^2 + y^2 + 2dxy, \quad K_N^* = \frac{N(N-1)}{2}.$$

We fix $\sqrt{x_i} = e^{iu_i}$. Vector components can involve these half-powers; scalar contractions below have integral powers. Identities at generic complex arguments extend as rational identities.

Lemma 2.3 (Homogeneous vacuum polynomial). *There is a homogeneous vector V_N of total degree K_N^* whose empty component is the symmetric polynomial*

$$h_N(x_1, \dots, x_N) = \prod_{i < j} \frac{D_0(x_i, x_j)}{x_i - x_j} \text{Pf} \left[\frac{x_i^2 - x_j^2}{D_0(x_i, x_j)} \right]. \quad (3)$$

For odd N , augment the skew matrix by a last column of ones and a last row of minus ones; set $h_0 = h_1 = 1$. In any slot k , the vacant part of V_N is polynomial in x_k of degree at most $N-1$, and its occupied part is $\sqrt{x_k}$ times a polynomial of degree at most $N-2$. In the last slot the leading vacant coefficient is V_{N-1} . The quotient $U_N = V_N/h_N$ has empty component one and satisfies the braid and splitting identities

$$U_N(\dots, u_i, u_{i+1}, \dots) = R_i(u_{i+1} - u_i)U_N(\dots, u_{i+1}, u_i, \dots), \quad U_N|_{u_{i+1}=u_i+j\lambda} = S_j U_{\text{red}}, \quad j = 2, 3.$$

The merge replaces $(q^{-1}x, qx)$ by x ; the deletion removes (x, q^3x) . Both V_N and U_N are cyclically covariant. Furthermore, U_N is covariant under reversing the row order, reflecting the pairing, and replacing u_i by $3\lambda - u_i$.

Proof. Translate every open-strip argument by $u_i \mapsto u_i - iB$. Then $a_i = \cos(2u_i - 2iB - 3\lambda) \sim rx_i$, where $r = e^{2B-3i\lambda}/2$. Define V_N, h_N as the limits of $r^{-K_N^*}W_N, r^{-K_N^*}p_N$. The scalar Pfaffian gives Equation (3). Its successive leading coefficients are h_{N-1} , so it is nonzero.

For the vector limit, use the open last-variable interpolation. At a deletion node its scalar factor is $O(r^{2N-3})$ and the size- $N-2$ vector is $O(r^{K_N^*-2})$ by induction. Transporting boxes, cardinal polynomials at the last argument, and sine ratios remain bounded for fixed generic offsets. The vacant interpolation also contains $W_{N-1} \prod_{i < N} (a_N - a(u_i + 3\lambda))$. Since $2N-3 + K_{N-2}^* = K_N^*$, all terms are $O(r^{K_N^*})$. Laurent polynomiality then gives the limit and homogeneity. Passing the interpolation to the limit yields the stated last-slot degrees and leading coefficient. Alternatively its coefficients are bounded by interpolation at fixed generic offsets. Braid transport moves any slot to last; the moving boxes stay bounded as that variable tends to zero. Thus no negative powers remain in any slot. The inherited upper exponent bounds and occupancy signs give all the asserted per-slot bounds.

The open braid identities pass to the limit because the boxes depend only on differences. The highest homogeneous terms of the open scalar quotient rules show that the splitting specializations of h_N are generically nonzero, so division gives the displayed identities for U_N .

For cyclic covariance, induct on N and compare the two cyclic orders in the same output slots. Their empty components agree by symmetry of h_N . On an ascending split among the first $N-1$

slots, both cyclic orders reduce to the same smaller vector: the splitter occupies the same output slots, and the remaining orders agree by induction. Braid transport within these slots therefore gives zeros at $u_b = u_a + j\lambda$ for every $a < b < N$ and $j = 2, 3$. Every nonempty component has an occupied slot $k < N$. For each of the $N - 2$ other earlier slots, these two split values give distinct generic roots in x_k . After dividing out $\sqrt{x_k}$, the component has degree at most $N - 2$, whereas it has $2(N - 2)$ such roots. This proves the claim for $N \geq 3$. For $N = 1$ only the empty component occurs; for $N = 2$ the cup component is a constant times $\sqrt{x_1 x_2}$, which is already cyclically invariant.

The Pfaffian also gives

$$h_N(x_1^{-1}, \dots, x_N^{-1}) = h_N(x_1, \dots, x_N) \prod_i x_i^{-(N-1)}. \quad (4)$$

Let J reverse the row order and reflect the pairing. The reflected quotient, after multiplication by the original h_N , is the vector

$$q^{-3K_N^*} \left(\prod_i x_i^{N-1} \right) JV_N(q^3/x_N, \dots, q^3/x_1).$$

In a vacant slot, inversion reverses exponents within $0, \dots, N - 1$. In an occupied slot it reverses exponents within $1/2, \dots, N - 3/2$. Thus this vector has exactly the original per-slot polynomial and parity bounds. Reversing order together with $u_i \mapsto 3\lambda - u_i$ preserves ascending splits and braid covariance, since the local operators are invariant under slot-order reflection. On each split the two cleared vectors agree by induction. Transporting the split to any other site gives two distinct generic roots for each of the $N - 1$ other arguments. These $2(N - 1)$ roots exceed the degree bound $N - 1$ (or $N - 2$ after an occupied square root is removed). The size-one case is immediate. $\square$

Lemma 2.4 (Algebraic stationarity). *Let $T(z)$ pass an auxiliary slot, vacant or occupied, through rows $1, \dots, N$ with boxes $R(z - u_i)$ and close the auxiliary output to its input on the annulus. With every closed loop given weight zero,*

$$T(z)U_N = U_N$$

as a rational identity in the row and auxiliary parameters.

Proof. At $z = u_i$ the corresponding box is the identity. Propagation through the other rows braids that row around the circle, so braid and cyclic covariance give stationarity at these N values. To obtain the second set of interpolation nodes, let J reflect the order of the physical slots and their pairing, and put $u'_i = 3\lambda - u_{N+1-i}$. Reversing the traversal of the closed auxiliary line gives the crossing relation

$$JT(z; u_1, \dots, u_N)J^{-1} = T(6\lambda - z; u'_1, \dots, u'_N). \quad (5)$$

Here the local crossing bends the auxiliary input and output in the opposite direction: it exchanges a single vertical connection with a cap or cup, exchanges two vertical connections with a cap–cup pair, and preserves the empty and single diagonal connections. Its coefficient permutation is therefore $(1, A, B, U, E, F) \mapsto (1, U, B, A, F, E)$. The displayed weights verify it without a scalar factor:

$$\mathcal{D}(3\lambda - v) = \mathcal{D}(v), \quad A(3\lambda - v) = U(v), \quad B(3\lambda - v) = B(v), \quad E(3\lambda - v) = F(v).$$

Apply this local move successively around the auxiliary circle; its encounter order reverses, which is exactly the relabeling by J . Every resulting closed component still has weight zero. Since $(6\lambda - z) - u'_i = 3\lambda - (z - u_{N+1-i})$, the operator obtained is precisely the right side of Equation (5). By the reflection covariance of the vacuum, $JU_N(u) = U_N(u')$. At $z = u_i + 3\lambda$ the reflected spectral

parameter is $3\lambda - u_i = u'_{N+1-i}$, an identity-box node for that state. This proves stationarity at the additional N nodes.

Multiply $(T(z) - I)U_N$ by $\prod_i \mathcal{D}(z - u_i)$. As a Laurent polynomial in e^{iz} it has exponents between $-2N$ and $2N$. All exponents are even, since occupancies on the closed auxiliary occur in pairs. There is one further leading-coefficient cancellation. As $e^{iz} \rightarrow \infty$, the local coefficients satisfy

$$A, U \longrightarrow 0, \quad B \longrightarrow 1, \quad E \longrightarrow k = e^{6i\lambda}, \quad F \longrightarrow k^{-1}.$$

Here the unsubscripted U denotes the local tile coefficient. A vacant auxiliary gives the identity. For an occupied auxiliary, skip vacant sites and pass it across an adjacent cup of the occupied-site pairing. Joining the two boxes gives coefficient $FE = 1$ for preserving the cup, and coefficient $E^2 + F^2 = k^2 + k^{-2} = 0$ for the other connection. Closed cycles already have weight zero. Removing cups successively leaves an auxiliary circle, again of weight zero. Thus the highest coefficient of the residual vanishes. Multiplication by e^{2Niz} now gives a polynomial in e^{2iz} of degree at most $2N - 1$, with $2N$ generic roots. This proves stationarity. $\square$

2.3 Convergence of the physical cylinder state

Algebraic stationarity does not yet identify U_N with a positive infinite-volume sum. For that identification we need decay at each fixed circumference. The next lemma also supplies the spanning-path estimate used by the annular pressure in Section 9.

Fix N and physical row parameters $u_i \in \{\lambda, 2\lambda\}$. The cylinder is the quotient by the period $\sum_i e^{iu_i/t}$. A column cut is the chain of successive upward rhombus sides; its ports are their midpoints. Let $b_M(i, j)$ be the critical mass of paths crossing M columns from west port type i to east port type j , meeting these two delimiting cuts only at their endpoints and remaining self-avoiding after projection to the cylinder.

Lemma 2.5 (Decay across a periodic slab). *For every such fixed cylinder there are constants $C_N < \infty$ and $0 < r_N < 1$ such that $b_M(i, j) \leq C_N r_N^M$ for all endpoint types and every $M \geq 1$.*

Proof. A cylinder path lifts to a planar self-avoiding path and uses at most $O(NM)$ vertices. Proposition 2.1 therefore bounds $b_M(i, j)$ and all same-side path sums by $\exp(o(M))$, with N fixed. A finite transfer on partial matching states, with one unmatched port attached to the source and every closed cycle discarded, computes $b_M(i, j)$. Such connectivity records all necessary information: distinct columns have disjoint vertices. Hence $\sum_M b_M(i, j)z^M$ is rational.

In fact $b_M(i, j)$ decays exponentially in M . Otherwise, because its coefficients are nonnegative and subexponential and its generating function is rational, some entry has $\sum_M b_M(i, j) = \infty$. We construct a projection-invalid connector for these precise endpoint types. A single column in the planar lift is an infinite chain of rhombi, with row labels periodic modulo N . Starting at its west port of type j , take the single west-to-north turn, then the single south-to-north connection through successive rhombi. Choose an integer k with $i + kN > j + 2N$ and take the south-to-east turn in row $i + kN$. The three kinds of one-arc tile used here have strictly positive weights at both physical parameters $\lambda, 2\lambda$; only one arc is placed in each tile. Since different rhombi have disjoint interiors, the lifted path is self-avoiding and meets the column's west and east cuts only at its endpoints. After projection, however, its intermediate south-to-north route repeats vertices in a row visited one period earlier. This is a width-one connector from type j to type i with some positive weight a , depending only on the fixed cylinder.

Concatenate lifts of bridges counted by the divergent entry with copies of this connector, always in successive disjoint column slabs. Their interiors are disjoint, so the resulting planar path is

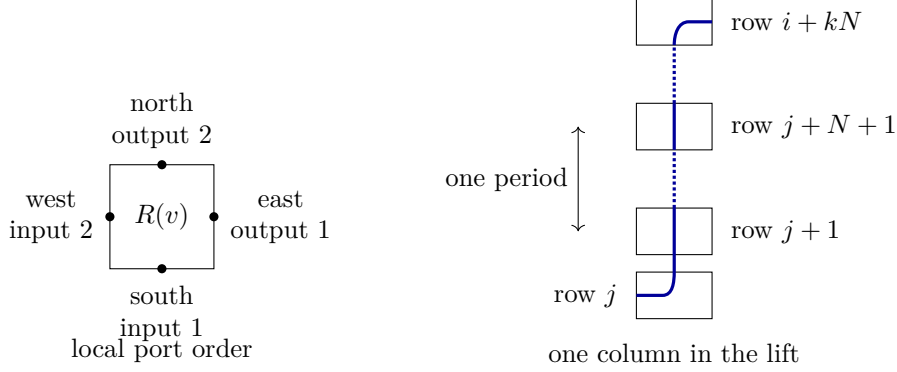


Figure 1: The boxes show cyclic port order, rather than physical angles. On the right, dotted portions contain the omitted intermediate rhombi. The two displayed south–north passages are one period apart, so they visit the same vertices after projection onto the cylinder. In the planar lift all rhombi used are distinct.

self-avoiding. To recover the concatenation, list all column cuts that the path meets exactly once. Between two adjacent column cuts in this list, a width-one subbridge inside any original valid slab is still projection-valid. The inserted connectors are precisely the projection-invalid width-one subbridges: they occupy one entire column and their two cuts are crossed only at their endpoints. Thus the connectors, the intervening slabs, and their original paths are uniquely recovered from the concatenated path.

Choose finitely many valid slab widths whose total mass is B with $aB > 1$. The width generating function of one valid-slab–connector block is $g(z) = a \sum_M b_M(i, j) z^{M+1}$, with the sum restricted to those widths. It satisfies $g(0) = 0$ and $g(1) > 1$. Hence the generating function $\sum_{r \geq 0} g(z)^r$ of recoverable concatenations has radius strictly less than one. Each slab has at most a fixed constant times its width in vertices, because its projection is self-avoiding on a fixed cylinder; the connector has bounded length. Consequently all these planar paths of total width w have length at most $C_N w$. Their total critical weight is bounded above by the sum of $c_n \rho^n$ for $n \leq C_N w$, up to fixed endpoint factors. This is $\exp(o(w))$ by Proposition 2.1, contradicting the radius just obtained. $\square$

Proposition 2.6 (Physical cylinder vacuum). *At physical row parameters $u_i \in \{\lambda, 2\lambda\}$, the column $T(3\lambda)$ restricted to vectors with zero empty component has spectral radius less than one. Its unique stationary state with empty component one is the regular continuation of U_N . It equals the absolutely convergent sum of half-cylinder configurations with vacant far boundary. The right-half state is $U_N(3\lambda - u_1, \dots, 3\lambda - u_N)$.*

Proof. A nonzero diagram propagated from a nonempty input pairing either has a west-to-east spanning path or closes a discarded cycle. The number of path segments is bounded at fixed N . Discarding their mutual avoidance bounds their weights by a product of an exponentially decaying spanning sum from Lemma 2.5 and subexponential same-side sums. Therefore the transfer on the subspace of zero empty component has spectral radius strictly below one. The empty component itself is invariant. Writing the transfer in these two blocks proves uniqueness and local analyticity of its normalized fixed vector, and convergence from a vacant far boundary proves the half-cylinder interpretation. The rational stationary vector V_N/h_N agrees with it wherever defined nearby, hence extends to the physical parameters. Horizontal reflection gives the stated right-half arguments. Vertical reflection also recovers the covariance proved algebraically above. $\square$

2.4 The finite separating-loop formula

Split the cyclic list into consecutive groups X, Y of sizes s, b , where $N = s + b$, and mark the two seams. Define $Z_{X,Y}(\chi)$ by contracting the left and right vacuum states, assigning a closed loop weight χ if it crosses the X portion of the cut an odd number of times, and weight zero otherwise. The empty configuration contributes one. If a group is empty all loops have weight zero, so $Z = 1$. At physical parameters the markers are face centers. Compactifying the cylinder ends gives a sphere, on which the odd-intersection rule is precisely separation of the markers. Therefore $Z_{X,Y}(\chi)$ counts disjoint polygon families all separating the markers, with weight $\prod_\ell \chi \rho^{|\ell|}$. Every such loop meets the cut; disk connectivity on its two sides suffices to test separation. The half-cylinder sums converge absolutely, and a family has a bounded number of loops at fixed N , so this interpretation is valid for every fixed χ .

For the balanced observable Z_N of the introduction, $N = 2m$ and the prescribed row list is m copies of 2λ followed by m copies of λ . Identify the triangular lattice with the complex plane and start its column cut at the face center $f_1 = 0$. Its successive edges have these row vectors, so the second marker and period are

$$f_2 = m e^{2\pi i/3} = -\frac{N}{4} + i \frac{N\sqrt{3}}{4}, \quad V = m(e^{2\pi i/3} + e^{\pi i/3}) = i \frac{N\sqrt{3}}{2}.$$

The two groups are the two portions of this prescribed cut from 0 to f_2 and from f_2 to V . Translation or reflection gives the same observable. In particular the markers are half a period apart in height; their longitudinal displacement is fixed by this construction.

Lemma 2.7 (Characterization of the contraction). *The rational functions $Z_{X,Y}(\chi)$ are uniquely determined by the following properties, with χ fixed.*

1. $H_{X,Y} = h_N(X, Y)^2 Z_{X,Y}$ is polynomial of degree at most $2N - 2$ in each argument, and Z is symmetric within each group.
2. An ascending merge or deletion within either group leaves Z unchanged after making the corresponding replacement or deletion.
3. Sending one argument to zero or infinity omits that site.
4. Sending $x \in X$ and $y \in Y$ to infinity at fixed generic ratio gives

$$Z_{X,Y} \longrightarrow Z_{\hat{X}, \hat{Y}} Z_{x,y}, \quad Z_{x,y} = 1 + \chi(2c)^2 \frac{xy}{(x+y)^2}.$$

5. At size zero, $Z = 1$.

All limits are rational limits at generic values of the remaining arguments. Hats mean omission of the indicated sites.

Proof. The right-half arguments are q^3/x_i . Equation (4), homogeneity, and the per-slot degree bounds for V_N give the polynomial bound for H : occupied half-powers occur in pairs in the contraction. To exchange sites within a group, move a symmetric braid box from one side of the contraction to the other by transposition. The two states braid in opposite directions because the starred arguments reverse differences. Loop parity is preserved: following the arcs in the box layer shows that each resulting loop has an even total number of X -hits on the two interfaces of that layer. Thus any cycle discarded on one side has zero gluing weight as well. This proves symmetry and

also permits a same-group splitting layer to be moved through the contraction. At the descending specialization on the starred side, apply $R(j\lambda) = S_j S_j^\dagger$ and the ascending splitting identity. To cancel S_j , project its output onto diagrams whose first new slot is vacant. For S_2 , omitting that slot leaves the diagonal map with coefficients 1 on a vacancy and ρ on an occupied strand. For S_3 , both new slots are then vacant; omitting them gives the identity. Thus S_j is injective, and $S_j U_{\text{red}} = S_j S_j^\dagger U_{\text{desc}}$ implies $S_j^\dagger U_{\text{desc}} = U_{\text{red}}$. This proves the reduction of Z .

Cyclic covariance and the last-slot leading coefficient show that sending one argument to infinity inserts a vacant site in the left vacuum. Reflection gives the same statement in the right vacuum. Horizontal reflection gives the limit at zero.

For completeness, the simultaneous two-site limit requires more than the individual limits. Reorder within groups so that x, y are adjacent across a marker, and rotate them to the last two positions. Write P for the other $N - 2$ arguments. We claim

$$\frac{V_N(P, x, y)}{(xy)^{N-2}} = V_{N-2}(P) \otimes V_2(x, y) + o(|x|), \quad V_2(x, y) = (x + y)\emptyset + 2c\sqrt{xy} \text{ cup}. \quad (6)$$

Fix square-root branches on the scaling ray. In the interpolation in y , the deletion nodes are $y_z = q^3 z$ for $z \in (P, x)$. For $N \geq 3$, nodes belonging to P have data $O(|x|^{N-1})$, cardinal factors $O(|x|^{N-3})$, and additional occupied square-root ratios at most $O(|x|^{1/2})$. After division by $(xy)^{N-2}$ they are $o(|x|)$. The leading vacant interpolation term becomes $(y - y_x)V_{N-2} \otimes \emptyset + o(|x|)$. At $y_x = q^3 x$ the data are

$$(x + y_x) \prod_{p \in P} (p - q^{-3}x)(p - q^6x) S_3 V_{N-2}(P).$$

The cardinal factor is asymptotic to $(y/y_x)^{N-2}$, with an extra $\sqrt{y/y_x}$ on occupied output. Since $q^{-3}q^6 = q^3$, these terms give exactly the remaining terms in Equation (6). The expression for V_2 follows directly from the size-two interpolation. Division by the empty components gives factorization of the normalized left vacuum, and reflection gives that of the right vacuum. The intersection-parity rule factors on the two subsystems, proving both the pair limit and the displayed $Z_{x,y}$.

To prove uniqueness, the zero-site value and omission already settle $N = 1$. At $N = 2$ with one site in each group, omission at zero and infinity leaves only a possible difference Cxy of the cleared numerators. Its quotient by $h_2^2 = (x + y)^2$ has a nonzero paired scaling limit unless $C = 0$, so the specified pair limit settles this case. For all other group sizes, induct on N and subtract two candidate cleared polynomials. The known scalar reductions of h_N turn the reductions of Z into identical values of these polynomials at each split. Omission at zero kills their constant coefficient, and omission at infinity kills their leading coefficient. In a variable of a group of size s , symmetry and the two reductions supply $4(s - 1)$ roots, namely $q^{\pm 2}$ and $q^{\pm 3}$ times every other same-group argument. If $s > b$, these roots, together with the zero constant coefficient, exceed the remaining degree $2N - 3$. If $s = b \geq 2$, the difference must be a constant multiple of

$$\left(\prod_i x_i \right) \prod_{\substack{i < j \\ \text{same group}}} (x_i^2 + x_j^2) D_0(x_i, x_j). \quad (7)$$

Indeed its degree in each variable is $1 + 4(s - 1) = 2N - 3$, exactly the remaining bound, so the quotient has degree zero in every variable. When one argument from each group tends to infinity, this factor has the same leading order as h_N^2 and a nonzero generic coefficient. The pair limit forces the rational difference to tend to zero, so its constant multiplier vanishes. The size-zero and size-two conditions cover the remaining balanced cases. $\square$

Using the fixed square roots, define the centered normalization $\tilde{h}_N = h_N \prod_i x_i^{-(N-1)/2}$. Its square is

$$\tilde{h}_N^2 = \frac{h_N^2}{\prod_{i < j} x_i x_j} = \frac{h_N^2}{\prod_i x_i^{N-1}}.$$

The three labels as formal site strings. Before writing the label sum, we construct its pair weights. Give a site in group $\sigma = +1$ on X or $\sigma = -1$ on Y a generic logarithmic coordinate a_i , with $x_i = e^{a_i}$. Here a_i denotes a logarithm, distinct from the earlier open-strip cosine coordinate. Attach a formal string of length $a \in \{0, 1, 2\}$ at

$$a_i + i\sigma\beta, \dots, a_i + i\sigma a\beta, \quad l = \sigma(1 - a).$$

These are prescribed spectral points, with no integration variables yet; at length zero the string is empty. Thus the three labels record the three possible string lengths. At this stage we assign weights only between distinct site strings; factors within one string, which have poles at these spectral points, will later be evaluated by residues. We will first verify the resulting finite algebraic formula, and then realize these strings as residues of a particle integral.

Put $P_j(r) = -2 \sinh((r + ij\beta)/2)$. Products below are evaluated at the coordinate difference of their arguments. For two objects in the same group, the site–site, particle–site, and particle–particle factors are, respectively,

$$B_s = P_2 P_{-2} P_3 P_{-3}, \quad F_s = (P_1 P_{-1})^{-1}, \quad E_s = P_0^2 P_2 P_{-2} F_s. \quad (8)$$

For different groups always use the plus coordinate minus the minus coordinate, and use

$$B_c = P_2^2 P_3^2, \quad F_c = P_1^{-2}, \quad E_c = P_0^2 P_2^2 F_c. \quad (9)$$

The same-group factors are even, so no ordering convention is needed there.

For ordered groups σ, ω and site labels l, k , let $a = 1 - \sigma l$ and $b' = 1 - \omega k$, with the plus group first when they differ. Let z be the difference of the two site coordinates. Multiply their site–site factor, all factors from a particle in either string to the other site, and all factors between particles in different strings. Denote this mutual product by $\mathcal{S}_{lk}^{\sigma\omega}(z) = \prod_j P_j(z)^{n_j}$. No factor internal to either string occurs. The exponents are encoded by the Laurent polynomial

$$\sum_j n_j \xi^j = B - F(A + D) + EAD, \quad A = \sum_{h=1}^a \xi^{\sigma h}, \quad D = \sum_{h=1}^{b'} \xi^{-\omega h}. \quad (10)$$

Here, for $\sigma = \omega$,

$$B = \xi^2 + \xi^{-2} + \xi^3 + \xi^{-3}, \quad F = \xi + \xi^{-1}, \quad E = 2 + \xi^2 + \xi^{-2} - F;$$

for $\sigma = 1, \omega = -1$,

$$B = 2\xi^2 + 2\xi^3, \quad F = 2\xi, \quad E = 2 + 2\xi^2 - F.$$

Labels l, r belong to $\{1, 0, -1\}$, also denoted $+, 0, -$, and $v_l = 2 + l^2$. For a pair of sites set $e = 1$ if they belong to the same group and $e = -1$ otherwise. The arrays below are indexed by $j \in \mathbb{Z}/8\mathbb{Z}$. A bracket $[a, b, \dots]$ denotes the sum of the corresponding unit vectors; repeated entries are counted

with their multiplicity. Rows and columns are ordered $+, 0, -$; a matrix entry 0 is the zero vector, whereas an entry $[0]$ is the unit vector at residue zero:

$$\begin{aligned} m^+ &= \begin{pmatrix} [2, 3, 5, 6] & [3, 5, 6] - [0] & [5, 6] - [0, 1] \\ [2, 3, 5] - [0] & [2, 3, 5, 6] - [1, 7] & [3, 5, 6] - [0] \\ [2, 3] - [0, 7] & [2, 3, 5] - [0] & [2, 3, 5, 6] \end{pmatrix}, \\ m^- &= 2 \begin{pmatrix} 0 & [5] & [5, 6] \\ [3] & [4] & [5] \\ [2, 3] & [3] & 0 \end{pmatrix}. \end{aligned} \quad (11)$$

Define

$$Q_{lr}^e(u) = q^{ev_l r} u^{-1-elr} \prod_{j=0}^7 (1 - uq^{-j})^{m_{lr}^e[j]}. \quad (12)$$

Substitution of $a, b' \in \{0, 1, 2\}$ gives $\sum_{j \equiv -h} (8) n_j = m_{lk}^{\sigma\omega}[h]$. Since

$$P_j(z) = e^{-(z+ij\beta)/2} (1 - e^z q^j),$$

the ratio of $Q_{lk}^{\sigma\omega}(e^z)$ to the string product is $q^{\sigma\omega v_l k + \sum_j j n_j / 2}$. Differentiating Equation (10) at $\xi = 1$ gives the pair-phase identity

$$\frac{Q_{lk}^{\sigma\omega}(e^z)}{\mathcal{S}_{lk}^{\sigma\omega}(z)} = q^{\sigma\omega(v_l k + v_k l)/2} \begin{cases} 1, & \sigma = \omega, \\ (-1)^{1-lk}, & \sigma = 1, \omega = -1. \end{cases} \quad (13)$$

Theorem 2.8 (Finite cylinder formula). *For $\chi = q^{-3}\tau + q^3\tau^{-1}$ and $\tau \neq 0$,*

$$\tilde{h}_N^2 Z_{X,Y}(\chi) = \sum_{\substack{l_i \in \{-1, 0, 1\} \\ \sum_{i \in X} l_i = \sum_{i \in Y} l_i = M}} \tau^M (-2c)^{\sum_i l_i^2} \prod_{i < j} Q_{l_i l_j}^{e(i,j)}(x_i/x_j). \quad (14)$$

The sum includes all allowed integers M . Apparent singularities at colliding arguments are evaluated by continuation.

Proof. We verify the characterization in Lemma 2.7. The arrays satisfy

$$\sum_j m_{lr}^e[j] = 2 + 2elr, \quad \sum_j j m_{lr}^e[j] \equiv e(v_l r - v_r l) \pmod{8},$$

and their exponents reverse under transposition. These identities give $Q_{lr}^e(u) = Q_{rl}^e(1/u)$ and hence within-group symmetry. The only possible finite nonzero poles are same-group poles at $u = 1, q, q^{-1}$. At $u = 1$ only unequal labels are singular, with simple poles, and exchanging these labels cancels the residues. At $u = q$ the singular labels are $(+, -)$ and $(0, 0)$. The first kernel divided by the second tends to $-1/(4c^2)$; the ratio of the corresponding spectator products for a spectator of label k and relative type e is q^{ek} . Neutrality makes the product of these spectator ratios one. The site weight ratio is $4c^2$, so the residues cancel. Reciprocity treats q^{-1} . Cross-group kernels have no such poles.

Here are the finite calculations that also verify the reductions. For a same-group pair x, y , put $u = x/y$ and $w = qy$:

u	(l, r)	nonzero value or pole coefficient
q (pole)	$(+, -), (0, 0)$	$-i(2 + 2\sqrt{2}), \quad 2i\sqrt{2}$
q^2	$(+, 0), (+, -), (0, -)$	$2q, \quad (2c^2)^{-1}, \quad 2q^{-1}$
q^3	$(+, -)$	1

In the first row the entries mean pole coefficients, obtained after multiplication by $1 - u/q$. At q^2 , set $L = l + r$ and $J = v_l + v_r - v_L - l + r$; the three surviving cases have $J = 1, 2, 1$. Their spectator ratios are

$$\frac{Q_{lk}^e(x/z)Q_{rk}^e(y/z)}{Q_{Lk}^e(w/z)} = q^{eJk} \frac{(w+z)^2}{wz}.$$

At q^3 the product of spectator factors is

$$Q_{+k}^e(x/z)Q_{-k}^e(y/z) = q^{3ek} \frac{(z - q^{-3}y)^2(z - q^6y)^2}{xyz^2}.$$

These identities require only the following identities of the exponent arrays, with indices taken modulo eight:

$$\begin{aligned} m_{+k}^e[j+1] + m_{-k}^e[j] &= m_{0k}^e[j+1] + m_{0k}^e[j], \\ m_{lk}^e[j+1] + m_{rk}^e[j-1] - m_{Lk}^e[j] &= 2\delta_{j,4} \quad (u = q^2), \\ m_{+k}^e[j+3] + m_{-k}^e[j] &= 2\delta_{j,2} + 2\delta_{j,3} \quad (u = q^3). \end{aligned}$$

The constants follow from $1 - q^a = -2ie^{ia\beta/2} \sin(a\beta/2)$. For the spectators, neutrality gives $\sum ek = -(l+r)$. Including site weights, the q^2 reduction is therefore multiplication by

$$2 \prod_z \frac{(w+z)^2}{wz},$$

and the q^3 reduction is multiplication by

$$(2c)^2 \prod_z \frac{(z - q^{-3}y)^2(z - q^6y)^2}{xyz^2}.$$

These are precisely the homogeneous scalar quotient rules for $\tilde{h}_N^2$; division verifies the two reductions of Z .

For a single argument x with label l , reciprocity and neutrality bound the product of its interactions by order x^{N-1-l^2} at infinity and $x^{-(N-1)+l^2}$ at zero. Only $l = 0$ contributes at leading order. Its phases multiply to one, leaving respectively $\prod_z x/z$ and $\prod_z z/x$, exactly the leading factors of $\tilde{h}_N^2$. Pole cancellation and these bounds also prove the required polynomial degree after removing the centering.

For $x \in X, y \in Y$ tending to infinity together, interactions with the remaining arguments have order $|x|^{2(N-2)-(l-r)^2}$. Only $l = r$ survives at leading order. Neutrality cancels the phases and ratio powers, leaving $\prod_z xy/z^2$ times the separate pair and remaining expressions. This agrees with Equation (6) for the scalar normalization. Finally, the two-site sum over the three equal labels is

$$\frac{(x+y)^2}{xy} + \chi(2c)^2,$$

and the zero-site sum is one. Lemma 2.7 now proves the formula. $\square$

2.5 Contour sectors and residue cancellation

For the asymptotic application, $s = b = m$, $N = 2m$, and the arguments in X, Y will be respectively $1, q^{-1}$. These are obtained from the physical row parameters 2λ on X and λ on Y , whose x -values are q^2, q , by the common rescaling $x_i \mapsto q^{-2}x_i$. Homogeneity of V_N and h_N leaves their quotient

and the contraction unchanged. We first keep the two rays apart and allow distinct real offsets. Denote the groups by $\sigma = +1, -1$, in that order, and set

$$\alpha_+ = 0, \quad \alpha_- = -\varepsilon\beta, \quad 0 < \varepsilon < 1.$$

Specialize the logarithmic site coordinates to $\Im a_i = \alpha_{\sigma_i}$. Eventually $\varepsilon = 1$ and the real offsets coalesce to zero.

A particle of group σ at shift j has coordinate $u + i\alpha_\sigma + i\sigma j\beta$, with $u \in \mathbb{R}$ and measure $du/(2\pi)$. There are three species in each group:

species	shifts of its elementary particles	activity
main singleton	0	p_σ
far singleton	11/4	$-p_\sigma$
far double	5/2, 7/2	$4c^2 p_\sigma^2$

The signed far species make the residue representation exact. When the main contours are moved outward, particles left on the far contour cancel against the negative far singletons; a particle attaching to one of those singletons cancels an original far double. The surviving configurations will therefore be strings of zero, one, or two particles attached to each site. The proof below establishes this cancellation with the factorials and phases retained.

The two constituents of a double share one real variable. For fixed numbers $n_\sigma, t_\sigma, D_\sigma$ of the three species, multiply all site–site factors, all particle–site factors, and all particle–particle factors between elementary constituents, omitting the internal E_s between the two constituents of a single double. Integrate over the real variable of every singleton and double. Multiply the integral by

$$\prod_{\sigma=\pm 1} \frac{p_\sigma^{n_\sigma} (-p_\sigma)^{t_\sigma} (4c^2 p_\sigma^2)^{D_\sigma}}{n_\sigma! t_\sigma! D_\sigma!}.$$

Let I denote the sum of these terms through total elementary degree $\sum_\sigma (n_\sigma + t_\sigma + 2D_\sigma) \leq N$. This finite truncation is sufficient for every coefficient used below; no convergence assertion for an infinite activity series is needed.

Theorem 2.9 (Contour-sector representation). *For the prescribed site rays, every $0 < \varepsilon < 1$, and every $\eta \in \mathbb{C}$, put $\chi = 2 \cos \eta$. Then*

$$\tilde{h}_N^2 Z_{X,Y}(2 \cos \eta) = (2d)^N \sum_{M=-\min(s,b)}^{\min(s,b)} (-1)^{sb+M} e^{i\eta M} [p_+^{s-M} p_-^{b+M}] I. \quad (15)$$

The coefficient integrals are absolutely convergent. The identity extends to $\varepsilon = 1$ and to coincident real site offsets by continuation and dominated convergence of the fused integrands.

Proof. Integrability and contour geometry. All coefficient terms on the right have exactly N elementary particles, because $(s - M) + (b + M) = N$. A particle at real coordinate u receives decay $O(e^{-N|u|})$ from the N sites; a pair interaction grows at most $O(e^{|u-u'|})$. The inequality $|u - u'| \leq |u| + |u'|$ therefore bounds the full integrand away from its prescribed poles by a constant times $\exp(-\sum |u|)$, counting elementary coordinates with multiplicity. The same estimate applies to smaller elementary degree and to fused particles or residues. Constants may depend on N , sites, and the fixed contours. This proves the convergence needed for contour movements and removal of their vertical ends.

For $\varepsilon < 1$, temporarily move the far singletons to a common shift h and the far doubles to $h - 1, h$, where

$$3 < h < \frac{7 - \varepsilon}{2}.$$

The offsets are measured outward from each group's site ray, in units of $i\sigma\beta$:

	fixed offsets	temporary offsets
main singleton	0	$0 \rightarrow h$
far singleton	$11/4$	h
far double	$(5/2, 7/2)$	$(h - 1, h)$
site residues	none initially	1 or $(1, 2)$.

A parenthesized pair shares one real coordinate. Only the far contours are moved in this first step; the main-contour arrow is the subsequent deformation. Moving the far contours crosses no pole. Indeed the interaction of a double with any particle has no poles after its two constituents are multiplied. Relative to the double midpoint, the same-group fused $E_s E_s$ has only positive exponents at shifts $\pm 1/2, \pm 5/2$; the cross-group product has exponent two at $-1/2, 5/2$. The same observation applies to pairs of doubles. Far particle-site poles are avoided because all far constituent shifts stay in $(2, 7/2]$. For opposite-group singletons the cross difference lies strictly between zero and seven in shift units.

Residues and cancellation of unbound particles. Move the main contours from shift zero to shift h . A same-group site can collect an ordered string of length $a = 0, 1, 2$, at its own coordinate plus $i\sigma\beta, \dots, i\sigma a\beta$. The first attachment occurs at shift one; interaction with it permits a second at shift two. An attempted third at shift three is canceled by the zero in its interaction with the first. Quadratic coincidence zeros likewise cancel attempts to attach twice at the same root. A main particle can also attach one unit below an original far singleton, converting it into a double. These are all possibilities: fused doubles have no particle poles, and $2h + \varepsilon < 7$ excludes cross-group poles.

Here is an ordering of the deformation that avoids any ambiguity from moving poles. Divide $[0, h]$ into generic layers of thickness less than $1/3$, separating the levels 2 and $h - 1$. In each layer move the remaining unbound variables one at a time; freeze every picked residue immediately, including during the rest of that layer. Unbound variables in a common group then never differ by a full unit of shift, so they do not create poles with one another. For fixed generic distinct real coordinates of original far particles, also distinct from site offsets, these are ordinary one-variable contour moves across simple poles. An attached far spectator appears as the complete double interaction. An unconverted far singleton is more than one shift unit beyond every site-bound root. Thus residual integrands have no unaccounted coincident-spectator singularity and satisfy the preceding decay bound. The identities can consequently be integrated over the original far spectators as well.

The internal weight of a string, including its factors with its own site and excluding activities, is

$$r_0 = 1, \quad r_1 = \frac{1}{2d}, \quad r_2 = \frac{c^2}{d^2}. \quad (16)$$

With measure $du/(2\pi)$, the oriented residue multiplier is $i\sigma$. The first F_s pole contributes $1/(2d)$. The second attachment contributes $-4c^2$ from the E_s pole at relative $i\sigma\beta$, and $F_s(2i\sigma\beta) = -1/(2d)$ from its site factor. Multiplication gives r_2 . Attachment at relative $-i\sigma\beta$, below a far singleton, instead contributes $4c^2$.

Fix a site-string pattern. The effective far singleton activity is $-p_\sigma + p_\sigma = 0$, from original and unpicked main singletons. The effective far double activity is $4c^2 p_\sigma^2 + 4c^2 (-p_\sigma) p_\sigma = 0$, from original doubles and attached original singletons. These cancellations respect all factorials. Explicitly, start

in one group with n, t, D particles of the three species. If r main particles fill the specified site strings and a attach to a of the original far targets, summing labeled assignments leaves factorial coefficient

$$\frac{1}{(n-r-a)!(t-a)!a!D!}.$$

Each assignment determines exactly one contour term through its choices at shifts $1, 2, h-1$. The remaining two kinds of singleton and two kinds of double can therefore be combined by the binomial formula. This argument is coefficientwise in p_+, p_- , so all canceling terms have the same elementary total. Only completely site-bound configurations survive. In particular every cancellation used for the degree- N coefficients takes place inside the finite truncation defining I .

Identification with the finite label formula. Assign label $l = \sigma(1-a)$ to a site in group σ with string length a . The coefficient constraints become $\sum_X l_i = \sum_Y l_i = M$. We check their mutual weights explicitly. Use the formal string weights and their pair-phase identity (13) established before the finite formula. To multiply these phases without a size restriction, put $\sigma_i = 1$ on X , $\sigma_i = -1$ on Y , and $v_i = 2 + l_i^2$. Neutrality gives $\sum_i \sigma_i l_i = 0$, while $l_i^3 = l_i$. Hence

$$\begin{aligned} \frac{1}{2} \sum_{i < j} \sigma_i \sigma_j (v_i l_j + v_j l_i) &= \frac{1}{2} \left[\left(\sum_i \sigma_i v_i \right) \left(\sum_i \sigma_i l_i \right) - \sum_i v_i l_i \right] \\ &= -\frac{3}{2} \sum_i l_i = -3M. \end{aligned}$$

The cross sign is $(-1)^{sb - (\sum_X l_i)(\sum_Y l_i)} = (-1)^{sb - M^2} = (-1)^{sb+M}$. Thus the product of all pair phases is exactly $q^{-3M}(-1)^{sb+M}$. Finally, since $\sum_i a_i^{\text{len}} = N$ for the string lengths a_i^{len} ,

$$(2d)^N \prod_i r_{a_i^{\text{len}}} = (-2c)^{\sum_i l_i^2}.$$

Indeed the per-site ratio of the right-hand weight to $(2d)^a r_a$ is $(-2c)^{1-a}$, whose product is one. Taking $\tau = q^3 e^{i\eta}$ in Theorem 2.8 now proves Equation (15).

Confluent physical limits. Return the far contours to the fixed shifts in the definition of I . Their fused pair products have no pole at $\varepsilon = 1$. On these fixed contours the same decay bound persists in a neighborhood of that value and of confluent real site offsets. This justifies both final limits. $\square$

Lemma 2.10 (Signs at the confluent rays). *At $\varepsilon = 1$ with confluent real site offsets, extract constant signs from the site factors and from each full pair of integration variables, multiplying double constituents before doing so, so that the value at zero real difference is positive, or its leading quadratic coefficient is positive when it vanishes. Their logarithms then satisfy conjugation under $u \mapsto -u$. They are regular on $\mathbb{R}$ except for same-species coincidences with singular part $2 \log |u - u'|$, including double-double coincidences. At fixed far counts, multiply the extracted signs by the explicit prefactor $(-1)^{sb+M}$ in Equation (15). The resulting sign has M -dependence exactly $(-1)^{fM}$, up to an M -independent sign, where $f = f_X + f_Y$ is the number of far elementary particles. The remaining explicit phase is $e^{i\eta M}$.*

Proof. For real r , $P_j(-r) = -\overline{P_j(r)}$. Each fused product has even total exponent, which gives the asserted conjugation after a real sign has been extracted. Real zeros of an elementary factor occur only at $r = 0$ and $j \in 8\mathbb{Z}$. Combine the exponents before testing for a zero, in particular for the two constituents of a double. Ordered by main singleton, far singleton, and far double, the extracted

pair signs and vanishing orders are

	sign matrix	order matrix at $r = 0$
same group	$\begin{pmatrix} + & - & + \\ - & + & + \\ + & + & + \end{pmatrix}$	$2I_3$
different groups	$\begin{pmatrix} - & - & + \\ - & - & + \\ + & + & + \end{pmatrix}$	0.

The site-particle signs are $(+, -, +)$ in the same group and $(-, -, +)$ in different groups, with zero vanishing order in both cases. These tables follow directly from $P_j(0) = -2i \sin(j\beta/2)$ and $P_{8k}(r) = -(-1)^k r + O(r^3)$: shift the exponents of $P_0^2 P_2 P_{-2}/(P_1 P_{-1})$ or $P_0^2 P_2^2/P_1^2$ by every constituent difference, add them, and cancel opposite powers. Thus there are no other real singularities.

Let $n_\sigma, t_\sigma, D_\sigma$ count main singletons, far singletons, and far doubles. Their elementary totals are $K_+ = s - M$ and $K_- = b + M$, with $K_\sigma = n_\sigma + t_\sigma + 2D_\sigma$. The cross-group site and pair signs in the table have parity

$$bK_+ + sK_- + K_+K_- \equiv sb + M \pmod{2}.$$

This cancels the prefactor in Equation (15). The remaining negative factors are a same-group main-far-single pair, a same-group far-single-site pair, and each far-single activity. Their total parity is

$$(n_+ + s + 1)t_+ + (n_- + b + 1)t_- \equiv M(t_+ + t_-) \equiv Mf \pmod{2},$$

because $t_\sigma^2 \equiv t_\sigma$ and $f = t_+ + t_- + 2D_+ + 2D_-$. Doubles contribute only positive signs in the tables. Any fixed extraction of the site-site signs is independent of M . This proves the required imbalance phase. $\square$

2.6 The Pfaffian denominator at confluence

The reference-kernel reduction uses Schur's Pfaffian identity and Cauchy's determinant identity; see [15, equations (1.1)–(1.2)]. We include their elementary forms in the present normalization. The confluent kernel and its determinant asymptotic then require the operator estimates below. This is an estimate for equal group sizes. The contour identity above holds for unequal sizes as well, but the present denominator theorem does not assert an asymptotic uniform in that larger parameter domain.

In this subsection $N = 2m$, the first m logarithmic sites are zero, and the last m are $-i\beta$, all understood as confluent limits. Set

$$\gamma = \frac{4}{3} = \frac{\pi}{3\beta}, \quad L = \frac{\log N}{\gamma}, \quad H_*(u) = \frac{e^{2u} + 2de^u + 1}{(e^u - 1)e^{u/2}} \tanh \frac{\gamma u}{2}.$$

At $u = 0$ use the removable continuation of H_* .

Theorem 2.11 (Balanced denominator asymptotic). *For $N = 2m \geq 2$, with exactly m sites at each of $0, -i\beta$, the normalization h_N is nonzero. As $N \rightarrow \infty$ through these even sizes,*

$$\log \left| \frac{\tilde{h}_N^2}{(\gamma/2)^N \prod_{i < j} H_*(a_i - a_j)^2} \right| = \frac{5}{24} \log N + o(\log N). \quad (17)$$

Proof. First take distinct sites. Write

$$K(u) = \frac{\sinh u}{\cosh u + d}, \quad K_0(u) = \tanh \frac{\gamma u}{2}.$$

Equation (3) and the definition of H_* give

$$\frac{\tilde{h}_N^2}{\prod_{i < j} H_*(a_i - a_j)^2} = \frac{\det[K(a_i - a_j)]}{\det[K_0(a_i - a_j)]}. \quad (18)$$

Here we used the Pfaffian identity

$$\text{Pf} \left[\frac{v_i - v_j}{v_i + v_j} \right] = \prod_{i < j} \frac{v_i - v_j}{v_i + v_j}, \quad v_i = e^{\gamma a_i}.$$

To prove it, clear denominators: the numerator is an alternating polynomial of Vandermonde degree and is therefore a constant times the Vandermonde. The residue at $v_{N-1} = -v_N$, followed by induction from size two, determines the constant to be one.

For an even-dimensional skew matrix A , adding the all-ones matrix does not change its determinant. Indeed the rank-one determinant formula gives the extra term $\mathbf{1}^\top \text{adj}(A) \mathbf{1}$, which is zero since $\text{adj}(A)$ is skew. Apply this to both matrices in Equation (18), then conjugate by the diagonal factors $e^{-\gamma a_i/2}$. The resulting reference kernel is

$$G_0(u) = e^{-\gamma u/2} (1 + K_0(u)) = \text{sech}(\gamma u/2),$$

and the actual kernel is $G(u) = e^{-\gamma u/2} (1 + K(u))$. With Fourier convention $\hat{f}(k) = \int_{\mathbb{R}} e^{-iku} f(u) du$ and $\zeta = \gamma/2 + ik$, their transform ratio is

$$\frac{\hat{G}(k)}{\hat{G}_0(k)} = \gamma \frac{\cos(\beta\zeta) \sin(\pi\zeta/\gamma)}{\sin(\pi\zeta)} = \frac{\gamma}{2} (1 + R(k)), \quad R(k) = \frac{1}{2 \cos(2\beta\zeta)}. \quad (19)$$

For the first equality, put $t = e^u$ and use

$$1 + K(u) = \frac{t}{t + e^{i\beta}} + \frac{t}{t + e^{-i\beta}}, \quad \int_0^\infty \frac{t^{p-1}}{1+t} dt = \frac{\pi}{\sin \pi p}.$$

The beta integral follows by a keyhole contour, initially for $0 < \Re p < 1$. The second equality in Equation (19) follows from $\pi/\gamma = 3\beta$ and $\pi = 4\beta$ by the triple-angle identity.

The determinant ratio is now explicit, but direct differentiation at coalescing sites would obscure its size. We instead view the reference matrix as a Gram matrix. Its orthogonal projection will occupy an interval of length $2\gamma^{-1} \log N$ in the real direction; the trace density on that interval will give the required logarithmic term.

We turn the determinant ratio into a finite-dimensional compression. Let $\mathcal{H}$ be the Hilbert space with norm

$$\|f\|_{\mathcal{H}}^2 = \int_{\mathbb{R}} |\hat{f}(k)|^2 v(k) \frac{dk}{2\pi}, \quad v(k) = \frac{1}{\hat{G}_0(k)} = \frac{\gamma}{2\pi} \cosh \frac{\pi k}{\gamma}.$$

Its functions have bounded evaluation at every z with $|\Im z| < \pi/(2\gamma)$. Write k_z for the evaluation vector, and $P = P_N$ for the orthogonal projection onto the span of evaluations and their derivatives of orders $0, \dots, m-1$ at $z = \pm i\beta/2$. Centering the two site rays changes no differences. The derivative evaluation vectors are linearly independent. Indeed, a linear relation among them, written in Fourier variables and multiplied by the common nonzero weight, has the form $p_+(k)e^{\beta k/2} + p_-(k)e^{-\beta k/2} = 0$

on $\mathbb{R}$, where p_+, p_- have degree at most $m - 1$. Sending k to $+\infty$ gives $p_+ = 0$, and then $p_- = 0$. Their Gram matrix is consequently positive definite.

Start with distinct paired real offsets and permute columns by complex conjugation; the reference matrix becomes the Gram matrix of these evaluation vectors. The actual matrix represents the sesquilinear form with the Fourier multiplier $\frac{\gamma}{2}(I + R)$ inserted. Passing to divided differences gives the derivative vectors at confluence. Hence

$$\frac{\tilde{h}_N^2}{(\gamma/2)^N \prod_{i < j} H_*(a_i - a_j)^2} = \det \left(I + PRP|_{\text{ran } P} \right). \quad (20)$$

The same column permutation is used in numerator and denominator, so it introduces no sign in their ratio. The reference Gram matrix is nondegenerate by the preceding independence argument. Moreover $\text{Re } R(k) \geq 0$ for real k : since $2\beta \text{Re } \zeta = \pi/3$, the real part of $\cos(2\beta\zeta)$ is $\frac{1}{2} \cosh(2\beta k) > 0$. Thus $I + PRP$ is strictly accretive and invertible. Since the removable factors $H_*(0)$ and $H_*(i\beta)$ are nonzero, the compression identity already proves $h_N \neq 0$ at these balanced confluent sites. The rest of the proof determines its logarithmic size.

The projection has a useful exact localization formula:

$$\|(I - P)k_z\|^2 = \|k_z\|^2 \prod_{\pm} \left| \frac{e^{\gamma z} - e^{\pm i\gamma\beta/2}}{e^{\gamma z} + e^{\pm i\gamma\beta/2}} \right|^{2m}. \quad (21)$$

To see this, augment the reference Gram matrix by k_z and take its Schur complement. Under $w = e^{\gamma z}$, its kernel is $1/(w + \overline{w})$ up to separate one-point factors. The Cauchy determinant formula follows by clearing denominators and using antisymmetry, with the constant fixed by residues. Its ratio after augmentation is exactly Equation (21). Divided differences give its confluent version and prove the nondegeneracy of the derivative Gram matrix.

For each fixed horizontal line in the evaluation strip, this implies

$$\|Pk_z\|^2 \leq C \min\{1, Ne^{-\gamma|\Re z|}\}. \quad (22)$$

Indeed, for $a = e^{\pm i\gamma\beta/2}$ and $w = e^{\gamma z}$,

$$1 - \frac{|w - a|^2}{|w + \overline{a}|^2} = \frac{4\Re w \Re a}{|w + \overline{a}|^2} \asymp e^{-\gamma|\Re z|}.$$

The product in Equation (21) is therefore within $CNe^{-\gamma|\Re z|}$ of one, and is at most $\exp(-cNe^{-\gamma|\Re z|})$. In particular, translated to the frame of any real x with $|x| \leq (1 - \delta)L$, the projections P_N converge strongly to the identity, uniformly for fixed $\delta > 0$. The complement tends uniformly to zero on every fixed evaluation vector, and these vectors span a dense subspace by Fourier uniqueness; contraction of the projections extends this to all of $\mathcal{H}$.

Integrating Equation (22) on horizontal lines near the two edges of the evaluation strip gives, for every fixed $\epsilon > 0$,

$$\text{Tr}(Pe^{-\epsilon|k|}P) \leq C_\epsilon L. \quad (23)$$

Here multipliers are written as functions of k . More explicitly, Plancherel on the line of height y gives the multiplier $v^{-1}e^{-2ky}$, and its trace against P is $\int_{\mathbb{R}} \|Pk_{x+iy}\|^2 dx$. Choose y near each strip edge. The sum of the two resulting positive multipliers dominates a constant multiple of $e^{-\epsilon|k|}$, while each integral is $O(L)$ by Equation (22).

We claim, for each integer $p \geq 1$,

$$\lim_{N \rightarrow \infty} \frac{\text{Tr}((PRP)^p)}{2L} = \int_{\mathbb{R}} R(k)^p \frac{dk}{2\pi}. \quad (24)$$

First truncate the exponentially decaying multiplier R to a bounded frequency interval. For any multiplier H ,

$$\|PHP\|_1 \leq \text{Tr}(P|H|P),$$

by factoring through $|H|^{1/2}P$ and using the Hilbert–Schmidt inequality. Equation (23) makes the normalized trace error tend to zero uniformly as the frequency interval grows; a telescoping product gives the same statement for each fixed power.

For a bounded compactly supported symbol S , cyclicity of the trace writes the corresponding trace as

$$\text{Tr}(v^{-1}(Sv)P(SP)^{p-1}).$$

The resolution $\int_{\mathbb{R}} |k_x\rangle\langle k_x| dx = v^{-1}$ turns this trace into $\int_{\mathbb{R}} F_N(x) dx$, where

$$F_N(x) = \left\langle (Sv)P(SP)^{p-1}k_x, k_x \right\rangle_{\mathcal{H}}.$$

The real-line evaluation vectors have a fixed norm. The multiplier Sv is bounded on the compact frequency support, and the rightmost operator acting on k_x contains P . Hence $|F_N(x)| \leq C\|Pk_x\|$. Translate k_x to k_0 and conjugate every P by the same real translation. The multipliers are unchanged. In the bulk $|x| \leq (1 - \delta)L$, strong convergence of these translated projections therefore gives, uniformly,

$$F_N(x) \longrightarrow \langle S^p v k_0, k_0 \rangle_{\mathcal{H}} = \int_{\mathbb{R}} S(k)^p \frac{dk}{2\pi}.$$

The transition region has length $O(\delta L)$; beyond $(1 + \delta)L$, Equation (22) has an integrable exponential tail of size $o(L)$. Divide by $2L$, then let $\delta \downarrow 0$. This proves the compact-frequency limit and hence Equation (24).

On the real frequency axis, $|R(k)| \leq 1$ and $\Re R(k) \geq 0$: here $2\beta\Re\zeta = \pi/3$. Moreover $\|PRP\|_1 = O(L)$ by Equation (23). For $0 < t < 1$, the logarithmic power series and Equation (24) yield

$$\lim_{N \rightarrow \infty} \frac{1}{2L} \text{Tr} \log(I + tPRP) = \int_{\mathbb{R}} \log(1 + tR(k)) \frac{dk}{2\pi}.$$

Accretivity gives $\|(I + tPRP)^{-1}\| \leq 1$ for $t \geq 0$, so the trace derivative is bounded by $\|PRP\|_1 = O(L)$. The same integrable bound holds for the limiting integral. Thus the limit extends to $t = 1$. Taking real parts is equivalent to taking the logarithm of the determinant modulus in Equation (20).

Finally shift the contour from $\Re\zeta = \gamma/2$ to $\Re\zeta = 0$. The function $\log(1 + 1/(2\cos(2\beta\zeta)))$ is analytic in the intervening strip, with the branch tending to zero at imaginary infinity; its horizontal-end contributions vanish exponentially. Therefore

$$\int_{\mathbb{R}} \log(1 + R(k)) dk = \int_{\mathbb{R}} \log\left(1 + \frac{1}{2\cosh(2\beta k)}\right) dk = \frac{5\pi^2}{72\beta}.$$

For the last evaluation use evenness and set $y = e^{-2\beta k}$ for $k > 0$. The logarithm becomes $\log[(1 - y^3)/((1 - y)(1 + y^2))]$. Termwise integration of its convergent logarithmic series, or monotone limits of their integrals, gives

$$\frac{1}{\beta} \int_0^1 \frac{\log(1 - y^3) - \log(1 - y) - \log(1 + y^2)}{y} dy = \frac{1}{\beta} \left(-\frac{\pi^2}{18} + \frac{\pi^2}{6} - \frac{\pi^2}{24} \right) = \frac{5\pi^2}{72\beta}.$$

Combining this with $2L/(2\pi)$ and $\beta\gamma = \pi/3$ gives the coefficient $5\pi/(72\beta\gamma) = 5/24$ in Equation (17). $\square$

3 Centering, coercivity, and the fluctuation spectrum

The finite contour formula expresses the cylinder partition function as an oscillatory sum of particle integrals. We first identify its exact mean density. Subtracting this density leaves an interaction whose real part is positive and therefore controls both the particle counts and excursions beyond a logarithmic wall. We then compare the interior interaction with a reference ensemble whose normalization and linear statistics can be evaluated. The energy estimates have two uses. The untruncated real energy estimate controls every valid particle count; the sharper imaginary trial fields identify the tilt cost in the fixed-fugacity exponent. Both use the same exact compensated-sector identity, Lemma 3.3. The determinant asymptotic in Theorem 3.10 supplies the remaining fluctuation contribution. Lemmas 3.11 and 3.12 then give the finite-dimensional approximation needed to realize that determinant by moving actual particle coordinates.

3.1 The exact mean density

Throughout this section we specialize Theorem 2.9 to $N = 2m$, $s = b = m$, $\varepsilon = 1$, all real site offsets zero. Fourier transforms in the real coordinate use e^{-iku} and inverse measure $dk/(2\pi)$. A particle is specified by its real coordinate u and its ray label; the imaginary part of its logarithmic coordinate is determined by that label. An interaction matrix kernel acts by convolution; symmetric means $H^{JI}(-u) = H^{IJ}(u)$, and forms with complex arguments are continued *bilinearly*. Write

$$\gamma = \pi/(3\beta), \quad L = (\log N)/\gamma, \quad p = 2\beta, \quad \kappa = (3\beta)^2.$$

Labels are $I = (\sigma, J)$, where $\sigma \in \{+1, -1\}$ and $J = 0, F, P$ denote the main, far-single and far-pair species. Their elementary shifts are, respectively, 0, $11/4$, and $(5/2, 7/2)$, as in Theorem 2.9. Their multiplicities w_I are 1, 1, 2; a pair still has only one real integration variable. Write H for minus the sign-normalized particle-particle log interaction, V for the log (without negation) of the sign-normalized site-particle interaction (particle row, site column). Differences in these kernels take the first real coordinate minus the second, e.g. the exponentiated factors use $H^{II'}(u - u')$ with minus sign for labels I, I' . More explicitly use linear combinations of

$$K_\theta(u) = -\log[2(\sin(\theta/2) \cosh(u/2) - i \cos(\theta/2) \sinh(u/2))], \quad K_0(u) = -\log|2 \sinh(u/2)|$$

reducing angles to $[0, 2\pi)$. Thus H uses the exponents of the log-sinh factors in E (fused for pair labels), and V the negative exponents in the site-particle product; cross terms in reverse order transpose and reflect the kernel. The exponents at angle zero in H are $2\delta_{IJ}$ by the shift lists; there are no angle-zero terms of V . These conventions agree with the sign normalization of the contour-sector identity.

Modulo distributions at zero frequency the transforms of K_θ are

$$\pi e^{(\pi-\theta)k}/(k \sinh(\pi k)), \quad \widehat{K}_0 = \pi \cosh(\pi k)/(k \sinh(\pi k)).$$

Indeed $K_\theta'' = \frac{1}{4}/\sinh^2((u + i\theta)/2)$; shifting by a period and taking the double pole residue gives $-\pi k e^{(\pi-\theta)k}/\sinh(\pi k)$. The second formula follows by even averaging as $\theta \downarrow 0$. Affine terms in comparisons below are fixed by tails: real parts $-|u|/2 + O(e^{-|u|})$, imaginary parts odd with limits given by $\frac{1}{2}(\pi - \theta) \operatorname{sign} u$ for $0 < \theta < 2\pi$. For the main (near) labels,

$$\widehat{H}_{nn}(k) = \frac{2\pi c_*(k)}{k \sinh(\pi k)} \begin{pmatrix} \cosh(3\beta k) & e^{pk} \\ e^{-pk} & \cosh(3\beta k) \end{pmatrix}, \quad c_*(k) = 2 \cosh(\beta k) - 1.$$

Define $\hat{\varrho} = 1/c_*$. Shifting by a period and taking simple residues gives

$$\varrho(u) = \frac{e^{-\gamma|u|} - e^{-5\gamma|u|}}{\sqrt{3}\beta(1 - e^{-6\gamma|u|})}.$$

The value at zero is its removable continuation. This is a positive, even, analytic probability density, and $\varrho(u) \asymp e^{-\gamma|u|}$. The lattice activity is denoted by ρ .

Let μ have density $m\varrho$ on each main label and vanish on the four far labels. We use (f, Hg) for the sum of the corresponding double integrals over labels and real coordinates.

Lemma 3.1 (Exact centering). *The particle-site kernel satisfies $V = H_n * \varrho$. Combining the site-site factors with $\exp((\mu, H\mu)/2)$ gives*

$$C_*^N \prod_{i < j} H_*(a_i - a_j)^2, \quad C_* = \frac{H_*(0)}{\sqrt{B_s(0)}} = \frac{(1+d)\gamma}{4d \cos(\beta/2)},$$

with the fixed signs prescribed by the contour-sector formula. Here the logarithmic site coordinates a_i are the balanced confluent sites: m copies of 0 and m copies of $-i\beta$. The function H_* is the removable analytic function used in Theorem 2.11.

Proof. On the main rows the identity follows from the displayed Fourier symbols. For an elementary far shift j in group σ , divide the H -row versus near columns (own group first) by $2\pi c_*/(k \sinh \pi k)$; it is $(\cosh(\beta k) e^{\sigma(4-j)\beta k}, e^{\sigma(p-j\beta)k})$, as follows using $2 < j < 5$. These entries give V after convolution for far shifts as well. Equality at nonzero frequencies suffices (matching real tails, with zero limiting difference, and bounded odd imaginary parts).

For a site pair, the sign-normalized log of B_s or B_c plus the corresponding entry of $\varrho * H_{nn} * \varrho$ at real log difference equals the normalized log of H_*^2 at the site log difference (with imaginary shift). For the same-group formula the symbol divided by $2\pi/(k \sinh \pi k)$ on the left is $-\cosh(2\beta k) - \cosh(\beta k) + \cosh(3\beta k)/c_*(k)$. On the right it is $-2\cosh(\beta k) + \cosh(4\beta k) - \sinh(\pi k) \tanh(3\beta k/2)$, the same expression. To check the second expression, factor $\cosh u + d$ and $\sinh(u/2)$ in H_* , and use the scaled $K_\pi - K_0$ identity

$$\log |\widehat{\tanh(\gamma u/2)}|(k) = -\frac{\pi}{k} \tanh \frac{\pi k}{2\gamma}.$$

Cross entries acquire the factor $e^{-\beta k}$ (extra $\sinh$ terms cancel since $\sinh(3\beta k)/c_* = \sinh(2\beta k) + \sinh(\beta k)$); H_* is analytic and nonzero through this shift, so its log acquires the same shift in the transform of the second derivative. Again real tails have matching slopes and zero constant, and imaginary parts are bounded odd. In particular the exponentiated half self term of $\varrho * H_{nn} * \varrho$ is

$$C_* = \frac{H_*(0)}{\sqrt{B_s(0)}} = \frac{(1+d)\gamma}{4d \cos(\beta/2)}.$$

In $B_s(0)$ the argument is the log difference. Thus all one-particle terms can be centered exactly using $H\mu$; site pairs together with the mean factor $\exp((\mu, H\mu)/2)$ yield $\prod_{i < j} H_*(a_i - a_j)^2$ times C_*^N (up to fixed signs). □

3.2 Positivity across the six species

Exact centering is useful only if the remaining energy controls every species, including far particles. We prove that control on arbitrary real fluctuations; no sign condition on the main components is needed. For a real fluctuation $x = (x_I)$ of zero *weighted total* mass, write

$$d_x = \sum_I w_I x_I, \quad D = \partial^{-1} d_x, \quad r = \sum_{I:\sigma=+1} w_I x_I, \quad B(x) = \frac{1}{2}(x, \operatorname{Re} H x)$$

with primitives starting at $-\infty$. Initially take smooth rapidly decreasing densities.

Lemma 3.2 (Coercivity of the real interaction). *Suppose that the components of x are initially smooth rapidly decreasing real densities and that $\int d_x = 0$. The tail $-w_I w_J |u|$ contributes exactly $\|D\|_2^2$; residual real kernels are integrable. We have*

$$B(x) \asymp \|D\|_2^2 + \sum_I \|x_I\|_{-1/2}^2, \quad \|x_I\|_{-1/2}^2 = \int (1+k^2)^{-1/2} |\hat{x}_I|^2 dk / (2\pi).$$

The form extends accordingly and controls $\|D\|_{1/2}^2 = \int (1+k^2)^{1/2} |\hat{D}|^2 dk / (2\pi)$ also.

Proof. Divide the real-kernel transform by $\pi/(k \sinh \pi k)$; the matrix in group blocks has diagonals S , off-diagonals T . With $h_j = \cosh(j\beta k/4)$, upper triangles listed by rows, they are

$$\begin{aligned} S : & 2(h_{16} - h_{12} + h_8), \quad h_{13} - h_1 + h_3 + 2h_5 - h_9, \quad h_{14} + h_2 + 2h_6; \\ & 2(h_{16} - h_{12} + h_8), \quad h_{13} + h_{15} + h_5 + h_7; \\ & 2(h_{12} + h_{16} + h_4 + h_8). \\ T : & 2(h_{12} + h_4 - h_8), \quad 2(h_1 - h_3 + h_7), \quad 2(h_{10} + h_2); \\ & 2h_{10}, \quad 2(h_{11} + h_9); \\ & 4(h_{12} + h_8). \end{aligned}$$

These follow by summing log-factor exponents with angles modulo 2π . Expand in $v = \cosh(\beta k/4) - 1$. At zero $S = T = 2ww^\dagger$, $w = (1, 1, 2)^\dagger$. All coefficient matrices of $S \pm T$ of positive degrees through 16 are strictly positive definite after division by the corresponding coefficient of h_{16} . Use the ratios

$$R_i(l) = i^2 \prod_{h=1}^{l-1} (i^2 - h^2) / (16^2 \prod_{h=1}^{l-1} (16^2 - h^2))$$

by the cosh polynomial differential equation $(2v + v^2)h_i'' + (1 + v)h_i' = i^2 h_i$ as polynomials in v . For clarity the first three normalized S, T triangles are:

l	S_{11}, S_{12}, S_{13}	S_{22}, S_{23}	S_{33}
1	11/8, 73/128, 17/16	11/8, 117/64	15/4
2	203/136, 483/1360, 849/1360	203/136, 1703/1360	939/340
3	685/408, 5849/22848, 23/51	685/408, 3163/3264	1133/476

l	T_{11}, T_{12}, T_{13}	T_{22}, T_{23}	T_{33}
1	3/4, 41/128, 13/16	25/32, 101/64	13/4
2	35/68, 19/272, 413/1360	165/544, 175/272	513/340
3	9/28, 293/22848, 55/476	55/476, 2035/7616	155/204

Both sum and difference have positive successive leading minors (they exceed $1/2, 1/4, 1/10$ respectively by substitution). For degrees $l \geq 4$, using $0 \leq R_i(l) \leq R_i(4)$ gives absolute off-diagonal bounds $0.21, 0.39, 0.90$ and diagonal lower bounds $1.60, 1.75, 1.59$, enough by diagonal dominance. This coefficient calculation gives the asserted norm comparison as follows. Near zero, the constant rank-one term contributes $\|D\|_2^2$, and the strictly positive degree-one coefficients control the remaining directions at constant scale. On each compact frequency interval avoiding zero, positivity and continuity give uniform upper and lower bounds. At large $|k|$, the $h_{16} = \cosh(4\beta k)$ term is diagonal and dominates all other entries. Since $\pi = 4\beta$, multiplication by $\pi/(k \sinh \pi k)$ gives the scale $|k|^{-1}$ there. These three ranges prove the two-sided Sobolev bound. Finally $\hat{d}_x = ik\hat{D}$, and $d_x = \sum_I w_I x_I$, so the same bound controls the stated $H^{1/2}$ -norm of D . $\square$

The full norm comparison controls counts. The leading low-frequency terms will also identify the optimal imaginary tilt. In the following calculation, Fourier transforms are implicit on the displayed variables. At small k the pointwise quadratic symbol of B , using variables D, r, x_{far} and suppressing hats for these variables, is

$$|D|^2 + Q(r, x_{\text{far}}) + O(|k|)(|D|^2 + |r|^2 + |x_{\text{far}}|^2).$$

Here Q is a constant positive definite real form and $Q(r, 0) = (\kappa - p^2)|r|^2$. Indeed insert $d_x = ikD$ in the matrices just given. With rows versus near columns, $\text{Im } H$ has a term proportional to $\text{sign } u$; the remaining odd kernel J_o has multiplier bounded by $C \min(|k|, (1 + |k|)^{-1})$ (integrable with moments, piecewise smooth of bounded variation). For the main rows the sign coefficients form the antisymmetric matrix with upper entry p . A far row differs from w_I times its corresponding main row; the additional coefficients against $+$, $-$ are sums over shifts j of

$$((4 - j)\beta, -j\beta) \quad (\sigma = +), \quad (j\beta, (j - 4)\beta) \quad (\sigma = -).$$

In particular the difference of the two additional coefficients is πw_I , with the same sign on both groups of far rays. Write g_I for the second (summed) additional coefficient.

3.3 Interior and explicit particles

We retain the collective fluctuations of particles near their mean and estimate the remaining particles separately. Split the main contours at $\pm L$. Keep as explicit particles ζ all main ones outside the wall and all far particles everywhere, regarded as positive counting measures by label (unordered, with the factorial measures of the sector formula). Write n_e for their total number of variables, $f = \sum_{I:\text{far}} w_I \int \zeta_I$, and let

$$m'_\sigma = m - \sigma M - \sum_{I:\sigma_I=\sigma} w_I \int \zeta_I.$$

The remaining interior variables are main-label particles with totals m'_σ . Splitting including factorials is exact by symmetry. The physical sector has nonnegative integer m'_σ . In variational estimates we also use signed near densities α_σ on $[-L, L]$ with these same prescribed totals, and we allow M to be complex. This enlargement concerns trial densities only; it does not change the integer dimensions of a particle integral.

We regularize only the interaction *between* interior and explicit particles (denoted by subscripts n, e), as follows. Fix $h > 0$ small; it stays fixed through $N \rightarrow \infty$. Take a positive convolution

probability $\psi = \psi^{(h)}$ with transform the average of $\cosh((\pi - ht)k)/\cosh(\pi k)$ for $0 < t < 1$, with density proportional to $\exp(-1/\sqrt{t})$. These kernels in real space before averaging are

$$\frac{\cosh(u/2) \sin(ht/2)}{2\pi(\sinh^2(u/2) + \sin^2(ht/2))}$$

by Fourier inversion (elementary sech integral by residues and imaginary translation). Thus ψ is even, bounded by C/h , with tail bounded by $Ch e^{-|u|/2}$ at $|u| \geq 1$, and $\int u^2 \psi \leq Ch$. Its transform is bounded above and below by $C \exp(-c(h|k|)^{1/3})$ and $c \exp(-C(h|k|)^{1/3})$ respectively, by optimizing $ht|k| + 1/\sqrt{t}$ (for the lower bound integrate a comparable interval); $1 - \hat{\psi} \leq Chk^2$.

Put $S_0 = 2K_0$. Then $S_0 \geq S_0 * \psi \geq S_0 * \psi * \psi$ on the real line, since convolution once replaces K_0 by averaged $\text{Re } K_{ht}$, by the transform formulas with matching tails. Let χ be an even smooth cutoff equal to 1 on $[-1/2, 1/2]$, zero off $[-1, 1]$, $0 \leq \chi \leq 1$, with derivatives bounded by $C^{l+1}(l!)^2$. Such cutoffs follow by integrating/rescaling bumps formed from $\exp(-1/x)$ on $x > 0$ zero-extended, by Cauchy estimates on circles of radius cx . Subtract from H_{ne} , only on same-label entries (thus involving an exterior main particle), the real nonnegative kernel

$$R_h(u) = \chi(u)(S_0 - S_0 * \psi)(u).$$

Call the resulting interaction $\tilde{H}_{ne}$, extending R_h by zero on other entries. The interaction $\tilde{H}_{ne}$ is smooth Gevrey (some finite order, for fixed h) with at most linear growth and bounded higher derivatives of each positive order. Here and below Gevrey bounds follow equivalently for bounded decaying terms from stretched exponential Fourier estimates; for example differentiating under a bound $C \exp(-c|k|^\tau)$ gives $C^{l+1}(l!)^{1/\tau}$ bounds, and multiplication of Gevrey functions preserves a finite order by Leibniz. For $S_0 * \psi$ one can first take two derivatives in the Fourier estimates.

For smooth trial densities define the complex energy

$$\mathcal{E}(\alpha, \zeta) = \frac{1}{2}(\alpha - \mu, H_{nn}(\alpha - \mu)) + (\alpha - \mu, \tilde{H}_{ne}\zeta) + \frac{1}{2}(\zeta, H\zeta)_{\text{off}}.$$

The last term omits all individual explicit self terms. One may fix generic distinct explicit real positions (null sets of coincidences never needed). Trial bounds will apply for example to smooth densities, or integrable densities of finite log energy with sufficient decay, by approximation. Point configurations have logarithmic self singularities, so this formula must be compensated before it is used in a particle integral. The following reference kernel makes that compensation explicit. Let $u \mapsto t$ be an increasing bijection $[-L, L] \rightarrow [0, \pi]$ smooth with strictly positive derivative in the open interval; set $p_L = dt/\pi$ as a probability there in u (we will specify the map). Use

$$H_{\text{ref}}(t, t') = -2 \log |2(\cos t - \cos t')|$$

separately on each main group, zero cross. Its p_L -mean vanishes by circle reflection (mean of $\log |2 \sin(t/2)|$ is zero by its Fourier series). Use this kernel for independent reference integrals with measures

$$\frac{1}{\nu_\sigma!} \prod_{i=1}^{\nu_\sigma} \frac{2 \sin t_i dt_i}{2\pi} \prod_{i < j} |2(\cos t_i - \cos t_j)|^2.$$

Under $x = \cos t$ this is the Legendre orthogonal-polynomial ensemble. Its normalization by polynomial norms uses Andréief's identity for the integral of a product of two determinants; see [22, equations (1.3), (3.1), and Section 5]. The same identity will give the tilted factorial densities in the fluctuation calculation. For interior configurations X_ν with these variables, put $q_\sigma = X_{\nu,\sigma} - \nu_\sigma p_L$. The smooth (near-diagonal compensated) observable uses

$$\Phi_M(q) = \mathcal{E}(m' p_L + q, \zeta) - \frac{1}{2}(q, H_{\text{ref}} q).$$

This notation denotes a single smooth-kernel pairing, not a difference of two infinite empirical self energies. More explicitly, put $a = m'p_L - \mu$ and

$$K = H_{nn} - H_{\text{ref}}, \quad b = H_{nn}a + \tilde{H}_{ne}\zeta.$$

The reference kernel is understood in the physical coordinates on each main component. Since its p_L -mean is zero,

$$\Phi_M(q) = \mathcal{E}(m'p_L, \zeta) + (b, q) + \frac{1}{2}(q, Kq).$$

All diagonal values in the last pairing are the finite continuous values of K ; the coordinate construction below proves the needed smoothness. This is also the meaning of Φ_M when $q = X_\nu - \nu p_L$ is a centered empirical distribution.

Lemma 3.3 (Compensated sector representation). *Fix an explicit configuration ζ with generic distinct real coordinates and a physical integer imbalance M for which $m'_\sigma \geq 0$. After division of the right-hand side of (15) by $(\gamma/2)^N \prod_{i < j} H_*(a_i - a_j)^2$, its contribution at M, ζ is*

$$c_\zeta e^{i(\eta + \pi f)M} \int A(X_{m'}, \zeta) \exp\{-\Phi_M(X_{m'} - m'p_L)\}, \quad A(X, \zeta) = \exp\{-(X, R_h\zeta)\}.$$

The integral includes the two unnormalized reference measures. The explicit integration measures and their species factorials are retained outside it. The prefactor c_ζ is independent of M , satisfies $|c_\zeta| \leq \exp(C_h n_e)$, and uses the fixed sign convention of Lemma 2.10. Moreover $0 \leq A \leq 1$. Sectors with a negative m'_σ contribute zero.

Proof. Restoring the omitted self terms of $H_{nn} - H_{\text{ref}}$ contributes per interior variable

$$(c/d) 2 \sin t t'(u),$$

since $E_s(u) \sim (d/c)^2 u^2$. Combined with $du/(2\pi)$, this is c/d times the one-particle reference measure. The scalar cancellation is

$$(2d)C_*(c/d) = \gamma/2.$$

Replacing $\tilde{H}$ by H gives exactly the additional factors A and $\exp((\mu, R_h\zeta))$. The latter is at most $\exp(C_h n_e)$: the cutoff can meet μ only within distance one of an exterior main particle, where $|u| \geq L - 1$, and its logarithmic singularity is integrable. Missing c/d factors and far activities each cost a fixed constant per explicit variable. Finally the sector sign calculation leaves precisely the phase $e^{i(\eta + \pi f)M}$, with all remaining signs independent of M at fixed ζ . This proves both the identity and the prefactor bound. $\square$

The identity is algebraic on distinct interior configurations, hence holds almost everywhere in the reference integrals. The coordinate choice below gives the smooth reflected extension of K needed for distributional estimates at the walls. For later interpolation we also *define* the same compensated integrand with complex M, m' and independent nonnegative integer reference dimensions ν , always setting $q = X_\nu - \nu p_L$. This extension keeps q centered at the actual integer reference dimension; only $\nu = m'(M)$ gives the physical identity of the lemma.

We now compare this regularized energy with the positive real form. The loss from omitting the explicit self interactions will be linear in the number of explicit variables.

Lemma 3.4 (Uniform smoothing inequality). *For the trial energy defined above, set*

$$x = \text{Re } \alpha - \mu + \psi * \zeta, \quad z = \text{Im } \alpha$$

with all species included for x , only near entries nonzero for z . These are weighted-total neutral. Then

$$\operatorname{Re} \mathcal{E}(\alpha, \zeta) \geq B(x) - B(z) - (x, \operatorname{Im} H z) - s(h)(B(x) + B(z) + 1) - C_h n_e, \quad s(h) \rightarrow 0. \quad (25)$$

Proof. At nonzero frequency,

$$|\hat{H}_{ne} - \hat{H}_{ne} \hat{\psi}| \leq C h^{c_0} (1 + |k|)^{-1} \hat{\psi}(k)$$

for some $c_0 > 0$, with no residual distributions at zero. Indeed nonsingular terms have transform $O(e^{-c|k|}(1 + k^{-2}))$, multiplied by $1 - \hat{\psi}$. For the singular S_0 term the difference in real space is $(1 - \chi)(S_0 - S_0 * \psi)$ with derivatives in L^1 bounded by $O(h^2)C^{l+1}(l!)^2$ by even averaging of imaginary shifts off the singularity, and the cutoff bounds. This gives $O(h^2) \exp(-c\sqrt{|k|})$ in frequency, sufficient by the lower bound for $\hat{\psi}$. Thus the mixed-kernel error is bounded by $C h^{c_0} \|\alpha - \mu\|_{-1/2} \|\psi * \zeta\|_{-1/2}$. For explicit-explicit pairs only the real kernel matters. Every smooth real summand (apart from diagonal S_0) changes under $\psi * \psi$ averaging by $O(h)e^{-c|u|}$, using exponentially decaying second derivatives and the tail and second moment of ψ . The diagonal singularity can only decrease. And the missing same-variable diagonals in the smoothed model cost $O_h(n_e)$.

To absorb these errors we need a quadratic count bound that is valid even when the interior trial densities are signed. We obtain it by testing far labels separately and testing main labels only outside the wall, where those trial densities vanish. Partition the explicit coordinates into unit-scale cells and let n_j denote their variable counts. We claim

$$\sum n_j^2 \leq C(1 + |\log h|)(B(x) + 1). \quad (26)$$

For far labels use translates of a nonnegative fixed bump covering a unit interval, to test separately each $x_I = \psi * \zeta_I$ positive. For outer main counts on either side of the wall use bumps supported *outside* the wall, adapted to cells starting from the wall (the grid may change by bounded overlaps). For the first cell use a bump rising from zero at the wall to 1 within distance h , plateauing to distance 2, then cutting off. This captures at least a constant smeared mass from each unsmeared point of the first cell, by the displayed real kernels. On all these outer tests $\operatorname{Re} \alpha$ does not intervene and μ has cell masses summably bounded. The $H^{1/2}$ norms squared of the bumps are $O(1 + |\log h|)$ even at the wall, by the kernel seminorm $\int |b(u) - b(v)|^2 |u - v|^{-2} du dv$ (equivalence by Plancherel); for the transition of length h , the translation differences in L^2 squared cost $O(\min(l, l^2/h))$ for translations by $0 < l < 1$. Bounded overlap gives the same squared-norm bound times $\sum a_j^2$ for linear combinations (sum differences pointwise has bounded multiplicity). Duality proves (26).

Also $\|\psi * \zeta\|_{-1/2}^2 \leq C(1 + |\log h|) \sum n_j^2$. Indeed the spatial kernel for multiplier $(1 + k^2)^{-1/2}$ is bounded by $C(1 + |\log |u||)e^{-c|u|}$, by writing the symbol as a gamma-integral of Gaussians. Convolved twice with ψ it is bounded by $C(1 + |\log h|)e^{-c'|u|}$ using the density and tail bounds. This and (26) control the mixed error just found (substitute $\alpha - \mu = x + iz - \psi * \zeta$), and the $O(h)e^{-c|u|}$ explicit pair errors, proving (25). $\square$

Corollary 3.5 (Control of explicit particles). *Constants may depend on the fixed smoothing parameter h . One has*

$$|\operatorname{Re} M|^2 + n_e^2 \leq C_h L(B(x) + 1), \quad \ell_e := \sum_{v \in \zeta} (|v| - L)_+ \leq C(B(x) + 1). \quad (27)$$

Proof. Outside the wall $\operatorname{Re} \alpha$ is absent. For $t > L$, the total count beyond t is at most twice its positive smeared cumulative there, hence its square integrates on $[L, \infty)$ to $O(B(x) + 1)$ by $\|D\|^2$ and the μ -tail. Same on the other side. The cumulative counts are nonnegative integers, so their squares dominate the counts themselves. Integrating gives the distance sum bound. It also bounds exterior counts away from the wall; (26) controls the remaining cells near the wall and the far-particle cells inside it.

Total *smeared* mass outside the wall is at most $C_h(1 + \sqrt{B(x)})$ also, by the cumulative test outside distance one from the wall and (26) convolved by ψ for the first cell. Finally $\int r = -\operatorname{Re} M$; test on the wall region with a smooth plateau extending outside by distance one, cost $O(\sqrt{L})$ in $H^{1/2}$, and bound the remainder by the positive tail masses. In particular, retaining a fraction of $B(x)$ in a trial lower bound gives exponential damping $\exp(-c_h \ell_e)$, sufficient for convergence and tail truncation of the explicit absolute integrals. Summing their factorial measures with factors $\exp(C_h n_e)$ costs at most $\exp(O_h(L))$: far variables have effective volume $O(L)$, and exterior main variables have effective volume $O(1)$. This is the coarse cost; part 4 of Proposition 3.6 will improve it for the configurations needed by the exponent. $\square$

3.4 The energy gain from an imaginary tilt

The cylinder parameter enters a sector through $e^{i(\eta + \pi f)M}$. We seek trial imaginary densities that convert this oscillation into a positive real cost while preserving the prescribed totals. The estimates here are variational statements; the particle contour deformation implementing them is proved in Section 5.

Let $a = \operatorname{Re} \eta$, $b_1 = \operatorname{Im} \eta$ (here b_1 is not a size); suppose $|a| \leq a_0 < \pi$, $|b_1| \leq b_0$ fixed. Allow Fourier aliases $j \in \mathbb{Z}$. In the real negative-log action $\mathcal{A}$, add to $\operatorname{Re} \mathcal{E}$ the negative log modulus of $e^{i(\eta + \pi f + 2\pi j)M}$. Use $D_z = \partial^{-1} \sum z_\sigma$, $z_r = z_+$, so $\int z_r = -\operatorname{Im} M$. Let f_R^ψ be the cumulative positive *weighted* smeared far count to the right, and f_R^0 the unsmeared version. The sign tails and (25) give

$$\begin{aligned} \mathcal{A} \geq & B(x) - B(z) - 2p \int (Dz_r + rDz) - 2 \sum_{I:\text{far}} g_I \int x_I D_z - (x, J_o z) \\ & - \int \Theta z_r - b_1 \int r - s(h)(B(x) + B(z) + 1) - C_h n_e, \end{aligned} \quad (28)$$

Here $\Theta = a + 2\pi(j + f_R^\psi)$. Indeed the baseline $+$ versus $-$ sign tails give the $2p$ terms by total neutrality and antisymmetry. The additional far rows give sign against $z_+ + z_-$ with coefficient g_I , and against z_r with coefficient πw_I ; the constant from $2f_R^\psi - f$ thus cancels the sign-bias πf .

The next proposition starts with the unshifted real action and then gives three uses of an imaginary trial field. Part 2 permits every Fourier alias. Part 3 suppresses aliases whose effective phase is large near an end of the wall. Part 4 gives the quadratic dependence on the tilt when those end phases are bounded.

We make the common assumptions and the end phases precise. An admissible near density $\alpha = (\alpha_+, \alpha_-)$ is supported on $[-L, L]$, has the prescribed totals $m'_\sigma(M)$, and has finite trial energy. We allow signed real parts. Put

$$x = \operatorname{Re} \alpha - \mu + \psi * \zeta, \quad z = \operatorname{Im} \alpha, \quad \mathcal{A} = \operatorname{Re} \mathcal{E}(\alpha, \zeta) - \log \left| e^{i(\eta + \pi f + 2\pi j)M} \right|.$$

Thus $\int z_\sigma = -\sigma \operatorname{Im} M$. For parts 2–4 below assume that p_L has uniformly bounded $H^{-1/2}$ norm. For each sufficiently small fixed $\delta > 0$, assume that it has a normalized smooth truncation p_L° , supported in $|u| \leq (1 - \delta/2)L$, whose difference from p_L is exponentially small in total variation

and $H^{-1/2}$. The derivatives of each fixed order of p_L° are to be polynomially bounded in L . The hard-wall coordinates constructed below supply these properties.

For a fixed δ , choose nonnegative smooth probes $\varphi_\pm \leq 1$, supported away from the ends of $\pm[(1-2\delta)L, (1-\delta)L]$, obtained by rescaling fixed bumps, with $\int \varphi_\pm \asymp \delta L$. For $a = \operatorname{Re} \eta$, define

$$\Theta(u) = a + 2\pi(j + f_R^\psi(u)), \quad \theta_\pm = \frac{\int \Theta \varphi_\pm}{\int \varphi_\pm}, \quad U = \max\{|\theta_+|, |\theta_-|\}.$$

Here f_R^ψ is the weighted smeared far count to the right, as in (28). A polynomial bound throughout the proposition means a fixed power of L with a fixed multiplicative constant.

Proposition 3.6 (Trial bounds with tilt). *Fix $a_0 < \pi$ and $b_0 < \infty$, and let $\eta = a + ib_1$ satisfy $|a| \leq a_0$, $|b_1| \leq b_0$. The following bounds hold uniformly for admissible real parts of α , with the imaginary fields specified below. In parts 1–3, $h > 0$ is fixed and sufficiently small.*

1. *If M is real and $z = 0$, then for sufficiently small fixed $h > 0$, with no truncation of the explicit counts,*

$$\mathcal{A} \geq cB(x) - C_h L - C_h n_e.$$

2. *Suppose n_e is polynomially bounded, and that p_L has the properties stated above. For any $v = \operatorname{Im} M \in \mathbb{R}$ and any $j \in \mathbb{Z}$, the choice $z_\sigma = -\sigma v p_L$ gives*

$$\mathcal{A} \geq cB(x) - C_h L - C_h n_e + 2\pi j v - C_h(v^2 + (1+f)|v|).$$

There is no polynomial restriction on j in this estimate.

3. *Suppose in addition that $|j|$ is polynomially bounded. There are a value $v = \operatorname{Im} M$, independent of $\operatorname{Re} M$, and an imaginary field z with the required totals such that*

$$\mathcal{A} \geq cB(x) + c_\delta U^2 L - C_h L - C_h n_e.$$

The value v is polynomially bounded. Each $z_\sigma + \sigma v p_L$ is the derivative of a smooth primitive supported in $|u| \leq (1-\delta/2)L$, with polynomial bounds on all required fixed-order derivatives. Moreover, the number of far particles in $[-(1-2\delta)L, (1-2\delta)L]$ is at most CU for large L .

4. *Under the same polynomial hypotheses, suppose $U \leq U_0$. Choose a desired error $\varepsilon > 0$, then h sufficiently small, then δ sufficiently small. The constant U_0 may be fixed arbitrarily after these choices. There are a value $v = \operatorname{Im} M$, independent of $\operatorname{Re} M$, and an imaginary field z , with the same support and polynomial regularity properties as in part 3, such that*

$$\mathcal{A} \geq c'_\varepsilon B(x) + 2L \frac{\operatorname{Re}(\eta^2)}{4\kappa} - C_\varepsilon L - o(L) - C_h n_e.$$

The selected v is polynomially bounded. The retained coefficient $c'_\varepsilon > 0$ can be taken independent of δ, U_0 . On these bounded- U configurations, integrating the explicit factorial measures with the factor $\exp(C_h n_e)$, while using a fixed portion of the retained $B(x)$ -damping, costs at most

$$\exp(C_{h,\varepsilon} \delta L + o(L)).$$

The same conclusion allows further fixed exponential factors per explicit particle after reducing δ as needed.

The constants are uniform once the polynomial bounds and the previously fixed parameters are fixed. In part 4 the asymptotic limit is taken only after $\varepsilon, h, \delta, U_0$ have been chosen in that order; constants depending on those fixed choices are allowed in $o(L)$.

Proof. For part 1, set $z = 0$ in (28). The remaining linear term satisfies

$$|b_1 \int r| \leq C_h \sqrt{L(B(x) + 1)} \leq \frac{1}{4} B(x) + C_h L$$

by Corollary 3.5; choose h so that the smoothing loss consumes at most another fixed fraction of $B(x)$. This proves the first bound without any count truncation.

For part 2 suppose that n_e is polynomially bounded and use the stated norm and truncation properties of p_L . For $z_\sigma = -\sigma v p_L$, $v = \text{Im } M$, (28) gives

$$\mathcal{A} \geq cB(x) - C_h L - C_h n_e + 2\pi j v - C_h(v^2 + (1 + f)|v|). \quad (29)$$

Here $D_z = 0$, and the mixed baseline and odd-remainder errors cost $C|v|\sqrt{B(x)}$. This bound uses no polynomial restriction on j .

For part 3 suppose also $|j| \leq \text{poly}(L)$, with the polynomial fixed. We select $v = \text{Im } M$, independent of $\text{Re } M$, together with z . In the choices below we first specify smooth z_r, D_z supported on $|u| \leq (1 - \delta)L$ (and $v = -\int z_r$); then we actually add $v(p_L^\circ - p_L)$ to z_r , leaving D_z unchanged. Thus each difference $z_\sigma + \sigma v p_L$ is a smooth density derivative with primitive compactly supported in $|u| \leq (1 - \delta/2)L$; all its required sizes and fixed-order derivatives are polynomially bounded. In evaluating (28) we can neglect the exponentially small correction, at cost $o(1)(1 + \sqrt{B(x)})$ by the norm bounds, total variation, and the polynomial count bounds.

Take smooth nonnegative probes $\varphi_\pm(u) \leq 1$, supported on $\pm[(1 - 2\delta)L, (1 - \delta)L]$, rescaled fixed bumps supported away from their interval edges with mass comparable to δL . Set $\theta_\pm$ equal to the corresponding weighted averages of Θ ; let $U = \max |\theta_\pm|$. A first, coarse tilt uses

$$D_z = 0, \quad z_r = -t_0 \sum_{\pm} \theta_{\pm} \varphi_{\pm}$$

with small fixed $t_0 > 0$. Then $-\int \Theta z_r = t_0 \sum \theta_{\pm}^2 \int \varphi_{\pm}$. The $B(z)$ cost is at most $C t_0^2 \sum \theta_{\pm}^2 \int \varphi_{\pm}$; baseline and odd mixed terms are absorbed similarly with a fraction of $B(x)$. The term with b_1 is absorbed as in the real bound. Thus

$$\mathcal{A} \geq cB(x) + c_{\delta} U^2 L - C_h L - C_h n_e. \quad (30)$$

By positivity, the number of far particles on $[-(1 - 2\delta)L, (1 - 2\delta)L]$ is $\leq CU$ for large L : their smeared masses contribute at least a constant each to the drop between the two probe averages.

For part 4 suppose $U \leq U_0$, where U_0 may be arbitrarily large but fixed. Given a desired small error, choose a small parameter $\varepsilon > 0$, then h sufficiently small (so $s(h) \ll \varepsilon$), then $\delta > 0$ sufficiently small relative to these. U_0 can be chosen after these parameters. Use a smooth mask $0 \leq w(u) \leq 1$ supported on $[-(1 - 3\delta)L, (1 - 3\delta)L]$, vanishing at distances $\leq \sqrt{L}$ of far positions, equal to 1 except in boundary buffers of total width $O(\delta L)$ inside $[-L, L]$ and holes of total width $O_{U_0}(\sqrt{L})$. Take fixed-order positive derivatives $O_{U_0, \delta, l}(L^{-l/2})$. This is possible by the count conclusion. Put $s(u) = \text{sign}(a + 2\pi(j + f_R^0(u)))$ and $A_0 = |a|/(2\kappa)$, $B_0 = -pb_1/(2\kappa)$. Use

$$z_r = -A_0 s(u)w(u), \quad D_z = B_0 w(u),$$

with the correction convention above. These are smooth before correction even at jumps. Since $|a| < \pi$, $|a + 2\pi(j + f_R^0)| \geq |a|$. On the mask support all far smearing tails from points outside their

own respective side of u are superpolynomially small. Consequently $-\int \Theta z_r \geq |a|A_0 \int w - o(1)$, and $\sum g_I \int x_I D_z = o(1)$. Moreover

$$B(z) \leq (B_0^2 + (\kappa - p^2)A_0^2) \int w + o(L), \quad |(x, J_o z)| \leq o(\sqrt{L})\sqrt{B(x)}.$$

Indeed before correction the scaled L^2 masses of z_r, D_z (squared divided by L) concentrate at low frequency since their positive-order derivatives have norms $o(\sqrt{L})$. Use the symbol expansion and the upper norm bound, and the odd-remainder multiplier bound.

We give details for the real-field optimization so that no slow-variation assumption on x is needed. Let subscript ℓ mean a smooth compact low-pass Fourier cutoff equal to 1 near zero, even real with values in $[0, 1]$, supported on sufficiently small fixed frequencies depending on ε . By the positive small-frequency expansion, reserving a fraction on all frequencies,

$$B(x) \geq c_\varepsilon B(x) + (1 - C\varepsilon) \int (D_\ell^2 + Q(r_\ell, x_{\text{far}, \ell})).$$

On the mask $x_{\text{far}, \ell}$ is superpolynomially small (Schwartz kernel tails, count bound). Thus in this lower bound we can use $(1 - C'\varepsilon) \int w(D_\ell^2 + (\kappa - p^2)r_\ell^2) - o(1)$ besides the reserve. The pairings against ws, w can replace D, r by D_ℓ, r_ℓ at $o(\sqrt{L})\sqrt{B(x)}$ cost by the derivative bounds and Sobolev norms. Also

$$|\int (1 - w)r| \leq (C\sqrt{\delta L} + o(\sqrt{L}))\sqrt{B(x)} + C_h.$$

Use the smooth wall plateau of (27) minus w , whose L^2 norm has the stated bound by the support estimate and whose derivative norm is $o(\sqrt{L})$, then positive outer tail bounds for the rest. Constants depending on U_0, δ, h in these estimates are allowed inside little-oh terms.

Absorb errors with a portion of the reserve, including (25), losing $C\varepsilon L + o(L)$ with δ sufficiently small. It remains on measure $w du$ to complete squares for D_ℓ, r_ℓ , whose linear coefficients in (28) are $2pA_0s, -(b_1 + 2pB_0)$. The lower cost from these optimizations per unit mass is at worst

$$-p^2 A_0^2 - \frac{(b_1 + 2pB_0)^2}{4(\kappa - p^2)} - O(\varepsilon).$$

Adding $-B_0^2 - (\kappa - p^2)A_0^2 + |a|A_0$ leaves $(a^2 - b_1^2)/(4\kappa)$ before errors. Thus

$$\mathcal{A} \geq c'_\varepsilon B(x) + 2L \frac{\text{Re}(\eta^2)}{4\kappa} - C\varepsilon L - o(L) - C_h n_e. \quad (31)$$

The retained constant can be taken independent of δ, U_0 . For these bounded- U estimates the explicit factorial measures with $\exp(C_h n_e)$ and a portion of the retained damping cost at most $\exp(C_{h, \varepsilon} \delta L + o(L))$: outer main points cost bounded grand mass by tail damping, only a bounded number of far points can be on the interior interval just counted, and the remaining far points use an effective volume $O(\delta L) + O_{h, \varepsilon}(1)$ per variable, with factorial divisors. This also works with more fixed exponential factors per particle, by further reducing δ . Bounds (29)–(31) are deterministic bounds on trial densities. Implementing them for the complex integrals, including fluctuation determinants and the discreteness of M , takes further work below. □

3.5 Coordinates that remove the diagonal singularity

The trial estimates require a probability coordinate concentrated near ϱ , while the reference ensemble uses a fixed compact interval. We choose a map that has both properties and makes the true-minus-reference interaction smooth through the endpoints.

Use a reference parameter $s \in [-A, A]$, $A = L + d_0$, and $u = W(s)$. This is odd and $W(A - v) = L - \omega(v)$ at the positive end, where ω is quadratic near 0, Gevrey, increasing with derivative comparable to $\min(v, 1)$, and $\omega(v) = v - d_0$ for $v \geq 1$. Construct it by integrating a cutoff interpolation of the derivative between v and 1 (fixes d_0). Thus $W = s$ except within bounded endpoint distance and W reflects evenly at the ends. Set

$$\lambda_L(s) = C_L \varrho(W(s)), \quad t = \theta(s) = \pi \int_{-A}^s \lambda_L, \quad C_L^{-1} = \int_{-A}^A \varrho(W(v)) dv = 1 + O(e^{-\gamma L}).$$

So p_L is $\lambda_L ds$ pushed to physical u . Its stated probability and truncation norm requirements hold: in the core $s = u$, and near each wall its density in u is bounded by $Ce^{-\gamma L}(L - |u|)^{-1/2}$; the log-kernel estimate for the $H^{-1/2}$ norm thus suffices. Cut smoothly in the core to form p_L° . Derivatives of λ_L are bounded by $C^{l+1}(l!)^P \lambda_L$ for a fixed P , and $\lambda_L \asymp e^{-\gamma|s|}$. Constants in Gevrey bounds below can change (including order P), but not with N . These elementary closure estimates under smooth Gevrey multiplication, composition on a bounded scale, and reciprocal of a function bounded away from zero follow by multiplying formal Taylor series with coefficients bounded by $C^l(l!)^P$; in composition of positive degrees totaling l , products of factorials are bounded by $l!$, and the number of compositions is at most exponential.

Let

$$K(s, v) = H_{nn}(W(s) - W(v)) - \mathbf{1} H_{\text{ref}}(\theta(s), \theta(v)), \quad \mathcal{K} = \partial_s \partial_v K,$$

where $\mathbf{1}$ denotes the identity matrix.

Lemma 3.7 (Regularity of the compensated kernel). *The kernel K is smooth, extending evenly by reflection in each endpoint variable. K and all derivatives of total order l satisfy bounds $C^{l+1}(l!)^P(1 + L)$, and $\mathcal{K}$ and its derivatives satisfy $C^{l+1}(l!)^P e^{-c|s-v|}$. For the latter estimates one only extends locally across endpoints. Here and below $\text{poly}(L)$ prefactors may replace $1 + L$ without harm.*

Proof. First suppose $|s - v| > c_0 > 0$. The $\sinh$ terms have the required estimates directly. Split the $\cos$ -difference logarithm into its two sine logarithms. Then $\sin(|\theta(s) - \theta(v)|/2)$ is bounded below by $c \max_{[s,v]} \lambda_L$, reversing the interval if needed, and $\sin((\theta(s) + \theta(v))/2)$ is at least as large. One derivative in each variable supplies the decaying ratio

$$\frac{\lambda_L(s)\lambda_L(v)}{(\max_{[s,v]} \lambda_L)^2}.$$

Higher derivatives preserve this bound, proving the off-diagonal estimate for $\mathcal{K}$.

Near the diagonal and away from the endpoints, divide both $2 \sinh((W(s) - W(v))/2)$ and $\cos \theta(s) - \cos \theta(v)$ by $s - v$. Their natural local scales are respectively 1 and λ_L^2 , since $\sin \theta \asymp \lambda_L$ there. Averaging derivatives along the joining interval gives uniform Gevrey bounds and lower bounds on those scales. Their logarithmic ratio is therefore smooth.

Near a common endpoint, let d_s, d_v be the distances from that endpoint. Both differences have the additional factor $d_s + d_v$: even reflection makes their zeros occur at both $s = v$ and the reflected diagonal. Divide by

$$(s - v)(d_s + d_v).$$

The division is smooth Gevrey, obtained by averaging a derivative of the first divided difference from its reflected zero. The quadratic fold $\omega'(d) \asymp d$ and the sine factors give positive lower bounds for both quotients on their respective scales. The same log-ratio argument now proves regularity through the corner. Even reflection in either variable follows from the construction of W and θ .

Finally, in variables at distance at least δL from the endpoints, the compensated kernels continue analytically for uniformly small complex displacements, with the same fixed-order derivative control in the remaining variables. Here $W(s) = s$ and $\theta' = C_L \pi \varrho$ is analytic with relative bounds on uniformly small disks, so the divided-difference arguments apply there as well. $\square$

The next statement supplies the linear and constant terms needed together with this kernel in the particle deformation and dimension interpolation. It explains why the exponentially large mean particle number does not make the compensated coefficients exponentially large.

Lemma 3.8 (Regularity of the compensated source). *Suppose $|M|, n_e, \ell_e$ have fixed polynomial bounds in L . In the s -coordinate, the linear source*

$$b = H_{nn}(m'p_L - \mu) + \tilde{H}_{ne}\zeta$$

in Φ_M is smooth and even under endpoint reflection. Its derivatives of order r are bounded by $\text{poly}(L)C^{r+1}(r!)^P$, for a fixed Gevrey order P , and it has the core analytic continuation just described. Writing

$$E_e = \frac{1}{2}(\zeta, H\zeta)_{\text{off}},$$

the difference $\mathcal{E}(m'p_L, \zeta) - E_e$ is polynomially bounded in absolute value. The unchanged explicit-pair term E_e , independent of M , has real part bounded below by a negative polynomial; no upper bound is required.

Proof. For each main component, decompose the centered mean as

$$m'_\sigma p_L - m\varrho = (m'_\sigma C_L - m)\varrho + m'_\sigma(p_L - C_L\varrho).$$

The coefficient in the first term is polynomially bounded, since $m'_\sigma - m$ is polynomial and $C_L - 1 = O(e^{-\gamma L})$. Its convolution is controlled by $H_{nn} * \varrho = V$.

The two measures in the second term agree in the core. Near or beyond the ends their densities, written in ds for p_L and du for ϱ , have derivative bounds with a factor $O(e^{-\gamma L})$ and exponentially bounded tails. This factor cancels $m'_\sigma = O(e^{\gamma L})$. Insert a fixed-width physical Gevrey cutoff there. For convolution of the first density with the diagonal logarithm, subtract $-2\log|s - v|$ and its reflected-image term using the divided differences of W . The remaining kernel is smooth. Transfer derivatives of the ordinary translated logarithm to the compactly cut off, evenly reflected density. For convolution with the second density, transfer derivatives in physical u and then compose with W . Both terms are analytic in the deep core by separation from their supports.

The source $\tilde{H}_{ne}\zeta$ has the same bounds for fixed h . Its regularization concerns only exterior main interactions and leaves an analytic kernel in the core. These arguments also give polynomial total variation and moments for $m'p_L - \mu$, hence the stated constant-term bound by log integrability. Finally, real logarithmic singularities of explicit pairs are repulsive; away from them the kernels have at most linear growth. The polynomial count and excursion bounds therefore give $\text{Re } E_e \geq -\text{poly}(L)$, including when distinct explicit particles approach each other. $\square$

3.6 The relative interaction operator

The compensated kernel is now smooth. It remains to compare its quadratic fluctuations with those of the reference law, uniformly as the interval grows. Let $Q = (Q_+, Q_-)$ be a primitive of a change of zero mass on each main label in the s -coordinate, initially smooth and zero at both endpoints. Define

$$h_0(Q, Q) = (Q', H_{\text{ref}}Q'), \quad H_{\text{form}}(Q, Q) = h_0(Q, Q) + \iint Q(s)^\dagger \mathcal{K}(s, v)Q(v) ds dv$$

where H_{form} is the true near interaction form. The second expression is complex symmetric; on complex fields both energy forms are extended bilinearly. The space $\mathcal{X}$ has squared norm h_0 ; in the t -coordinate the fields extend as odd functions at both ends, completed from smooth ones. Per label

$$h_0 = \pi^2 \sum_{n \geq 1} n |b_n|^2, \quad Q = \sum b_n \sin(nt)$$

for the real form/norm, since $H_{\text{ref}}(t, t') = 4 \sum_{n \geq 1} \cos(nt) \cos(nt')/n$ as an integrable-kernel Fourier expansion and $\int \cos(nt) dQ = n\pi b_n/2$ on Dirichlet fields. For symmetric energy calculations on complex fields $h_0(\cdot, \cdot)$ extends bilinearly; norms and operator adjoints use the ordinary complex Hilbert convention. The inverse kernel using ds pairing is

$$G(s, v) = \frac{1}{2\pi^2} \log \left(\frac{\sin((\theta(s) + \theta(v))/2)}{\sin(|\theta(s) - \theta(v)|/2)} \right).$$

In particular $j : \mathcal{X} \rightarrow L^2(ds)$ the inclusion has bounded norm uniformly, $jj^* = G$ separately per label: the log singularity is square integrably bounded locally by $C(1 + |\log|s - v||)$ using $\sin \theta \leq C\lambda_L$; off diagonal the kernel and derivatives decay exponentially, by writing the squared ratio as $1 + \sin \theta(s) \sin \theta(v) / \sin^2((\theta(s) - \theta(v))/2)$. Schur's integral bound applies.

The next estimate controls the number of nearly degenerate directions, not just the smallest eigenvalue. This distinction is needed when passing from trace estimates to logarithmic determinants.

Lemma 3.9 (Trace bound and small eigenvalues). *On the complexification of $\mathcal{X}$, let*

$$T = I + j^* \mathcal{K} j, \quad A_T = \frac{1}{2}(T + T^*).$$

There are constants independent of L such that $\|T - I\|_1 \leq CL$, $\|T\| \leq C$, and

$$A_T \geq cL^{-2}I, \quad N_{A_T}(t) := \dim \mathbf{1}_{(-\infty, t)}(A_T) \leq CL\sqrt{t} \quad (0 < t < t_0). \quad (32)$$

The same lower gap and counting estimate hold for the singular values of T , and for every orthogonal compression of T .

Proof. Partition $[-A, A]$ into intervals I_a of lengths in $[1, 2]$, and use normalized cosine bases $e_{a,n}$, $n \geq 0$, on these intervals. The mixed-kernel estimates imply

$$|\langle e_{a,n}, \mathcal{K} e_{b,m} \rangle| \leq \frac{Ce^{-c|a-b|}}{(1+n^2)(1+m^2)}.$$

For a positive frequency, integrate twice by parts: the first boundary term vanishes because the sine is zero at either endpoint, and the second boundary term involves only the first derivative of the kernel. Doing this in both variables gives the displayed estimate, including all derivative boundary terms. Summing the trace norms of these rank-one matrix entries gives $\|\mathcal{K}\|_1 \leq CL$. Schur's test applied directly to its exponentially decaying kernel gives $\|\mathcal{K}\| \leq C$. The bounds for T follow from the uniform bound for j .

It remains to control the lower edge of the spectrum. For a two-component field, sum over components in the following definition:

$$\mathfrak{e}(Q) = \iint_{|s-v| \leq 2} \frac{|Q(s) - Q(v)|^2}{|s-v|^2} ds dv + \int_{d(s) \leq 2} \frac{|Q(s)|^2}{d(s)} ds, \quad d(s) = A - |s|.$$

We first establish the two comparisons

$$h_0(Q, Q) \leq C(\mathfrak{e}(Q) + \|Q\|_{L^2(ds)}^2), \quad \langle Q, A_T Q \rangle_{\mathcal{X}} \geq c\mathfrak{e}(Q).$$

The circle difference formula for the odd extension in t expresses h_0 , up to fixed constants, using the two kernels

$$\frac{\theta'(s)\theta'(v)}{\sin^2((\theta(s) - \theta(v))/2)}, \quad \frac{\theta'(s)\theta'(v)}{\sin^2((\theta(s) + \theta(v))/2)},$$

paired respectively with $|Q(s) - Q(v)|^2$ and $|Q(s) + Q(v)|^2$. When $|s - v| \geq c_0$, the first kernel is exponentially integrable in the separation. The second kernel has the same property except at a common endpoint; there it is bounded by a constant times the inverse square of the sum of the endpoint distances. Its integral in one variable is at most $C/d(s)$. These assertions follow from $\lambda_L(s) \asymp e^{-\gamma|s|}$ and $\sin \theta(s) \asymp \lambda_L(s) \min(1, d(s))$. On the remaining diagonal region the first kernel is bounded by $C|s - v|^{-2}$. This proves the upper comparison.

For the lower comparison, push the primitive forward to physical coordinate $u = W(s)$ and extend it by zero outside $[-L, L]$. Lemma 3.2 bounds its Fourier seminorm with multiplier $k^2/(1 + |k|)$. For any fixed sufficiently large C_0 , this multiplier is comparable to the multiplier

$$\int_{|z| \leq C_0} \frac{|e^{ikz} - 1|^2}{z^2} dz \asymp \frac{k^2}{1 + |k|}$$

of the truncated difference seminorm

$$\iint_{|u - u'| \leq C_0} \frac{|Q(W^{-1}(u)) - Q(W^{-1}(u'))|^2}{|u - u'|^2} du du',$$

where the zero extension is understood. For an interior point at distance $\ell < C_0/2$ from a wall, integration against the adjacent exterior interval gives weight $1/\ell - 1/C_0 \asymp 1/\ell$. Thus interactions with the exterior zeros give a wall weight comparable to $W'(s)/(L - |W(s)|) \asymp 1/d(s)$ near an endpoint. When the two endpoint distances are comparable, $W'(s)W'(v)/|W(s) - W(v)|^2 \asymp |s - v|^{-2}$. The part with noncomparable distances is bounded by the wall term: for example, integrate $|s - v|^{-2}$ over $d(v) < d(s)/2$ or $d(v) > 2d(s)$, using $|Q(s) - Q(v)|^2 \leq 2|Q(s)|^2 + 2|Q(v)|^2$. A fixed enlargement of the endpoint region causes no change in these bounds. This proves the second comparison.

For completeness, divide any interval of length comparable to $b \geq 1$ into cells of lengths in $[1/2, 1]$, and let q_a be the mean of Q on cell a . The local seminorm controls

$$\sum_a \int_{I_a} |Q - q_a|^2 + \sum_a |q_{a+1} - q_a|^2.$$

The first bound is the elementary variance identity on a cell; the second follows by averaging the difference over two adjacent cells, whose mutual distances are at most 2. If the mean on the whole block vanishes, the discrete Poincare inequality gives $\sum_a |q_a|^2 \leq Cb^2 \sum_a |q_{a+1} - q_a|^2$. Thus $\|Q\|_2^2 \leq Cb^2 \epsilon(Q)$ for fields with zero mean on each such block. On the whole interval, the wall term controls the first and last cell means; summing the successive differences then gives the same estimate with $b \asymp L$, without any constraints. The two comparisons now give the gap cL^{-2} .

For t above a fixed multiple of L^{-2} , choose $b = c_1 t^{-1/2}$ with c_1 sufficiently small. Vanishing block means impose at most CL/b linear constraints, and on their common kernel the Rayleigh quotient of A_T is at least t . The min-max principle proves the counting bound. Below the global gap the count is zero; the intervening range is handled by changing the constants. Compressing a positive Hermitian form cannot increase this count. Finally, if a subspace satisfies $\|TQ\| < t\|Q\|$, then on it $\langle Q, A_T Q \rangle \leq \|Q\| \|TQ\| < t\|Q\|^2$. The variational characterization of singular values therefore gives the same count and gap for T and its compressions. $\square$

3.7 The determinant contribution

Away from the center and the endpoints, the compensated kernels approach translation-invariant kernels. Their Fourier symbol determines the leading determinant. Lemma 3.9 supplies the uniform control near zero needed to integrate its logarithm.

Theorem 3.10 (Relative determinant asymptotics). *The Fredholm determinant satisfies*

$$-\frac{1}{2} \log |\det T| = (7\gamma/8)L + o(L). \quad (33)$$

Proof. Accretivity gives an injective operator with closed range. Since $T - I$ is trace class, T has Fredholm index zero and is therefore invertible. Its Fredholm determinant is defined by the convergent exterior-power series. Equivalently, take determinants of finite orthogonal compressions converging strongly to the identity, then pass to the limit in trace norm. The same procedure gives $\log |\det T| = \frac{1}{2} \text{Tr} \log(T^*T)$. We compute this trace through the limiting symbol

$$T_\infty(k) = \frac{k^2 \widehat{H}_{nn}(k)}{2\pi k \coth(\pi k/(2\gamma))}.$$

First consider a nonconstant word in $T - I$ and $T^* - I$. By cyclicity its trace is an integral alternating kernels G and $\mathcal{K}$ or $\mathcal{K}^*$. The local bound $C(1 + |\log |s - v||)$ for G , and exponential off-diagonal bounds for all kernels, provide an integrable majorant for the relative coordinates in this cyclic integral. More explicitly, truncate every separation at a fixed radius R . The discarded integral is bounded by CLe^{-cR} times a fixed polynomial in R . Each local logarithmic singularity can be integrated first against an adjacent bounded $\mathcal{K}$ kernel. With R fixed, the central region and the two endpoint regions of width $O(R)$ contribute $O(R)$, which vanishes after division by $2L$.

Under translation into either remaining tail, with distance from both the center and the endpoints tending to infinity, one has $W(s) = s$ and $\theta(s)$ or $\pi - \theta(s)$ asymptotic to a constant times $e^{-\gamma|s|}$, locally with every fixed derivative. Consequently

$$G(s, v) \longrightarrow \frac{1}{2\pi^2} \log \coth \frac{\gamma|s - v|}{2}.$$

The mixed derivative of the reference kernel tends to that of $-2 \log |2 \sinh(\gamma(s - v))|$; its additional one-variable terms disappear upon taking mixed derivatives. The Fourier transforms of these two limiting kernels are reciprocal, respectively

$$\frac{\tanh(\pi k/(2\gamma))}{2\pi k}, \quad 2\pi k \coth \frac{\pi k}{2\gamma}.$$

It follows from the logarithmic transform identities that multiplication by the limiting $\mathcal{K}$ symbol gives $T_\infty - I$. Dominated convergence, followed by $R \rightarrow \infty$, proves that the trace of each word, divided by $2L$, converges to the integral of the corresponding matrix trace against $dk/(2\pi)$.

We pass from words to the logarithm without losing uniformity at zero. For singular values $s_i(T)$, Equation (32) gives

$$\sum_{s_i < \tau} \log \frac{\tau}{s_i} = \int_0^\tau \frac{\#\{s_i < t\}}{t} dt \leq CL\sqrt{\tau}.$$

Thus clipping $\log(T^*T)$ below τ^2 changes its normalized trace by at most $C\sqrt{\tau}$. On the compact interval containing its spectrum, approximate the clipped logarithm by polynomials vanishing at one, with error at most $\epsilon|y - 1|$: apply polynomial approximation to the continuous quotient by

$y - 1$. The resulting trace error is bounded by $\epsilon \|T^*T - I\|_1 \leq C\epsilon L$. The symbol has the same integrable control: its smaller singular value is comparable to k^2 near zero, its other singular value stays positive there, and it tends exponentially to the identity at infinity. These assertions follow directly from the displayed 2×2 symbol and its positive Hermitian part. Hence clipping costs $O(\sqrt{\tau})$ on the symbol side as well. Passing successively $L \rightarrow \infty$, $\epsilon \downarrow 0$, and $\tau \downarrow 0$ yields

$$\lim_{L \rightarrow \infty} \frac{\log |\det T|}{2L} = \int_{\mathbb{R}} \log |\det T_{\infty}(k)| \frac{dk}{2\pi}.$$

Set $x = \beta|k|$ and $y = e^{-x}$. The determinant is positive, and its positive square root equals

$$\sqrt{\det T_{\infty}(k)} = \frac{c_*(k) \sinh(3x)}{\sinh(4x) \coth(3x/2)} = \frac{(1 - y^3)(1 - y^6)}{(1 + y)(1 - y^8)}.$$

Integrating the absolutely integrable logarithmic series gives

$$\begin{aligned} \int_0^{\infty} \log \sqrt{\det T_{\infty}(x/\beta)} dx &= -\frac{\pi^2}{18} - \frac{\pi^2}{36} - \frac{\pi^2}{12} + \frac{\pi^2}{48} \\ &= -\frac{7\pi^2}{48}. \end{aligned}$$

Here $\int_0^{\infty} \log(1 - e^{-ax}) dx = -\pi^2/(6a)$ and $\int_0^{\infty} \log(1 + e^{-x}) dx = \pi^2/12$, obtained by termwise integration; $\sum n^{-2} = \pi^2/6$ follows, for example, from Parseval's identity. Accounting for both signs of k , the square of the displayed square root, and $\gamma = \pi/(3\beta)$, we obtain

$$\log |\det T| = -\frac{7\pi}{12\beta} L + o(L) = -\frac{7\gamma}{4} L + o(L),$$

which is Equation (33). □

3.8 A finite-dimensional core for particle transport

A determinant calculation on the completed Hilbert space does not yet provide a deformation of a finite particle integral. We next approximate the relative interaction by smooth fields supported where the mean particle density is exponentially large. The remaining directions have a negligible normalized logarithmic cost. Fixing the approximation tolerance before taking the large-size limit is essential.

Lemma 3.11 (Smooth core approximation). *For every sufficiently small fixed $\xi > 0$ and all sufficiently large L , there are $\delta_* > 0$, independent of L , and a real orthogonal projection P in $\mathcal{X}$, of rank $O_{\xi}(L)$, whose range consists of smooth fields supported in $|s| \leq (1 - \delta_*)L$, such that*

$$\|T - I - P(T - I)P\|_1 \leq \xi L. \tag{34}$$

An orthonormal basis of its range can be chosen with every required fixed derivative bounded by a polynomial in L . The images of these basis fields under h_0 and H_{form} , regarded as test densities for the ds pairing, have the same bounds and smooth odd reflections at the endpoints. Constants may depend on ξ and on the derivative order.

Proof. We first approximate $\mathcal{K}$ on $L^2(ds)$. Truncate the cosine basis in each unit-scale cell at frequency J . The coefficient estimate in Lemma 3.9 gives trace-norm error at most CL/J , since $\sum_{n>J} (1 + n^2)^{-1} \leq C/J$ and the sum over cell separations converges. Approximate the finitely

many retained cosines in each cell by smooth functions compactly supported inside that cell, with L^2 -error at most η , and orthonormalize. For fixed J and sufficiently small η , their Gram matrices are uniformly invertible. Replacing a rank-one term $u \otimes v$ by its smoothed counterpart changes its trace norm by at most $\|u - \tilde{u}\| \|v\| + \|\tilde{u}\| \|v - \tilde{v}\|$. Summing the absolutely convergent cell expansions therefore adds at most $C_J \eta L$.

Discard cells meeting $|s| > (1 - 2\delta_*)L$. There are $O(\delta_* L + 1)$ such cells, and the total trace-norm contribution of terms with either index in these cells is of that same order. Denote the resulting real bump projection by Q_b . Then

$$\|\mathcal{K} - Q_b \mathcal{K} Q_b\|_1 \leq C(J^{-1} + C_J \eta + \delta_*)L + O(1).$$

Choose J , then η , then δ_* , so the coefficient on the right is as small as required. There are $O_\xi(L)$ bumps; each has uniformly bounded derivatives of every fixed order, with constants allowed to depend on these fixed choices.

Let $b_1, \dots, b_d$ be the orthonormal bump basis and put $v_i = j^* b_i$. As a function in s , $v_i = G b_i$. Their Gram matrix in $\mathcal{X}$ is uniformly conditioned. The upper bound follows from $\|j\| \leq C$. For the lower bound, every b in the bump span is a permissible smooth field and satisfies $\|b\|_{\mathcal{X}} \leq C_\xi \|b\|_2$, by the local upper norm estimate and its uniformly bounded H^1/L^2 ratio. Duality gives

$$\|j^* b\|_{\mathcal{X}} \geq \frac{|\langle j^* b, b \rangle_{\mathcal{X}}|}{\|b\|_{\mathcal{X}}} = \frac{\|b\|_2^2}{\|b\|_{\mathcal{X}}} \geq c_\xi \|b\|_2.$$

Thus the map $b \mapsto Gb$ loses no linear independence or uniform conditioning.

Choose a smooth cutoff equal to one on a neighborhood of the bump support and zero outside $|s| \leq (1 - \delta_*)L$, leaving a separation at least $c\delta_* L$ between the bumps and its nonconstant region. Multiply each v_i by this cutoff, obtaining $\tilde{v}_i$. The off-diagonal derivative bounds for G imply

$$\|v_i - \tilde{v}_i\|_{\mathcal{X}} \leq C_\xi L^C e^{-c\delta_* L}.$$

To check the endpoint term, the explicit sine ratio gives

$$G(s, v) = O(d(s) e^{-c|s-v|})$$

when s is close to an endpoint and v lies in a retained cell. Every fixed positive-order derivative has the same exponential bound, without necessarily retaining the factor $d(s)$. The difference field vanishes at the endpoint. In particular, the elementary estimate

$$\int_0^2 \frac{|q(d)|^2}{d} dd \leq 2 \int_0^2 |q'(d)|^2 dd, \quad q(0) = 0,$$

controls its wall contribution. The upper local norm estimate, or its consequence $h_0(q, q) \leq C \|q\|_{H^1(ds)}^2$ for such fields, then proves the claim. The Gram matrix remains uniformly invertible because $d = O_\xi(L)$. If P_0, P are the orthogonal projections onto the spans of $v_i, \tilde{v}_i$, respectively, the formula $V(V^*V)^{-1}V^*$ for a span projection yields

$$\|P - P_0\| \leq C_\xi L^C e^{-c\delta_* L}.$$

The operator $B = j^* Q_b \mathcal{K} Q_b j$ has both its range and its adjoint range in $\text{ran } P_0$; hence $B = P_0 B P_0$. Writing $A_1 = T - I$, we obtain

$$\|A_1 - P A_1 P\|_1 \leq 2\|A_1 - B\|_1 + 2\|P - P_0\| \|B\|_1 \leq C(J^{-1} + C_J \eta + \delta_*)L + o(1).$$

This proves (34) after the parameter choices above.

Finally, Gb_i has controlled derivatives of any fixed order in the core. Locally subtract the ordinary diagonal logarithm from G ; the remaining divided differences are smooth with uniform derivative bounds. For the logarithm itself, transfer derivatives to the smooth bump, and use the exponentially decaying off-diagonal estimates on the rest. Multiplication by the cutoff preserves these bounds. Passing to an orthonormal basis uses the inverse square root of a uniformly conditioned d -dimensional Gram matrix; its operator norm is bounded, so pointwise derivative bounds increase by at most $O(\sqrt{d})$. For the action of h_0 , write the test density as the derivative of H_{ref} applied to the derivative of the core field. Subtract its local logarithm and transfer derivatives to that smooth input. The even endpoint reflections of the kernel give odd reflections of the resulting test density. Adding the smooth operator $\mathcal{K}$ gives the same assertions for H_{form} . $\square$

The finite core carries almost all of the determinant. Eliminating its coordinates also leaves a positive interaction on its orthogonal complement. The following statement quantifies both assertions using the small-value count, rather than the gap alone.

Lemma 3.12 (Determinants and Schur complements). *Let $S = \text{ran } P$, $R = S^\perp$, and*

$$T_e = T_{RR} - T_{RS}T_{SS}^{-1}T_{SR}.$$

The Hermitian part $\text{Re } T_e = (T_e + T_e^)/2$ has the gap and small-eigenvalue count in (32). Moreover,*

$$|\log |\det T_{SS}| - \log |\det T|| = o_\xi(L), \quad \sum_{\substack{\lambda < 1 \\ \lambda \in \text{spec}(\text{Re } T_e)}} -\log \lambda = o_\xi(L). \quad (35)$$

Eigenvalues are counted with multiplicity. Here the notation means

$$\lim_{\xi \downarrow 0} \limsup_{L \rightarrow \infty} \frac{|o_\xi(L)|}{L} = 0.$$

It does not assert an $o(L)$ error for a fixed ξ . One may take errors bounded respectively by $C\xi^{1/5}L$ and $C\xi^{1/3}(1 + |\log \xi|)L$, after the core approximation is made with tolerance ξ .

Proof. The gap for T_{SS} makes it invertible. Set

$$Jy = (-T_{SS}^{-1}T_{SR}y, y), \quad y \in R.$$

Then $TJy = (0, T_e y)$, so

$$J^*(\text{Re } T)J = \text{Re } T_e, \quad \|Jy\| \geq \|y\|.$$

The gap follows immediately. If $E \subset R$ is a subspace on which the Rayleigh quotient of $\text{Re } T_e$ is less than t , the quotient of $\text{Re } T$ on JE is also less than t . Since J is injective, min-max proves the count. These identities use Hermitian adjoints; the matrices of the bilinear energy are, separately, complex symmetric in a real basis.

We make explicit how the counting bound controls logarithms. If the positive eigenvalues or singular values of any operator under consideration have counting function $N(t) \leq CL\sqrt{t}$, then

$$\sum_{s_i < \tau} \log \frac{\tau}{s_i} = \int_0^\tau \frac{N(t)}{t} dt \leq 2CL\sqrt{\tau}.$$

The lower gap justifies the identity even for the finite-size operators. Put $B = T_{SS} \oplus I_R$. Equation (34) gives $\|T - B\|_1 \leq \xi L$, and $\|T\|, \|B\| \leq C$, whence $\|T^*T - B^*B\|_1 \leq C\xi L$. The function $y \mapsto$

$\log \max(y, \tau^2)$ has Lipschitz constant τ^{-2} . The trace difference of this function on two positive operators is bounded by that constant times their trace-norm difference; this follows first for finite matrices by the eigenvalue variation inequality and then by finite compressions. Removing the clipping costs $CL\sqrt{\tau}$ by the preceding integral identity. Thus

$$|\log |\det T| - \log |\det B|| \leq CL(\sqrt{\tau} + \xi\tau^{-2}).$$

Take $\tau = \xi^{2/5}$ to obtain the first assertion.

For the second assertion choose $0 < \alpha < t_0$ and split the singular-value expansion of T_{SS}^{-1} as $B_{hi} + B_{lo}$, where

$$\text{rank } B_{hi} \leq CL\sqrt{\alpha}, \quad \|B_{lo}\| \leq \alpha^{-1}.$$

This is precisely the count for singular values of T_{SS} below α . Every block of $T - B$ has trace norm at most ξL , while the off-diagonal blocks have operator norm at most C . It follows that

$$\text{Re } T_e = I + F + E, \quad \text{rank } F \leq 2CL\sqrt{\alpha}, \quad \|E\|_1 \leq C\xi L(1 + \alpha^{-1}),$$

with F, E Hermitian. The factor two in the rank bound accounts for taking a Hermitian part. Since $\text{Re } T_e > 0$, interlacing and integration of its counting function below one imply

$$\text{Tr}(I - \text{Re } T_e)_+ \leq \text{rank } F + \|E\|_1.$$

Indeed a rank- r perturbation changes a count by at most r , and integration over $(0, 1)$ then bounds the contribution of F by r ; the contribution of E is bounded by its trace norm. For $0 < \tau < t_0$, the clipped negative logarithm satisfies

$$-\log \max(\lambda, \tau) \leq \frac{\log(1/\tau)}{1 - \tau}(1 - \lambda)_+.$$

Combining this bound with the logarithmic tail integral gives

$$\sum_{\lambda < 1} -\log \lambda \leq 2CL\sqrt{\tau} + \frac{\log(1/\tau)}{1 - \tau} (CL\sqrt{\alpha} + C\xi L(1 + \alpha^{-1})).$$

Taking $\alpha = \tau = \xi^{2/3}$ proves the stated estimate. The same argument with $\|\text{Re } T - I\|_1 \leq CL$ and fixed τ also gives $\sum_{\lambda < 1} -\log \lambda = O(L)$ for $\text{Re } T$. $\square$

The centered interaction has now supplied three pieces of information: a positive cost for explicit particles, the leading interior determinant, and a smooth finite core that captures that determinant up to a vanishing normalized loss. The next section supplies the remaining probabilistic input: bounds for linear and compensated quadratic statistics under the normalized reference ensemble.

4 Reference fluctuations and compensated quadratic integration

The preceding section reduced each electric sector to an integral against an explicit positive reference measure. We now prove the integration bounds used to turn its deterministic energy estimates into bounds on that integral. The first bound applies at every integer dimension and will also be useful for the uniform fugacity bound. The sharper estimate uses the electric coordinate and a polynomial bound on the empirical primitive. Both estimates refer to normalized expectations; the reference partition factors must be restored when computing cylinder weights.

4.1 The normalization and the circle comparison

Denote each reference partition integral more generally by $J_n(w)$ with weight $w(t)$ in $dt/(2\pi)$; the weight to normalize is $w_0 = 2 \sin t$. Its mass is

$$J_n(w_0) = \prod_{i=0}^{n-1} \frac{2 \cdot 16^i (i!)^4}{\pi (2i+1) ((2i)!)^2} = \exp \left(-\frac{1}{4} \log(n+1) + O(1) \right).$$

Indeed in $x = \cos t$ use Gram orthogonalization for dx/π with the Legendre polynomials $P_i = (2^i i!)^{-1} (d/dx)^i (x^2 - 1)^i$; integration by parts gives leading coefficient $2^{-i} (2i)! / (i!)^2$ and norm squared $2/(2i+1)$ for dx . Include the Vandermonde scaling 2^{2i} . The asymptotic follows by Stirling (or the exact adjacent-factor ratios plus Wallis/Stirling leading order fixing limit 1 of the factors).

The circle estimate is the finite upper bound obtained from monotonicity of the centered Toeplitz determinants and the strong Szegő limit; see [21, Theorems 1.1 and 2.4]. We give a Schur-polynomial proof to retain the exact coefficient and finite-size normalization. The power-sum and Schur orthogonality method appears in the trace-moment calculation of Diaconis and Shahshahani [4] and its development by Diaconis and Evans [3, Section 2]. The contraction argument below applies in every degree.

Lemma 4.1 (Circle comparison). *For a real smooth test $f(t) = f_0 + 2 \sum_{k \geq 1} f_k \cos kt$, with smooth even reflection, put $w_{\pm} = 2(1 \pm \cos t)$. Then*

$$J_n(w_+ e^f) J_n(w_- e^f) e^{-2nf_0} \leq \exp \sum_{k \geq 1} k |f_k|^2. \quad (36)$$

Proof. Before centering, the product equals the circle Gram determinant of e^f of size $2n$: shift its consecutive frequencies to centered half-integers, then split into cosines and sines by evenness. The two evaluation determinants factor $\sqrt{2} \cos(t/2)$ respectively $\sqrt{2} \sin(t/2)$ per variable times a polynomial Vandermonde with leading coefficients 2^j , yielding exactly the two measures (by determinant expansion/integration).

For completeness this circle bound follows already on polynomials f and then by approximation. In the Vandermonde-squared probability on m circles (here $m = 2n$), the centered integral is the squared norm of $\exp(\sum_{k \geq 1} f_k p_k)$ where $p_k = \sum_{l=1}^m z_l^k$. Symmetric holomorphic polynomials have the orthonormal alternant-quotient basis (strictly increasing nonnegative monomial powers in the determinant numerator, ordinary Vandermonde denominator), by integrating determinants. Its reproducing kernel summed over all degrees, as a formal graded identity, is $\prod_{i,j} (1 - z_i \bar{u}_j)^{-1}$ by Cauchy–Binet then the elementary Cauchy determinant formula. This kernel equally expands using $\exp(\sum p_k(z) \bar{p}_k(u)/k)$. Thus the functions $\prod_k (p_k/\sqrt{k})^{\alpha_k} / \sqrt{\alpha_k!}$ in each degree have synthesis norm at most 1 from coefficient ℓ^2 (matrix times its adjoint equals the identity in the orthonormal basis). The indicated exponential series hence has squared norm at most $\exp(\sum k |f_k|^2)$; on polynomial input this series converges to the function also pointwise/absolutely, or take radial limits. This proves (36). □

Write $\mathbb{E}_0$ for the two-label *normalized*, independent reference law at nonnegative integer dimensions $\nu_{\sigma} \leq C e^{\gamma L}$. We keep the two coordinate roles explicit. In the physical parameter $s \in [-A, A]$,

$$q_{\sigma} = Q'_{\sigma} = X_{\nu, \sigma} - \nu_{\sigma} \lambda_L(s) ds, \quad \ell(Q) = (f, q) = - \sum_{\sigma} \int_{-A}^A f'_{\sigma}(s) Q_{\sigma}(s) ds.$$

Here $\lambda_L(s) ds = dt/\pi$, and Q_σ vanishes at both endpoints. The centering measure is the reference equilibrium measure, not the finite-dimensional expected empirical measure. For a test written in the s -coordinate, define its cosine coefficients by

$$f_\sigma(\theta^{-1}(t)) = f_{\sigma,0} + 2 \sum_{k \geq 1} f_{\sigma,k} \cos(kt).$$

The squared dual h_0 -norm of ℓ is consequently $\|\ell\|_*^2 = \sum_{\sigma,k} k |f_{\sigma,k}|^2$. Regularity of a physical test below refers to s , whereas this norm uses its cosine coefficients in t . The following estimate keeps the reference partition masses visible.

Theorem 4.2 (Normalized reference source bound). *For two independent normalized reference ensembles with dimensions $0 \leq \nu_\sigma \leq C_0 e^{\gamma L}$, $L \geq 1$, and every real smooth reflected test f_σ , one has*

$$\mathbb{E}_0 e^{\ell(Q)} \leq \prod_{\sigma} J_{\nu_\sigma}(w_0)^{-1} \exp(\|\ell\|_*^2/2) \leq \exp(CL + \|\ell\|_*^2/2). \quad (37)$$

The constant depends only on C_0 and γ ; a zero-dimensional ensemble consists of the empty configuration and has partition mass one.

Proof. For each label, Cauchy–Schwarz in the unnormalized integral and $w_0 = \sqrt{w_+ w_-}$ bound the tilted integral by the geometric mean of the two tilted $w_\pm$ integrals. Lemma 4.1 bounds that mean after subtracting $\nu_\sigma f_{\sigma,0}$. Divide by $J_{\nu_\sigma}(w_0)$ and multiply the two estimates. The exact normalization above gives the last inequality, including $\nu_\sigma = 0$. $\square$

Additive constants in f do not matter. Use $\|\cdot\|_{-2,A}$ on primitives for the Sobolev norm with squared weights $(1 + (n/A)^2)^{-2}$ in the s -interval orthonormal sine basis (odd endpoint reflection). These empirical primitives need not be in $\mathcal{X}$. Pairings against smooth odd-reflected tests and mixed kernels are nevertheless defined by distributions. On a polynomially bounded $\|Q\|_{-2,A}$ event, dual Sobolev estimates control these pairings by the physical derivatives of the tests and kernels.

The loss CL in (37) is suitable for the untruncated count bound, but is of the same order as the exponent we want to determine. The next estimate removes that loss for dimensions comparable to m . Throughout its statement and the sharp quadratic estimate below, polynomial derivative bounds mean the following: for every fixed derivative order r , the relevant C^r -norm is at most $C_r L^{a_r}$, with C_r, a_r fixed before $L \rightarrow \infty$. For a kernel this convention applies to derivatives in both variables. Errors are uniform over families with these fixed bounds.

Theorem 4.3 (Uniform source inequality). *If $\nu_\sigma \asymp m$, $\mathcal{G} = \{\|Q\|_{-2,A} \leq L^D\}$ for fixed D , and the tests arise from real smooth even-reflected $f_\sigma(s)$ with polynomial derivative bounds in the s -coordinate as specified above, then uniformly,*

$$\mathbb{E}_0[\mathbf{1}_{\mathcal{G}} e^{\ell(Q)}] \leq \exp(o(L) + \|\ell\|_*^2/2) \quad (38)$$

(where the normalized excess limsup is at most zero). The error is uniform once D , the polynomial derivative bounds, and the constants implicit in $\nu_\sigma \asymp m$ are fixed.

Proof. We compare the two Jacobi weights w_+ and w_- after opposite small transports of the particles. Their geometric mean cancels the terms linear in the source and in the empirical fluctuation. The surviving gain will be $\gamma \int (a - 2a^2) + o(L)$, where a is the transport primitive. Its maximizing value $1/4$ gives $\gamma L/4$ per label, the logarithmic normalization loss up to the small boundary error. We now justify this comparison, including the Jacobian and its uniform error on the event.

The transport and its smooth error. Fix $0 < \delta < 1/4$. Choose a smooth primitive a with $0 \leq a \leq 1/4$, equal to $1/4$ on $|s| \leq (1-2\delta)L$, and zero for $|s| \geq (1-\delta)L$. It may be chosen monotone in each transition region, with $\|a^{(r)}\|_\infty \leq C_{r,\delta} L^{-r}$ for $r \geq 1$. For each label and $\epsilon \in \{+1, -1\}$, define

$$F_{\sigma,\epsilon}(s) = s - \epsilon \frac{a(s)}{\nu_\sigma \lambda_L(s)}.$$

These are increasing diffeomorphisms, equal to the identity near the endpoints. For every fixed derivative order,

$$\|F_{\sigma,\epsilon} - \text{id}\|_{C^r} \leq C_{r,\delta} L^{C_r} e^{-c_\delta L}.$$

Indeed $\nu_\sigma \lambda_L \geq c e^{\gamma \delta L}$ on the support of a , and all relative derivatives of λ_L there are bounded. Write $m_\sigma = \nu_\sigma \lambda_L ds$. Taylor expansion of pushforward against a test function gives

$$d_{\sigma,\epsilon} := (F_{\sigma,\epsilon})_* m_\sigma - m_\sigma = \epsilon a' ds + r_{\sigma,\epsilon}(s) ds, \quad \|r_{\sigma,\epsilon}\|_{C^r} \leq C_{r,\delta} L^{C_r} e^{-c_\delta L}.$$

For example, the first-order term is $-\nu_\sigma \lambda_L (F_{\sigma,\epsilon} - s)' = \epsilon a'$; every higher term contains one additional exponentially small displacement. The remainder is supported in a slightly enlarged core, has zero integral, and obeys the same bounds after integration.

We prove the comparison for one label, temporarily suppressing σ . Put

$$h(s, v) = H_{\text{ref}}(\theta(s), \theta(v)), \quad K_\epsilon(s, v) = h(F_\epsilon(s), F_\epsilon(v)) - h(s, v), \quad g(s) = \log \cot(\theta(s)/2).$$

The ratio of the two cos-differences defining K_ϵ extends smoothly across the diagonal. Its diagonal value is

$$K_\epsilon(s, s) = -2 \log \frac{\sin \theta(F_\epsilon(s)) \theta'(F_\epsilon(s)) F'_\epsilon(s)}{\sin \theta(s) \theta'(s)}.$$

The one-particle base density in s is $w_0(\theta(s))\theta'(s)/(2\pi) = \sin \theta(s) \lambda_L(s)$. Consequently its Jacobian and density ratio cancel exactly the half-diagonal contribution omitted from the Vandermonde interaction. For $X = \sum_i \delta_{s_i} = m + q$, the logarithm of the pulled-back base-density ratio is therefore exactly

$$-\frac{1}{2}(X, K_\epsilon X).$$

Since $hm = 0$, its mean-mean and mixed terms satisfy the identities

$$(m, K_\epsilon m) = (d_\epsilon, h d_\epsilon), \quad (q, K_\epsilon m) = (q, (h d_\epsilon) \circ F_\epsilon).$$

It follows that, uniformly on $\mathcal{G}$,

$$-\frac{1}{2}(X, K_\epsilon X) = -\epsilon(q, h a') - \frac{1}{2}h_0(a, a) + o(1).$$

Here is the precise uniform error estimate. If $q = Q'$, then

$$|(q, b)| \leq \|Q\|_{-2,A} \|b'\|_{2,A}, \quad |(q, Kq)| \leq \|Q\|_{-2,A}^2 \|\partial_s \partial_v K\|_{2,A;2,A},$$

where the positive Sobolev norms are dual to the negative norm in the statement, and the last norm is its tensor product. Derivatives through order three in each variable suffice to bound the last expression, up to a polynomial factor from the interval length. The divided difference argument for K_ϵ , together with the sine bounds away from the diagonal, bounds all these derivatives by $L^C e^{-c_\delta L}$. The kernel has even reflections at both ends, so its mixed derivatives have the odd reflections required for this Sobolev pairing. Convolution of h with the smooth core-supported remainder r_ϵ has

the same bound up to a polynomial factor: subtract the diagonal logarithm and transfer derivatives to r_ϵ , as in Lemma 3.7. The assumptions $\|Q\|_{-2,A} \leq L^D$ and fixed polynomial source bounds thus make every error in the preceding formula $o(1)$. This argument also justifies the expansions that follow.

The change of the source contributes

$$(f \circ F_\epsilon - f, X) = \epsilon(f, a') + o(1).$$

Since $w_\epsilon/w_0 = e^{\epsilon g}$, the extra weight contributes

$$\epsilon(g \circ F_\epsilon, X) = \epsilon(g, X) + (g, a') + o(1).$$

Only the core derivatives of g are involved in this expansion; although g diverges at the endpoints, the map is the identity there. If $D_\epsilon(X)$ denotes the pulled-back unnormalized density with weight $w_\epsilon e^f$, and $D_0(X)$ the original density with weight $w_0 e^f$, we have obtained

$$\log \frac{D_\epsilon(X)}{D_0(X)} = \epsilon((g, X) + (f, a') - (g, ha')) + (g, a') - \frac{1}{2}h_0(a, a) + o(1).$$

The terms odd in ϵ cancel in the geometric mean. Define $G_a = (g, a') - h_0(a, a)/2$. Then on the same event $\mathcal{G}$,

$$D_0(X) \leq e^{-G_a + o(1)} \sqrt{D_+(X)D_-(X)}.$$

The gain and the normalization. Since $g' = -\theta'/\sin\theta$, $-g'$ is bounded on the core and tends to γ under translations into either tail whose distance from the endpoints tends to infinity. Thus

$$(g, a') = - \int g' a = \gamma \int a + o_\delta(L).$$

On the support of a' , two points in the same tail satisfy

$$h(s, v) = 4\gamma \min(|s|, |v|) + O(1 + |\log \min(1, |s - v|)|),$$

and opposite-tail values are bounded. To verify this, use $\theta(s) \asymp e^{\gamma s}$ in the left tail and $\pi - \theta(s) \asymp e^{-\gamma s}$ in the right tail, factor the difference of the corresponding squared distances, and take its logarithm. The additive bounded error is uniform on these transition regions. The logarithmic error has bounded integral against $|a'(s)a'(v)|$, since $\|a'\|_1 \leq C$ and $\|a'\|_\infty \leq C_\delta/L$. For an absolutely continuous b of compact support on $[0, \infty)$,

$$\iint_{[0, \infty)^2} \min(r, t) b'(r) b'(t) dr dt = \int_0^\infty b(r)^2 dr;$$

this follows by writing $\min(r, t) = \int_0^\infty \mathbf{1}_{u < r} \mathbf{1}_{u < t} du$. Apply it on each half-line to obtain

$$h_0(a, a) = 4\gamma \int a^2 + O_\delta(1), \quad G_a = \gamma \int (a - 2a^2) + o_\delta(L) \geq \frac{\gamma L}{4} - C_\delta L - o_\delta(L).$$

Finally apply the geometric-mean inequality simultaneously to the two labels, using one common sign ϵ ; this permits the good event to couple both configurations. Cauchy–Schwarz, removal of that event from the two positive integrals, and change of variables back to their original domains give

$$\mathbb{E}_0[\mathbf{1}_{\mathcal{G}} e^{\ell(Q)}] \leq \exp \left(-2G_a + o(1) - \sum_\sigma \log J_{\nu_\sigma}(w_0) + \frac{1}{2} \|\ell\|_*^2 \right)$$

by (36); the constant Fourier coefficients cancel because ℓ is centered. Uniformly for $\nu_\sigma \asymp m$, $\log J_{\nu_\sigma}(w_0) = -\gamma L/4 + O(1)$. Hence the excess over $\|\ell\|_*^2/2$ is at most $C_\delta L + o_\delta(L)$. Taking first $L \rightarrow \infty$, then $\delta \downarrow 0$, proves (38) with the stated uniformity. $\square$

4.2 Integrating a compensated quadratic cost

The source inequality bounds linear statistics. The next lemma converts it into a bound for a coercive quadratic perturbation of the reference law. It is essential to complete the square before dropping the positive part of the perturbation: that preserves the correct dependence on the linear source and the minimizing energy.

Lemma 4.4 (Integration of a compensated quadratic cost). *Let F be a bounded real symmetric operator on $\mathcal{X}$, with $F \geq gI$, $g > 0$, and let $F - I$ be trace class. Let $l \in \mathcal{X}$ be real. Suppose that there are a smooth real vector test $b(s)$, odd under endpoint reflection, and a smooth real matrix kernel $\mathcal{C}(s, v)$ satisfying $\mathcal{C}(s, v) = \mathcal{C}(v, s)^\dagger$, odd under reflection in either variable, such that*

$$h_0(l, Q) = \int b(s)^\dagger Q(s) ds, \quad h_0(Q, (F - I)Q) = \iint Q(s)^\dagger \mathcal{C}(s, v) Q(v) ds dv.$$

Let $c \in \mathbb{R}$. For the compensated cost

$$V(Q) = c + h_0(l, Q) + \frac{1}{2}h_0(Q, (F - I)Q), \quad m_F = c - \frac{1}{2}h_0(l, F^{-1}l),$$

one has

$$\log \mathbb{E}_0 e^{-V(Q)} \leq -m_F + CL + \frac{1}{2} \sum_{\lambda < 1, \lambda \in \text{spec}(F)} -\log \lambda. \quad (39)$$

Here and below smooth compensated expressions are evaluated on empirical primitives by distributional pairing; their individual singular h_0 -energies are not defined. For the sharp form used below, assume $\nu_\sigma \asymp m$, $g^{-1} + \|F\| + \|l\|_{\mathcal{X}} \leq CL^a$ for fixed C, a , and polynomial physical derivative bounds for b and $\mathcal{C}$ in the convention above. Then for every fixed D , (39) holds with CL replaced by $o(L)$ and expectation restricted to $\|Q\|_{-2, A} \leq L^D$. The error is uniform once these bounds and D are fixed. The constant term c is unrestricted: its factor e^{-c} occurs on both sides of the estimate.

Proof. First suppose that all data are generated by finitely many physical sine tests. Put $Q_* = -F^{-1}l$, and complete the square before discarding any positive quadratic terms:

$$-V(Q) = -m_F + h_0(Q_*, Q) - \frac{1}{2}\|Q_*\|^2 - \frac{1}{2}h_0(Q - Q_*, (F - I)(Q - Q_*)).$$

With $J = (I - F)_+$, the last term is at most $\frac{1}{2}h_0(Q - Q_*, J(Q - Q_*))$. Let g_J be a real Gaussian vector of covariance J . Gaussian linearization and (37) give

$$\begin{aligned} \mathbb{E}_0 e^{-V(Q)} &\leq e^{-m_F - \|Q_*\|^2/2} \mathbb{E}_{g_J} \left[e^{-h_0(g_J, Q_*)} \mathbb{E}_0 e^{h_0(Q_* + g_J, Q)} \right] \\ &\leq e^{-m_F + CL} \mathbb{E}_{g_J} e^{\|g_J\|^2/2} = e^{-m_F + CL} \det(I - J)^{-1/2}. \end{aligned}$$

The cancellation of the mixed and Q_* -terms in the second line is exact. The nontrivial eigenvalues of $I - J$ are precisely those eigenvalues of F below one, which proves the formula.

For general smooth data, truncate in the physical s -sine basis. The resulting kernels converge with enough derivatives to converge on every fixed empirical primitive. Their operators converge in trace norm, and the linear Riesz vectors converge in $\mathcal{X}$; these statements follow by summing the Fourier coefficient tails and using the bounded embedding $\mathcal{X} \hookrightarrow L^2(ds)$. For sufficiently large cutoffs the gap is at least $g/2$, so the infima converge. To control the determinant term, on any finite-dimensional negative-excess subspace compare a truncated operator to the corresponding compression of F . For positive operators bounded below by $g/2$, the integral identity for log, or differentiation along their line segment, bounds the change of log determinant by $2g^{-1}$ times the

trace-norm error. Interlacing bounds the negative log sum of a compression by that of F . Fatou's lemma now proves (39).

The coarse argument also allows the additional cost $-t\|Q\|_{-2,A}^2$. Indeed its operator B_A , defined by $h_0(Q, B_A Q) = \|Q\|_{-2,A}^2$, has bounded norm and trace $O(L)$: factor it through the bounded $L^2(ds)$ embedding and sum $(1 + (k/A)^2)^{-2}$. Its physical sine truncations also converge on empirical fields. Choosing $2t\|B_A\| \leq g/2$ preserves a gap.

For the sharpened estimate, take a physical sine cutoff $K = L^B$. Smoothness gives errors smaller than any prescribed inverse power of L , by choosing a sufficiently high fixed derivative order and then B . Dual Sobolev estimates make the cost error $o(1)$ on $\|Q\|_{-2,A} \leq L^D$; trace-norm error divided by the gap and norm error in the completed square are also $o(1)$. The Riesz vectors of the first K physical sine tests have a Gram matrix with inverse-polynomial lower bound. To verify this, test the linear combination of their functionals on the same sine combination as a primitive. Its L^2 -pairing is the Euclidean squared norm, whereas the local derivative and boundary estimate for h_0 bounds its $\mathcal{X}$ -norm by a polynomial times that norm. Duality gives the claimed lower bound. Consequently every polynomially bounded vector in this finite-dimensional space has a smooth dual test with polynomial derivative bounds.

Apply (38) in the displayed Gaussian argument when $\|g_J\| \leq L^{B'}$. Outside this ball use (37). After the factor $e^{\|g_J\|^2/2}$, the normalized tilted Gaussian has covariance $J(I - J)^{-1}$, of polynomial norm, in polynomial dimension. For covariance norm at most L^c and dimension at most L^c , its tail outside radius $L^{B'}$ is bounded, for large B' , by $\exp(-L^{2B'-c}/4)$ times an exponential of a fixed polynomial. Choose B' to absorb the factor e^{CL} . This gives the stated $o(L)$ loss. $\square$

5 The electric contour integral

The quadratic integration estimate is real. Electric sectors also carry an oscillatory imbalance factor, and obtaining its sharp contribution requires a complex displacement. We first implement the deterministic trial fields by moving the actual particles, while keeping their integer dimensions fixed. The next section then varies the dimensions through an explicitly constructed holomorphic approximation.

5.1 Reference dimensions and compensated quadratic costs

Freeze the positions, labels, and counts of the explicit particles ζ from Section 3. For a pair of nonnegative integers $\nu = (\nu_+, \nu_-)$ and an independent parameter $M \in \mathbb{C}$, define

$$F_\nu(M) = \int A(X_\nu, \zeta) \exp(-\Phi_M(q)), \quad q_\sigma = X_{\nu,\sigma} - \nu_\sigma p_L,$$

with the two unnormalized reference measures from Lemma 3.3. Thus F_ν includes their masses $J_{\nu_+}(w_0)J_{\nu_-}(w_0)$, but not the extracted site product, the explicit-particle prefactor, or the imbalance phase. The energy Φ_M retains the prescribed trial totals $m'(M)$; the empirical field is centered at the actual integer dimensions ν . These two quantities need not agree. Their independence will be needed for interpolation in the next section.

Write

$$d_e = \ell_e + \frac{(\operatorname{Re} M)^2 + n_e^2}{L}, \quad H_0 = H_{\text{ref}}.$$

We use H_0 in the s -coordinate on both groups. Derivatives of primitives are densities or distributions in that coordinate and are pushed forward by W when inserted in $\mathcal{E}$.

Proposition 5.1 (Bounds at integer reference dimensions). *Fix $|\operatorname{Re} \eta| \leq a_0 < \pi$ and $|\operatorname{Im} \eta| \leq b_0 < \infty$, with the parameter conventions of Proposition 3.6. For nonnegative integer dimensions ν_σ , put*

$$\mathcal{L}_{\nu,j}(M) = \log |e^{i(\eta+\pi f+2\pi j)M} F_\nu(M)|, \quad v = \operatorname{Im} M.$$

The following estimates hold.

1. *If M is real and $0 \leq \nu_\sigma \leq Ce^{\gamma L}$, then, without any restriction on the explicit counts,*

$$\mathcal{L}_{\nu,j}(M) \leq C_h L + C_h n_e - c_h d_e.$$

This bound is independent of the integer alias j .

2. *Suppose n_e is polynomially bounded, as in part 2 of Proposition 3.6. In the same dimension range, for every real v and integer j ,*

$$\mathcal{L}_{\nu,j}(M) \leq C_h L + C_h n_e - c_h d_e - 2\pi j v + C_h(v^2 + (1+f)|v|). \quad (40)$$

3. *Assume the common hypotheses of parts 3 and 4 of Proposition 3.6, polynomial bounds on $n_e, |j|, |M|, \ell_e$, and $\nu_\sigma \asymp m$. Choose the imaginary imbalance and trial density from part 3 of that proposition. Then*

$$\mathcal{L}_{\nu,j}(M) \leq C_h L + C_h n_e - c_h d_e - c_\delta U^2 L + o(L). \quad (41)$$

If $U \leq U_0$, the choice from part 4 instead gives

$$\mathcal{L}_{\nu,j}(M) \leq \frac{3\gamma L}{8} - 2L \frac{\operatorname{Re}(\eta^2)}{4\kappa} + C\varepsilon L + o(L) + C_h n_e - c_{h,\varepsilon} d_e. \quad (42)$$

Here δ can be taken sufficiently small without degrading the displayed damping and per-particle constants, and U_0 may then be fixed arbitrarily, as in part 4 of that proposition.

The estimates are uniform once the parameters in the trial bounds and the polynomial size powers are fixed.

We prove Proposition 5.1 using the real integration bound of Lemma 4.4 and the particle deformation constructed below.

Throughout this section, a polynomial bound means a bound by a fixed power of L , whose exponent may depend on previously fixed parameters. It never refers to the exponentially large reference dimensions. Energy expressions use the complex bilinear extension of h_0 , whereas norms and adjoints use the Hermitian convention. Angle brackets against physical test functions denote distributional pairing.

For smooth real Hilbert fields Y , and a real smooth primitive Z_0 compactly supported in the core $|s| \leq (1 - \delta_c)L$ for some fixed $\delta_c > 0$, use the functional

$$C_z(Y) = \mathcal{E}(m' p_L + (Y + iZ_0)', \zeta) - i(\eta + \pi f + 2\pi j)M = c_z + h_0(u_z, Y) + h_0(Y, TY)/2$$

with complex bilinear convention; write C_0 for $Z_0 = 0$, so $C_z(Y) = C_0(Y + iZ_0)$. In (41),(42), the pushforward of Z'_0 by W is the density $z_\sigma + \sigma v p_L$ chosen in parts 3 and 4 of Proposition 3.6. Its required fixed-order derivative bounds are polynomial. Everywhere in the polynomial regimes u_z is a bounded Hilbert vector of polynomial norm, obtained by duality from a smooth odd-reflected test in s with polynomial fixed-order derivatives. Indeed use Lemma 3.8, and the action of H_{form}

on smooth core fields (localized diagonal log convolved by their smooth derivatives, as in the core-subspace proof). Also c_z is polynomially bounded except for the unchanged explicit-explicit constant $(\zeta, H\zeta)_{\text{off}}/2$, whose real part is at least $-\text{poly}(L)$; the equality in polynomial size for c_z here uses $|j|$ polynomial as well. Without polynomial truncation we will only use $Z_0 = 0$; the forms are still continuous at fixed sizes and generic frozen positions. The infimum

$$m_z = \inf_{Y \text{ real in } \mathcal{X}} \text{Re } C_z(Y)$$

satisfies the indicated trial lower bounds (with $B(x)$ replaced using (27) by positive damping in d_e , minus a constant). Indeed apply them first on dense smooth Dirichlet fields, then use continuity. Every such variation preserves the two specified totals. Primitives correspond to s -densities here, so integrable square-root singularities at physical endpoints do not impede the energy bounds.

The real reference bound already handles parts 1 and 2. For the sharp estimates in part 3, the imaginary trial density and the relative determinant must both be realized inside the original particle integral. The following deformation does this while leaving its integer dimensions unchanged.

Lemma 5.2 (Particle contour deformation). *Under the hypotheses of part 3 of Proposition 5.1, choose the core projection P of (34), with $S = P\mathcal{X}$, $R = S^\perp$, and Schur complement T_e . For every prescribed fixed tail exponent, a sufficiently large fixed D gives an exact contour deformation of the reference integral. On $\|Q\|_{-2,A} \leq L^D$, its compensated cost is $C_0(Q + a(Q)) - h_0(Q, Q)/2 + o(1)$, where*

$$Q + a(Q) = iZ_0 + Q_R + T_{SS}^{-1/2}Q_S - T_{SS}^{-1}Pu_z - T_{SS}^{-1}T_{SR}Q_R.$$

The extra Jacobian, after the one-coordinate reference-density cancellation, has modulus $(1 + o(1))|\det T_{SS}|^{-1/2}$. The full cost $\text{Re } C_0(Q + a(Q))$ has Hessian I on S and $\text{Re } T_e$ on R , and its infimum is at least m_z . The complementary contribution is smaller than the prescribed stretched exponential.

Proof. We first construct an affine path of smooth fields with a positive real Hessian. We then implement this path by particle displacements, compute the compensated cost and the remaining Jacobian, and estimate the region where a cutoff switches off the deformation.

The affine path and its real cost. Choose P, S, R, T_e as in (34), (35) with ξ sufficiently small (here $R = S^\perp$). After reducing $\delta_c > 0$, all core basis and shift functions are supported in $|s| \leq (1 - \delta_c)L$. The frozen affine shifts $Q \mapsto Q + a(Q)$ along the successive stages are:

1. Add $i\tau Z_0$, $0 \leq \tau \leq 1$.
2. Keeping iZ_0 added, replace $Q_S = PQ$ by $Q_S - \tau(T_{SS}^{-1}Pu_z + B_s Q_R)$, where $B_s = T_{SS}^{-1}T_{SR}$ here denotes a block map, $Q_R = (I - P)Q$.
3. In the fully shifted coordinates of stage 2 replace the free input Q_S of that expression by $D_\tau Q_S$, with

$$D_\tau = ((1 - \tau)I + \tau T_{SS})^{-1/2}.$$

Thus the output primitive in the last stage equals $iZ_0 + Q_R + D_\tau Q_S - T_{SS}^{-1}Pu_z - B_s Q_R$. Use a single smooth path parameter in $[0, 1]$ by flattening at stage boundaries. Matrix functions and their required derivatives here have polynomial norms by (32). For instance the symmetric inverse square root is given by the positive-resolvent integral $\pi^{-1} \int_0^\infty y^{-1/2}(yI + (1 - \tau)I + \tau T_{SS})^{-1} dy$, using accretivity (its identity follows from the scalar formula and derivatives on Jordan blocks). All nontrivial input uses of Q in a are through smooth dual tests with polynomial bounds.

The real Hessian of the full cost $\text{Re } C_0(Q + a(Q))$ has an inverse-polynomial gap throughout this path. To check this without treating complex changes as real orthogonal maps, write $A_S = T_{SS}$, $C_S = T_{SR}$, and $B_s = A_S^{-1}C_S$. The linear part of stage 2 sends (s, r) to $(s - \tau B_s r, r)$. Direct expansion of the complex bilinear quadratic form gives

$$T[(s - \tau B_s r, r), (s - \tau B_s r, r)] = T[(s, (1 - \tau)r), (s, (1 - \tau)r)] + (2\tau - \tau^2)T_e[r, r].$$

For real s, r , both terms have positive real part, and the coefficients on r add to one. Stage 1 leaves the Hessian unchanged. At stage 3 it is block diagonal, with blocks T_e and $A_S H_\tau^{-1}$, where $H_\tau = (1 - \tau)I + \tau A_S$. The identity

$$H_\tau^* \text{Re}(A_S H_\tau^{-1}) H_\tau = (1 - \tau) \text{Re } A_S + \tau A_S^* A_S$$

and the polynomial norms and inverse norms of these matrices give the required gap. Here D_τ commutes with A_S and is symmetric, so $D_\tau^t A_S D_\tau = A_S H_\tau^{-1}$.

All residual kernels, linear dual tests, and their path derivatives have polynomial physical derivative bounds, by the core-subspace construction. Their trace norms are polynomial as well, since the change from $T - I$ has polynomial norm and rank $O(L)$. At the endpoint the Hessian is $I \oplus \text{Re } T_e$. For the infimum comparison, fix a real r . The stationary point of $C_z(s + r)$ in the complex S -variable is $s_* = s_1 + is_2 = -A_S^{-1}(Pu_z + C_S r)$. Completing the complex square at s_* yields

$$\text{Re } C_z(s_* + r) = \text{Re } C_z(s_1 + r) + \frac{1}{2}h_0(s_2, \text{Re } A_S s_2) \geq m_z.$$

The remaining endpoint input contributes $\|s\|^2/2$. This proves the infimum claim even though the final contour need not have the initial imaginary trial profile.

Realizing the path as an exact contour deformation. Choose a C^∞ function $\chi : [0, \infty) \rightarrow [0, 1]$, equal to one on $[0, 1]$ and zero on $[4, \infty)$. If the flattened path just constructed is a_τ , use the particle-dependent parameter

$$\tau(Q) = \chi(\|Q\|_{-2,A}^2/L^{2D}), \quad a(Q) = a_{\tau(Q)}(Q),$$

and homotope from zero by replacing $\tau(Q)$ with $u\tau(Q)$, $0 \leq u \leq 1$. The empirical primitive, as a function of a particle at s_i , has first derivative $-\delta_{s_i}$ and second derivative δ'_{s_i} . Both belong continuously to the physical H^{-2} space: in the sine basis their squared norms are bounded respectively by constant multiples of $A^{-1} \sum_k (1 + (k/A)^2)^{-2}$ and $A^{-1} \sum_k (k/A)^2 (1 + (k/A)^2)^{-2}$. Thus the deformation is C^2 in the real particle variables, which is sufficient for Stokes' theorem.

Move a particle of label σ at s to

$$\Psi_\sigma(s; Q) = s + \Delta_\sigma(s; Q), \quad \Delta_\sigma(s; Q) = -\frac{a_\sigma(Q)(s)}{\nu_\sigma \lambda_L(s)}.$$

For any fixed derivative order, on the support of this motion,

$$\|\partial_s^k \Delta\|_\infty \leq L^{C_{D,k}} e^{-\gamma \delta_c L}.$$

Indeed a and its required derivatives are polynomial on $\|Q\|_{-2,A} \leq 2L^D$, and $\nu_\sigma \lambda_L(s) \geq ce^{\gamma \delta_c L}$ in its fixed core. This estimate concerns spatial differentiation at frozen Q .

For completeness, the exact change of contour is a change of an integration chain, not an assertion that the map Ψ is holomorphic or globally injective. Partition every original real coordinate into a slightly enlarged core interval and its complement, and fix the complementary coordinates. The

factor A is independent of all active core coordinates. The remaining density times the coordinate volume form is a holomorphic top form Ω in a neighborhood of the active intervals and their images. Its reference part is a squared Vandermonde times the holomorphic one-body density, so coincidences are zeros rather than poles. The regular exponent is holomorphic there by the core regularity estimates of the preceding section. Hence $d\Omega = 0$. On a side face of the homotopy prism, the corresponding coordinate lies at an interval endpoint and its displacement vanishes in a neighborhood; its differential is zero on that face, so the pullback of Ω vanishes. Stokes' theorem therefore equates the two integrals. Apply this argument for almost every frozen configuration and sum the finitely many partitions. The estimates below give absolute integrability; configurations with coincident explicit particles may be omitted as null sets.

The compensated cost after displacement. The contour identity is exact. We now estimate its pullback uniformly on $\|Q\|_{-2,A} \leq 2L^D$. Write $X = X_\nu$, $\Psi = \text{Id} + \Delta$. Separate the fixed-field single-coordinate Jacobians $1 + \Delta'_\sigma$. The diagonal omission in the Vandermonde log change contributes half the diagonal of

$$H_0(\Psi(s), \Psi(v)) - H_0(s, v)$$

per particle (kernels here in s , with appropriate labels). Its exponential cancels the transformed $2 \sin \theta'$ ratio and these single-coordinate Jacobians exactly. The difference kernel is defined by the log of the cos-difference ratio, with analytic branch near 1, and its required derivatives are exponentially small with odd/even reflections as appropriate. Indeed near a moved diagonal use the divided-difference bounds in the analytic core and small derivatives of Δ ; off diagonal use the sine separation bounds with displacements relative to θ' . Thus the resulting base negative-log change is half the difference kernel on X, X . Combining with Φ_M evaluated after movement and including the phase in the cost, one obtains uniformly up to $o(1)$

$$C_0(Q + a(Q)) - \frac{1}{2}h_0(Q, Q). \quad (43)$$

This notation always means the compensated smooth expansion, not the two singular forms separately on an empirical field.

We justify the error in (43) by a single scalar Taylor expansion, which avoids any dimension-dependent multivariable remainder. Freeze Q and a , and replace Δ by $z\Delta$. On the disk $|z| \leq R_L = \exp(\gamma\delta_c L/4)$, all required spatial derivatives of $z\Delta$ remain exponentially small. Core analyticity and the divided-difference estimates keep every logarithmic ratio in its branch near one, including at coincident particle positions. After removing the unchanged explicit-explicit constant, the logarithmic cost on this disk is bounded by $L^C e^{2\gamma L}$: every term is linear or quadratic in measures of total variation $O(e^{\gamma L})$, with polynomially bounded kernels on the disk. Cauchy's estimate bounds the remainder after order K , at $z = 1$, by

$$CL^C \exp([2\gamma - (K+1)\gamma\delta_c/4] L).$$

Choose the fixed integer K so that $K+1 > 8/\delta_c$.

Each retained coefficient is a polynomial in the components of the frozen displacement in the real span of the core basis and Z_0 . It is therefore enough to compute it for real displacements and extend the resulting polynomial identity to complex coefficients. For real displacements, the pushforward of the smooth mean has coefficients

$$d_k = \frac{(-\partial_s)^k}{k!}(\nu_\sigma \lambda_L \Delta_\sigma^k).$$

Here $d_1 = a'_\sigma$, while all fixed-order derivatives of d_k , $k \geq 2$, are exponentially small. The useful cancellation is $\nu_\sigma \lambda_L \Delta_\sigma^k = -a_\sigma \Delta_\sigma^{k-1}$; using the total variation of the mean alone would not prove this assertion near the edge of the core. All these densities are supported in the core. For the centered distribution $q = Q'$, every positive-order push coefficient, tested against a smooth f , is $\langle q, \Delta^k f^{(k)} \rangle / k!$. Integration by parts and duality with $\|Q\|_{-2,A}$ make it exponentially small, with all fixed test derivatives needed below.

These rules give the substitution $q \mapsto q + a'$ directly for the smooth kernel K and the smooth linear term in Φ_M . For the base-kernel difference, its q, q term is exponentially small by the ratio-kernel bounds and Sobolev duality in both variables. Terms containing means use $H_0(\lambda_L ds) = 0$. The one-mean coefficients reduce to a pushed q paired with $H_0 d_k$, and the two-mean coefficients pair positive-order d_k, d_l . These identities first hold for real diffeomorphisms. Differentiation is legitimate because the changed mean has smooth core density; local convolution with the diagonal logarithm transfers derivatives to that density. It gives polynomial smooth norms for $H_0 d_k$. Consequently the base half-difference is

$$(q, H_0 a') + \frac{1}{2}(a', H_0 a') + o(1).$$

Polynomial continuation of the finite jets, followed by the scalar remainder estimate, proves the same identity for complex displacements. No pushforward density on the real line is asserted for a complex contour.

Explicitly $\Phi_M(q)$ is a constant plus its smooth linear test on q plus $(q, Kq)/2$. Hence replacing q by $q + a'$ there (and subtracting the phase log) gives exactly (43) minus $(q, H_0 a') + (a', H_0 a')/2$, without ever needing a bare self energy of q . All the Taylor estimates are uniform even close to particle coincidences: on the growing scalar disk the shifted cos-difference over the original one in a same-label base ratio remains near 1, by the simultaneous divided difference (the same frozen shift function is used on the two arguments and has tiny spatial derivatives). The huge constant from a possible very close explicit-explicit pair never participates in these jets.

The Jacobian from the dependence on the configuration. The preceding cancellation includes each single-coordinate Jacobian at fixed Q . It remains to estimate the part arising from the dependence of a on the whole configuration. On the full-shift region write

$$D_Q a = \sum_{b=1}^r e_b \otimes \phi_b, \quad r = \dim S = O(L),$$

where ϕ_b are smooth physical dual tests, and put $J_i = 1 + \partial_s \Delta_{\sigma_i}(s_i; Q)$. The sign $\partial_{s_i} Q = -\delta_{s_i}$ cancels the minus sign in $\Delta = -a/(\nu \lambda_L)$. The matrix determinant lemma gives exactly, after factoring $\prod_i J_i$, the reduced determinant

$$\det(I + V), \quad V_{bb'} = \sum_i \frac{\phi_{b, \sigma_i}(s_i) e_{b', \sigma_i}(s_i)}{J_i \nu_{\sigma_i} \lambda_L(s_i)}.$$

For the summand test $g = \phi_b e_{b'}/(J \nu \lambda_L)$, centered quadrature is quantified by

$$\left| \sum_{i: \sigma_i = \sigma} g(s_i) - \nu_\sigma \int \lambda_L g ds \right| = |\langle Q_\sigma, g' \rangle| \leq \|Q_\sigma\|_{-2,A} \|g'\|_{H^2} \leq L^C e^{-cL}.$$

The support of $e_{b'}$ is inside the core, so all norms on the right have the required exponential factor. Replacing J by one in the mean term has the same error. Thus $V_{bb'} = \int \phi_b e_{b'} ds + O(L^C e^{-cL})$, with

the labels summed. On S , the derivative of the endpoint affine map is D_1 ; hence $I + V = D_1 + E_L$, $\|E_L\| \leq L^C e^{-cL}$. Since $\|D_1^{-1}\|$ is polynomial,

$$|\log \det(I + D_1^{-1} E_L)| \leq 2r \|D_1^{-1} E_L\| = o(1).$$

The principal matrix square root satisfies $|\det D_1| = |\det T_{SS}|^{-1/2}$, giving the asserted modulus.

The transition region and the outer tail. In the transition region, differentiation of the cutoff adds a single rank-one term with output $\partial_\tau a_\tau \in S + \mathbb{C}Z_0$. The total rank is at most $r + 1$. The cutoff differentials on point deltas have polynomial bounds on this region, and every entry of the reduced matrix is at most $e^{C_D L}$, even if bounded without centering. Hadamard's inequality therefore bounds its modulus by $\exp(O_D(L^2))$. This estimate uses the reduced matrix of size $O(L)$, never the exponentially large particle matrix. Outside the cutoff region the extra determinant is one.

Finally fix a path parameter τ . Its compensated real cost has inverse-polynomial gap and polynomial linear and trace-norm bounds, with constant bounded below by minus a polynomial. These bounds are independent of the cutoff exponent D . Choose E once so large that subtracting $L^{-E} \|Q\|_{-2,A}^2$ preserves half the gap. Lemma 4.4 and the elementary inequality $\mathbf{1}_{\{\|Q\| > L^D\}} \leq \exp(L^{-E} \|Q\|^2 - L^{2D-E})$ yield

$$\mathbb{E}_0[\mathbf{1}_{\{\|Q\|_{-2,A} > L^D\}} e^{-V_\tau(Q)}] \leq \exp(-L^{2D-E} + L^{E_1}),$$

with E_1 fixed independently of D . For example, the negative log sum is at most a polynomial because $-\log \lambda \leq g^{-1}(1 - \lambda)$ on $[g, 1]$, while the trace norm and inverse gap are polynomial.

The parameter in the transition region depends on Q , so a fixed-parameter estimate alone is insufficient. On $\|Q\|_{-2,A} \leq 2L^D$, the physical kernel bounds give $|\partial_\tau V_\tau(Q)| \leq L^{C+2D}$, after factoring out any unchanged explicit-pair constant. A parameter grid of mesh $L^{-C-2D-1}$ approximates every such cost within one. Bound its exponential by e times the sum over grid costs and apply the preceding tail estimate at every grid point. The additional logarithmic cost is only $O_D(\log L)$. The undeformed outer tail already has fixed parameter zero. Combining this estimate with the transition Jacobian bound and bounded reference masses gives

$$\exp\{-L^{2D-E} + L^{E_1} + O_D(L^2)\}.$$

Increasing D makes this smaller than any prescribed $\exp(-L^{E_2})$. The factor A , which depends only on fixed coordinates, remains in $[0, 1]$ in all these absolute estimates. $\square$

Proof of Proposition 5.1. For the estimates without a shift, take absolute values, use $0 \leq A \leq 1$, and apply Lemma 4.4 with $F = \operatorname{Re} T$ and infimum m_0 . The negative log sum is $O(L)$ by (32) and the trace bound; the reference masses are bounded above. The real trial estimate and (29), with (27), prove the first two assertions.

On the full-shift range use the sharp version of (39), (35), and the infimum comparison to m_z . Including the full Jacobian and both $J_{\nu_\sigma}(w_0)$ gives the log bound there

$$-m_z + (7\gamma/8)L - \gamma L/2 + o_\xi(L) + o(L)$$

by (33). Taking ξ small, then applying the deterministic lower bounds on m_z , and the separate negligible bad-region estimate (choose its power after the polynomial restrictions), proves (41) and (42). $\square$

6 Continuing the imbalance sum

The number of reference particles must next be varied with M . There is no integral of noninteger dimension to which Cauchy's theorem could be applied directly. We construct a holomorphic approximation from polynomial moments of the integer-dimensional integrals. The construction has two outputs: arbitrarily small absolute error on the integer grid, and a coarse growth bound on a complex dimension polydisk. The grid-transfer lemma at the end of the section uses both outputs to carry the sharp integer-dimension estimates to the complex dimensions needed for the imbalance contour shift.

Proposition 6.1 (Interpolation of the reference dimension). *Fix polynomial bounds on $n_e, \ell_e, |M|$. For every fixed $b > 0$ there is a jointly holomorphic function $\hat{F}(\nu, M)$, with $\nu = (\nu_+, \nu_-) \in \mathbb{C}^2$, on a neighborhood of the indicated M -disk and the polydisk $|\nu_\sigma - m| \leq \sqrt{m}$, such that*

$$|\hat{F}(\nu, M)| \leq \exp(L^C), \quad |\hat{F}(\nu, M) - F_\nu(M)| \leq e^{-L^b} \quad \text{for } \nu \in \mathbb{Z}_{\geq 0}^2 \text{ in the polydisk.} \quad (44)$$

Here C is fixed after the truncation bounds and b are fixed. The bounds are uniform in the real explicit-particle positions on their labeled lines. The integrals include their reference masses $J_{\nu_+}(w_0)J_{\nu_-}(w_0)$, but no external phase factor.

The next three subsections prepare the proof of Proposition 6.1: polynomial approximation, a finite endpoint-subset expansion, and continuation of the retained factorial moments. The proof is assembled after these constructions.

6.1 Polynomial approximation under the integer laws

We first make all errors that will be estimated under a positive integer-dimensional probability law. In Φ_M , truncate the physical sine expansions of the linear dual test and of the primitive kernel $\mathcal{K}$. The degrees can be polynomial in L , with error $e^{-L^{B_0}}$ in any prescribed fixed smooth norm. Indeed a Gevrey bound $A^k(k!)^P$ and k integrations by parts bound the n -th Fourier coefficient by $CA^k(k!)^P n^{-k}$; optimizing k gives $C \exp[-c(n/A)^{1/P}]$. The Gevrey scale here is polynomial in L . Fixed Fourier cutoffs preserve holomorphy in M .

The total variation of each particle or mean measure is $O(e^{\gamma L})$, so the exponent error is bounded by $e^{-L^{B_0} + O(L)}$ uniformly on configurations. Independently of this Fourier cutoff, Lemma 4.4 bounds the absolute expectation of the original exponential by $\exp(L^{C_0})$, using the gap, polynomial norm of the linear term, and polynomial lower bound on the real constant. Since $|e^{-z} - 1| \leq |z|e^{|z|}$, choosing B_0 larger than the required precision and C_0 makes the error in expectation negligible. Multiplication by $A \in [0, 1]$ does not alter this bound. Thus this first precision choice precedes all moment constructions.

Orthogonalize the Riesz vectors of the retained physical sine tests. The subspace and its real orthogonalization are independent of M, ν . Let $Y \in \mathbb{R}^d$ be the vector of resulting tests on Q . The truncated exponential is

$$g(Y) = \exp(-c_1 - l^\dagger Y - \tfrac{1}{2} Y^\dagger D Y), \quad I + \operatorname{Re} D \succeq \delta I, \quad \delta = L^{-r_0},$$

with D complex symmetric, $d, \|D\|, \|l\|$ polynomial, and $\operatorname{Re} c_1 \geq -L^C$. Operator-norm convergence of the truncated kernel preserves this gap. Each coordinate of Y is a centered sum of a physical cosine polynomial $f(s)$, with polynomial degree and coefficients. The inverse-polynomial Gram bound established in Lemma 4.4 justifies this last assertion after orthogonalization.

Lemma 6.2 (Weighted polynomial approximation). *For the preceding Gaussian exponential, choose $a = 1 - \delta/2$. Given any fixed precision exponent, there is a polynomial $P(Y)$ of polynomial total degree K_1 such that*

$$\sup_{y \in \mathbb{R}^d} e^{-a|y|^2/2} |g(y) - P(y)| \leq e^{-L^{B_1}}.$$

Its coefficients are holomorphic in M and bounded by an exponential of a fixed polynomial. The degree and coefficient bounds are fixed before any endpoint expansion.

Proof. Use the Gaussian probability measure $d\mu_a = (a/\pi)^{d/2} e^{-a|y|^2} dy$, and its orthonormal Hermite polynomials defined by

$$e^{\sqrt{2a}z^t y - z^t z/2} = \sum_{\alpha} H_{\alpha}(y) \frac{z^{\alpha}}{\sqrt{\alpha!}}.$$

Gaussian integration proves orthogonality. For completeness, polynomials are dense: an orthogonal complement would have an entire Gaussian-weighted Laplace transform with every derivative at zero equal to zero; Fourier uniqueness then makes the complement zero. The integrability needed here follows from Gaussian decay.

Let $c_{\alpha} = \int g H_{\alpha} d\mu_a$. Completing the square gives

$$\begin{aligned} \sum_{\alpha} c_{\alpha} \frac{z^{\alpha}}{\sqrt{\alpha!}} &= e^{-c_1} \det(I + D/(2a))^{-1/2} \\ &\times \exp\left\{\frac{1}{2}l^t(2aI + D)^{-1}l - \sqrt{2a}z^t(2aI + D)^{-1}l - \frac{1}{2}z^t D(2aI + D)^{-1}z\right\}. \end{aligned}$$

The square root is the analytic branch specified by the Gaussian integral. Accretivity ensures its existence and polynomial inverse norms. Set $B = D(2aI + D)^{-1}$. The identity

$$(2aI + D)^*(2aI + D) - D^*D = 4a(aI + \operatorname{Re} D) \succeq 2a\delta I$$

shows $\|B\| \leq 1 - L^{-C}$. The other factors in the generating function have size at most $\exp(L^C)$. Therefore, for some $r = 1 + L^{-C_1} > 1$, integrating the squared generating function at rz against $\pi^{-d} e^{-|z|^2} dz$ on $\mathbb{C}^d$ gives

$$\sum_{\alpha} r^{2|\alpha|} |c_{\alpha}|^2 \leq e^{L^C}.$$

This follows from monomial orthogonality; convergence follows from $r^2\|B\| < 1$, with inverse-polynomial margin.

The Hermite generating identity also gives Mehler's diagonal formula [17, equation 18.18.28]

$$\sum_{\alpha} t^{|\alpha|} |H_{\alpha}(y)|^2 = (1 - t^2)^{-d/2} \exp\left(\frac{2at}{1+t}|y|^2\right), \quad 0 < t < 1.$$

One direct derivation writes each one-variable unnormalized Hermite polynomial as the moment of $\sqrt{2a}y + iG$, for a standard real Gaussian G , and integrates the summed exponential for two independent copies. Taking $t = r^{-1}$ bounds the right side by $(1 - r^{-2})^{-d/2} e^{a|y|^2}$. Cauchy-Schwarz now bounds the Hermite tail above degree K , after multiplying by $e^{-a|y|^2/2}$, by $e^{L^C} r^{-K/2}$. The prefactor includes the polynomial-dimensional Mehler factor. A polynomially large $K = K_1$ attains any specified B_1 . Completeness and local uniform convergence identify the series with g . Expanding the retained Hermite polynomials in monomials gives the asserted coefficient bound, since the dimension and degree are polynomial. The Gaussian coefficient integrals, uniformly dominated on compact M -sets, prove holomorphy. $\square$

Gaussian linearization and (37) imply

$$\mathbb{E}_0 e^{a|Y|^2/2} \leq e^{CL}(1-a)^{-d/2}.$$

Consequently the weighted error in Lemma 6.2 gives an error in expectation smaller than e^{-L^b} by fixing B_1 after d, r_0 . From this point onward K_1 and all polynomial coefficient bounds are fixed.

We next replace each constituent $f(s(t))$ by

$$f(s(t)) = g_f(\cos t) + e_f(t) + r_f(t), \quad \deg g_f \leq d_g = \lfloor m/L^{D_1} \rfloor, \quad \|r_f\|_\infty \leq e^{-L^{B_2}},$$

where e_f is polynomially bounded and supported on $\min(t, \pi - t) \leq L^{D_E}/m$. Choose D_1 first so that $K_1 d_g = o(m)$, with ample fixed margin, then choose B_2 after the degree and coefficient bounds of P , and finally choose D_E . To obtain this decomposition, multiply $f(s(t))$ by an even-reflected Gevrey mask which is zero up to distance $L^{D_E}/(2m)$ from an endpoint and one beyond L^{D_E}/m . The omitted part is e_f . Outside the omitted region, integration of $\lambda_L(s) \asymp e^{-\gamma|s|}$ places $s(t)$ at a growing distance from the folded endpoint buffer. There $\theta' \asymp \min(t, \pi - t)$, with relatively bounded analytic derivatives on fixed small s -disks. The analytic inverse theorem therefore extends $s(t)$ to a disk of radius $c \min(t, \pi - t)$, with bounded displacement. Shrinking this radius by a polynomial to handle the physical sine bandwidth gives masked derivative bounds

$$L^C (C' m / L^{D_E - C_2})^k (k!)^{P'},$$

with fixed C_2, P' independent of sufficiently large D_E . The resulting cosine Fourier tail above d_g is at most a polynomial times $e^{O(L)} \exp[-cL^{(D_E - C_2 - D_1)/P'}]$. Increasing D_E proves the required sup error. Both the real sup norm of g_f and its individual cosine coefficients are polynomially bounded.

Keep the original centering constants of Y . The uncentered sums of all retained tests and edge functions have magnitude at most $e^{CL} L^C$. A polynomial of total degree K_1 , with the coefficient bounds already fixed, has derivative at most $e^{L^{C'}}$ on this range. The total perturbation of its arguments is at most $e^{CL - L^{B_2}}$. Thus the mean-value formula bounds the error in P uniformly by $e^{L^{C'} + CL - L^{B_2}}$, which was made negligible when choosing B_2 . This estimate is completed before any subset of endpoint particles is introduced.

6.2 A finite endpoint-subset expansion

Set $x = \cos t$. For each label the normalized reference law is the Legendre projection process for Lebesgue measure on $[-1, 1]$. Let E_L be the two endpoint intervals corresponding to $\min(t, \pi - t) \leq L^{D_E}/m$; their total length is $O(L^{2D_E}/m^2)$. Enlarge D_E , if necessary, so that every factor of A differs from one only on E_L . This is possible because those factors require physical distance bounded from a wall. They are bounded in $[0, 1]$ and independent of M, ν .

Let S_1 denote the vector of global sums of the polynomials $g_f(x)$, and let B be the actual labeled endpoint-particle set. At fixed S_1 and centerings, define $H(D)$, for $D \subseteq B$, by using only points of D in the edge sums and in the product of endpoint factors of A . Define

$$I(B') = \sum_{D \subseteq B'} (-1)^{|B'| - |D|} H(D).$$

Möbius inversion on subsets gives the exact identity $H(B) = \sum_{B' \subseteq B} I(B')$: the coefficient of $H(D)$ in the sum is $(1 - 1)^{|B \setminus D|}$. Each $I(B')$ is a polynomial in S_1 of total degree at most K_1 . Uniformly on integer configurations,

$$|I(B')| \leq 2^{|B'|} e^{L^{C'_2}}.$$

The exponent here is independent of $|B'|$, since every possible subset contains at most $O(m)$ points and the polynomial degree has already been fixed.

For the normalized Legendre polynomials $p_i = \sqrt{(2i+1)/2} P_i$, one has $|P_i(x)| \leq 1$ on the real interval. A direct verification uses

$$P_i(x) = \frac{1}{\pi} \int_0^\pi (x + i\sqrt{1-x^2} \cos \phi)^i d\phi.$$

Summing the geometric series proves its generating function $(1 - 2xz + z^2)^{-1/2}$, hence its equality with the polynomials from the Legendre recurrence. The integrand has modulus at most one. Therefore $\Pi_n(x, x) = \sum_{i < n} p_i(x)^2 \leq n^2/2$. The factorial densities are minors of this positive kernel; the Gram determinant bound gives, for the total endpoint count N_E of both labels,

$$\mathbb{E}_0 \binom{N_E}{k} \leq \frac{\beta_L^k}{k!}, \quad \beta_L = CL^{2D_E}.$$

For two labels this follows either from the direct-sum kernel or by summing the binomial convolution of their separate bounds. Thus truncating the subset expansion at $|B'| \leq K_e$ has error at most

$$e^{L^{C'_2}} \sum_{k > K_e} \frac{(2\beta_L)^k}{k!}.$$

If $K_e + 1 \geq 4e\beta_L$, this tail is at most $2e^{L^{C'_2}} (2e\beta_L/(K_e + 1))^{K_e+1}$. Choose the polynomial K_e large enough to attain the desired precision. Crucially, this does not increase K_1 : the endpoint subset size and the global polynomial degree are separate orders.

6.3 Continuation of the retained factorial moments

The determinant formulas here are the standard consequences of Andréief's identity for an orthogonal-polynomial ensemble [22, equations (1.3), (3.1), and Section 5]. We give the expansion because the subsequent continuation uses its precise finite degrees.

We describe one label, with integer dimension n , later continued to ν . Introduce formal source variables $\mathbf{t} = (t_j)$, and $G(x) = \sum_j t_j g_j(x)$, where the g_j are the global polynomial tests. The order- k factorial density, tilted by $e^{\sum_i G(x_i)}$ and without the divisor $k!$, is

$$\mathcal{D}_n(\mathbf{t}) \det[\mathcal{C}_n(x_h, x_{h'}; \mathbf{t})]_{h, h' \leq k},$$

where

$$\mathcal{D}_n = \det_{\Pi_n}(\Pi_n e^G \Pi_n), \quad \mathcal{C}_n = \Pi_n (\Pi_n e^G \Pi_n|_{\Pi_n})^{-1} \Pi_n e^G.$$

These identities are formal at $\mathbf{t} = 0$, where the inverse exists. To prove them, insert an additional factor $\prod_i (1 + h(x_i))$ in the tilted expectation. Integration of the two polynomial evaluation determinants gives the Gram determinant with weight $e^G(1 + h)$. Factor out its value at $h = 0$, and expand the remaining determinant in its minors. The coefficient of $h(x_1) \cdots h(x_k)$ is exactly the formula above.

For a labeled endpoint set write

$$I(B') = \sum_{|\alpha| \leq K_1} c_\alpha(B'; M, \nu) S_1^\alpha.$$

The coefficients are holomorphic in M, ν : they are obtained from P , centerings linear in ν , and sums or products of bounded real endpoint functions evaluated at the specified points. Their magnitudes are at most e^{L^C} . For each split of $|B'| = k$ into k_+, k_- , integrate these coefficients against $\partial_{\mathbf{t}}^\alpha|_{\mathbf{t}=0}$ of the product of the two tilted factorial densities, and divide by $k_+!k_-!$. This is precisely the expected subset sum. Only source degrees at most K_1 are required, regardless of $k \leq K_e$.

In the polynomial-index basis, multiplication by x is the tridiagonal Jacobi matrix J , with off-diagonal coefficients

$$a_i = \frac{i}{\sqrt{(2i-1)(2i+1)}} = (4 - i^{-2})^{-1/2}.$$

At source degree r , multiplication by e^G has bandwidth at most rd_g . In

$$\log \mathcal{D}_n = \text{Tr}_{\Pi_n} \log(I + \Pi_n(e^G - 1)\Pi_n),$$

a closed index path of degree r has total displacement at most rd_g . If its starting index lies farther than rd_g below n , it never meets the upper projection boundary; every inserted Π_n can be removed. The complete coefficient then equals that of $\log e^G = G$, and vanishes for $r \geq 2$. This argument is an identity of formal series, including the cancellations among the finitely many logarithm terms. All higher cumulants therefore involve only indices within $2K_1d_g + O(1)$ of n .

The kernel correction has the same property. Exactly, $\mathcal{C}_n\Pi_n = \Pi_n$ and $(I - \Pi_n)\mathcal{C}_n = 0$, so $\mathcal{C}_n - \Pi_n$ has rows below n and columns at least n . Its degree- r inverse-series coefficient has bandwidth rd_g , forcing both indices within rd_g of n . All these assertions concern polynomial kernels coefficient by coefficient; pointwise evaluation at endpoints is legitimate.

Continue the required finite blocks by replacing a_{n+j} with

$$a_{\nu+j} = (4 - (\nu + j)^{-2})^{-1/2},$$

using the branch near $1/2$, on $|\nu - m| \leq cm$ for a small fixed c . Choose a fixed offset block, for example $|j| \leq m/8$, and let the projection test the sign of the offset. Our choice $K_1d_g = o(m)$ ensures that no retained coefficient reaches its artificial endpoints. The block differs in norm by $O(m^{-2})$ from the real constant tridiagonal matrix with entries $1/2$. The latter has spectrum in $[-1, 1]$. On a contour at a sufficiently large constant times m^{-2} from this interval, a resolvent identity bounds the continued resolvent by $O(m^2)$. For the Chebyshev polynomial $T_p(x) = \cos(p \arccos x)$, the quadratic-root formula gives $|T_p(x)| \leq Ce^{Cp/m}$ on this contour. Expanding each g_j in its cosine coefficients and applying the resolvent formula thus bounds $\|g_j(J)\|$ by $e^{CL}L^C$. There are at most exponentially many indices but only polynomially many factors in any retained term; all products, traces, and formal coefficients therefore have bounds e^{L^C} . This resolvent estimate is essential because an absolute enumeration of paths of length d_g would give a much larger bound.

The degree-one trace $T_j(n) = \text{Tr}(\Pi_n g_j \Pi_n)$ requires a separate construction. Its forward difference $T_j(n+1) - T_j(n) = (g_j(J))_{nn}$ is a near-index entry, hence has an analytic continuation $h_j(\nu)$ on $|\nu - m| \leq cm$ with bound $e^{CL}L^C$. If $h_j(m+w) = \sum_{p \geq 0} h_{jp} w^p$, Cauchy's estimate gives $|h_{jp}| \leq e^{CL}L^C(C/m)^p$. Let S_p be the polynomial with $S_p(0) = 0$ and $S_p(w+1) - S_p(w) = w^p$. Define

$$\hat{T}_j(m+w) = T_j(m) + \sum_{p=0}^{P_3} h_{jp} S_p(w).$$

This agrees, at integer w , with summing the truncated forward differences in either direction from zero. The generating function

$$\sum_{p \geq 0} S_p(w) \frac{t^p}{p!} = \frac{e^{tw} - 1}{e^t - 1}$$

and Cauchy's estimate on a circle of radius comparable to $(p+1)/(p+1+|w|)$ give $|S_p(w)| \leq [C(1+|w|+p)]^{p+1}$. For $|w| \leq \sqrt{m}$ and any fixed polynomial P_3 , the continued trace is consequently bounded by $e^{CL}L^C$, uniformly in the chosen truncation power for sufficiently large L . Its integer-grid error is at most $e^{CL}L^C\sqrt{m}(C/\sqrt{m})^{P_3+1}$. Increasing the polynomial order gives any required $e^{-L^{B_3}}$ precision.

The same construction continues the reference masses. The exact norm product gives

$$\Delta^2 \log J_n(w_0) = \log \left(1 + \frac{1}{(2n+1)(2n+3)} \right).$$

Keep the exact $\log J_m$ and $\Delta \log J_m = O(m^{-1})$, Taylor-expand this analytic second difference at m , and sum twice by polynomial antidifferences. Its Taylor coefficients are $O(m^{-2})(C/m)^p$. The polynomial twice-antidifference with values zero at $w = 0, 1$ has generating function

$$\frac{e^{tw} - 1 - w(e^t - 1)}{(e^t - 1)^2}$$

and bound $[C(1+|w|+p)]^{p+2}$. The resulting analytic logarithm stays $O(L)$ on $|w| \leq \sqrt{m}$. Integer errors acquire at most a factor m when summed twice; after exponentiation they still attain arbitrary $e^{-L^{B_3}}$ accuracy. These uniform size bounds do not grow with the final polynomial Taylor order.

Endpoint evaluations are continued directly. The constant kernel is represented by Christoffel–Darboux [17, equation 18.2.12]:

$$\Pi_n(x, y) = a_n \frac{p_n(x)p_{n-1}(y) - p_{n-1}(x)p_n(y)}{x - y},$$

with the removable diagonal filled by its limit. For x near $+1$, continue a wavefunction with index $z = \nu + j$ by

$$p_z(x) = \sqrt{(2z+1)/2} \sum_{r=0}^{\infty} \frac{\prod_{l=0}^{r-1} (z-l)(z+l+1)}{(r!)^2} \left(\frac{x-1}{2} \right)^r.$$

This is the Rodrigues expansion at integer z . For $|j| \leq m/8$, $|\nu-m| \leq \sqrt{m}$, and $|x-1| \leq CL^{2D_E}/m^2$, the terms with $r \leq 4m$ are bounded by $(CL^{D_E})^{2r}/(r!)^2$, apart from the square-root prefactor. For $r > 4m$, the ratio of successive absolute majorants is at most $C|x-1| < 1/2$. Thus the function is bounded by e^{L^C} , also on slightly larger complex endpoint disks. Cauchy estimates give the same type of bound for its first derivative, including the factor m^2 from the disk radius. This controls same-endpoint diagonal quotients; opposite-endpoint denominators are bounded away from zero.

Near -1 , use the value at $-x$ and multiply by $(-1)^j$, omitting any putative $(-1)^\nu$. At integer n this prescription gives $\tilde{p}_{n+j}(x) = d_n(x)p_{n+j}(x)$, where $d_n(x) = 1$ near $+1$ and $d_n(x) = (-1)^n$ near -1 . Consequently both the Christoffel–Darboux kernel (with its original $x-y$ denominator) and every retained correction kernel are changed by $K(x, y) \mapsto d_n(x)K(x, y)d_n(y)$. For any endpoint set their determinant is unchanged, because the row and column multipliers contribute $\prod_h d_n(x_h)^2 = 1$. This proves agreement at integer dimensions even for configurations using both endpoints, while providing a holomorphic continuation in ν .

Proof of Proposition 6.1. Use the polynomial and endpoint-subset approximations just constructed. In each retained subset integral, replace the higher cumulants and kernel corrections by their continued near-index formulas, the constant kernel by the continued Christoffel–Darboux expression, and the linear traces and masses by their polynomial-antidifference approximants. Reconstruct $\mathcal{D}_n$ by formal exponentiation through source degree K_1 , and take the indicated source derivatives. The two labels are combined by independence.

Every expression before endpoint integration is holomorphic in ν, M , with a uniform bound e^{L^C} . Indeed source types, source orders, and determinant sizes are polynomial. Near-index blocks have length $O(m)$, so an expression with polynomially many index sums has at most $\exp(\text{poly}(L))$ terms. Permutation expansions, source coefficient extractions, and factorials of polynomial orders satisfy the same bound. Coefficients c_α include only the fixed polynomial P , centerings, and bounded endpoint data. Integration over fixed real endpoint regions preserves these bounds and holomorphy by dominated convergence (or Morera's theorem). No analyticity of the endpoint factors themselves as functions of their real positions is needed.

We record the order of the error choices to exclude a circular precision requirement. First fix B_0 using the original integer-law expectation bound. This fixes the dimension and gap in the Gaussian approximation. Next fix B_1 and K_1 ; then D_1, B_2, D_E , in that order; finally choose K_e for the factorial tail. All errors to this point have already been bounded on the actual integer laws by e^{-L^b} , with a freely reserved margin. They are not passed through an analytic continuation formula. The remaining formulas, through degree K_1 , are polynomial in the linear trace coefficients and in the kernel entries. Their sensitivity to the final trace and normalization errors is at most $e^{L^{C_*}}$, for an exponent C_* depending only on the choices already made. The preceding size bounds for trace and mass approximants are independent of their final polynomial Taylor order. Choose $B_3 > \max(b, C_*) + 1$, and the corresponding orders, last. The resulting errors are negligible after this final amplification. Taking the initially reserved margins, for example replacing each requested error by e^{-2L^b} , makes their finite sum at most e^{-L^b} for large L . This proves both assertions in (44). $\square$

Lemma 6.3 (Transfer from the integer grid). *Suppose M is in the fixed polynomial box and $|F_\nu(M)| \leq B$ at all integer grid points of $|\nu_\sigma - m| \leq \sqrt{m}$, with $B \geq e^{-L^{b'}}$, $b' < b$. If $\nu^* - m$ has polynomial size, then*

$$|\hat{F}(\nu^*, M)| \leq B \exp(o(L)). \quad (45)$$

The estimate is uniform under the stated bounds; in particular it applies to $\nu^ = m'(M)$.*

Proof. Taylor-expand $\hat{F}$ in its two dimension variables at (m, m) . Its bound e^{L^C} on the radius- $\sqrt{m}$ polydisk implies that a polynomial P of polynomial degree K approximates it on the half-radius polydisk to an error $o(B)$, by choosing K so that $e^{L^C} 2^{-K} = o(B)$. Put $R = \lfloor \sqrt{m}/3 \rfloor$, and let S be the sup norm of P on the real square $[m - R, m + R]^2$. The rescaled one-variable Chebyshev expansion has coefficients at most twice its sup norm and $\|T'_k\|_{[-1,1]} \leq k^2$. Holding the other variable real therefore bounds each partial derivative by $CK^3 S/R$. Comparing a maximizer with a nearest integer point gives

$$S \leq \max_{\text{integer square}} |P| + CK^3 S/R.$$

Since R is exponential and K polynomial, the second coefficient is $o(1)$. Proposition 6.1 and $b > b'$ then give $S \leq (1 + o(1))B$.

The tensor Chebyshev coefficients are at most $4S$, and there are at most $(K + 1)^2$ of them. At the target, each rescaled coordinate has magnitude $\text{poly}(L)/R$. Its complex inverse-cosine angle differs from $\pi/2$ by the same order, so every Chebyshev basis value up to degree K is bounded by $\exp(CK \text{poly}(L)/R) = 1 + o(1)$. Thus $|P(\nu^*)| \leq C(K + 1)^2 B$, which is $B \exp(o(L))$. The Taylor error is negligible relative to B . $\square$

For fixed explicit data, the prescribed totals $m'_\sigma(M)$ are affine in M . Under the retained polynomial restrictions, $m'(M) - m$ has polynomial size and lies in the dimension polydisk. Consequently

$$H(M) = \hat{F}(m'(M), M)$$

is holomorphic throughout the chosen polynomial M -disk. At every valid integer M in that disk, it approximates the original integer-dimensional integral to the precision in (44). The grid-transfer lemma supplies the sharper bounds on H away from those integer values; they do not follow from the coarse holomorphic bound alone.

7 The cylinder exponent

We can now sum the contour sectors. The interpolation just proved is needed only during the summation over imbalance: it gives a holomorphic function approximating the fixed-dimension integrals to arbitrarily high exponential precision. After moving that sum, we restore the scalar normalization and obtain the physical lower bound from positivity.

Take the balanced physical cylinder with $N = 2m$. Its row parameters are 2λ on the first group and λ on the second. Their multiplicative site values q^2, q become $1, q^{-1}$ under the common rescaling by q^{-2} , which leaves the contraction unchanged. Thus the logarithmic sites used below are $0, -i\beta$, as in Section 2.

Define the scalar

$$\Delta_N = \frac{\tilde{h}_N^2}{(\gamma/2)^N \prod_{i < j} H_*(a_i - a_j)^2}, \quad G_N(\eta) = \Delta_N Z_N(2 \cos \eta). \quad (46)$$

Theorem 2.11 proves that $\Delta_N \neq 0$ and $\log |\Delta_N| = (5/24) \log N + o(\log N)$. Dividing the exact contour-sector identity by the same extracted site product expresses G_N as the sum and integral over explicit configurations ζ of a frozen prefactor times

$$\sum_M e^{i(\eta + \pi f)M} F_{m'(M)}(M). \quad (47)$$

Only valid integer indices and dimensions occur. The explicit measures include their species factorials; the prefactor is independent of M and has modulus at most $\exp(C_h n_e)$.

Proposition 7.1 (Summed contour bound). *Locally uniformly in the strip $|\operatorname{Re} \eta| < \pi$,*

$$\log |G_N(\eta)| \leq 3\gamma L/8 - 2L \operatorname{Re}(\eta^2)/(4\kappa) + o(L) \quad (48)$$

Proof. Fix $|\operatorname{Re} \eta| \leq a_0 < \pi$, $|\operatorname{Im} \eta| \leq b_0$. Choose the small parameters for (41),(42) as in (30),(31). We first truncate the damped real sectors and replace their integrals by the holomorphic approximation. A rectangle summation formula then separates the integer sum into Fourier aliases, each of which can be moved to the height selected by its trial bound.

Truncation and holomorphic replacement. First the untruncated real bound in part 1 of Proposition 5.1 lets us restrict $n_e, \ell_e \leq L^3$ and $|M| < T$, with T a half-integer of a sufficiently large fixed polynomial scale to be increased below. Indeed $m'(M)$ on valid indices is $O(m)$; summing the reserved part of damping in M costs only $O(\sqrt{L})$. A reserved exponential damping in ℓ_e makes the summed explicit factorial measures, even with the per-particle factors, cost $\exp(O_h(L))$, just by integration of each variable. Thus these tail errors after prefactors are, say, $O(\exp(-c_h L^2))$. Under the resulting restrictions all integers inside the M -cut are valid for large L ; $\nu = m'(M)$ lies within a polynomial distance of m .

At each retained ζ use (44) to replace the integral in (47) by $H(M) = \hat{F}(m'(M), M)$. The precision power and domain power in (44) can be taken sufficiently large after all polynomial bounds below; hence the summed error is negligible, even with the phase, prefactors and integration. For

instance even the undamped truncated factorial measures with fixed per-particle factors cost at most $\exp(\text{poly}(L))$. For each bound (40)–(42) at a particular M below we can transfer it from all independent integer ν near m to $H(M)$ by (45), losing only $o(L)$ in the log. Here truncate j polynomially when transferring (41),(42); in (40) just transfer for $j = 0$ and then insert any alias phase exactly. Indeed the quantitative upper majorants used for $|F_\nu(M)|$ (after removing phases) are bounded below by $\exp(-\text{poly}(L))$ with powers fixed independently of the precision choice. All such transfers are uniform.

The rectangle formula and alias shifts. Write $f_\eta(M) = e^{i(\eta+\pi f)M} H(M)$, and let $\mathcal{R}_T$ be the counterclockwise rectangle with real endpoints $\pm T$ and imaginary heights ± 1 . Since the residue of $(1 - e^{2\pi i M})^{-1}$ at an integer is $-1/(2\pi i)$, the exact summation formula is

$$\sum_{k \in \mathbb{Z}, |k| < T} f_\eta(k) = - \int_{\partial \mathcal{R}_T} \frac{f_\eta(M)}{1 - e^{2\pi i M}} dM.$$

At the half-integer vertical sides, the kernel has modulus at most one. On the two horizontal sides its absolutely convergent expansions are

$$\frac{1}{1 - e^{2\pi i M}} = \begin{cases} \sum_{j \geq 0} e^{2\pi i j M}, & \text{Im } M = 1, \\ - \sum_{j < 0} e^{2\pi i j M}, & \text{Im } M = -1. \end{cases}$$

Equation (40) bounds the alias tails by a geometric factor $e^{-2\pi J}$, times an exponential of a fixed polynomial from the explicit truncations. Thus it suffices to retain $|j| \leq J = L^{C_J} + O(1)$, choosing C_J sufficiently large. This choice can precede the final power of T : the Gaussian damping in $\text{Re } M$ is integrable on the whole line, and even any later fixed polynomial interval length is absorbed by $e^{-2\pi J}$.

For each remaining alias, move its horizontal integration at fixed endpoints' real parts to the height v selected in (30) if $U > U_0$, or (31) if $U \leq U_0$. The choice depends only on alias, bias, and explicit data, and has a polynomial bound independent of T . The connecting verticals and the original end integrals are all negligible by (40): there the loss $c_h T^2/L$ can dominate all other powers, including possible growth from $|jv|$, by taking the power of T sufficiently large after C_J . These choices only require a fixed polynomial box in (44); the precision there is now chosen as just described. All movements of H times its phases are thus within its holomorphy domain.

Summing the shifted bounds. On the shifted lines (41),(42) apply, up to $o(L)$ by the transfer. In the large- U case integrate the bound (41) using exponential ℓ_e -damping and factorials, and absorb its $O_h(L)$ costs by taking U_0 sufficiently large (after δ and h). In the bounded- U case the bound (42) costs after prefactors and integration at most an additional $C_{h,\varepsilon} \delta L + o(L)$ in the log, as following (31). More explicitly this case is supported on explicit configurations with at most C_{U_0} far particles on $|u| \leq (1 - 2\delta)L$, no restriction needed for near outer particles other than being exterior. Thus this same integrated bound works for each alias and there are only polynomially many; likewise horizontal length costs at most polynomially. This pointwise split does not require any contour movement of explicit variables. All constants allow δ small after ε, h , then U_0 arbitrarily large fixed, little-oh errors then for these fixed choices. Sending the small errors to zero proves (48). $\square$

Proof of Theorem 1.1. By (46), the physical exponent is the contour numerator exponent minus the scalar exponent. The deformation determinant contributes $7/8$, the two reference masses contribute

$-1/2$, and the scalar normalization contributes $-5/24$, all as coefficients of $\log N$. Thus the leading constants are

$$\frac{7}{8} - \frac{1}{2} - \frac{5}{24} = \frac{1}{6}, \quad \frac{1}{2\gamma\kappa} = \frac{2}{3\pi^2},$$

since $\gamma = 4/3$ and $\kappa = (3\pi/4)^2$. Thus Proposition 7.1 gives, locally uniformly in $|\operatorname{Re} \eta| < \pi$,

$$u_N(\eta) := \frac{\log |Z_N(2 \cos \eta)|}{\log N} \leq \mathcal{H}(\eta) + o(1), \quad \mathcal{H}(\eta) = \frac{1}{6} - \frac{2}{3\pi^2} \operatorname{Re}(\eta^2).$$

Propagation from the empty-configuration value. The function $\mathcal{H}$ is harmonic. Since Z_N has nonnegative coefficients and $Z_N(0) = 1$, the interior point $\eta = \pi/2$ satisfies $u_N(\pi/2) = \mathcal{H}(\pi/2) = 0$. We show that every fixed disk inside the strip contains points η_N with $u_N(\eta_N) \geq \mathcal{H}(\eta_N) - o(1)$.

Join the anchor to the disk by a compact path in the strip and choose a connected neighborhood with positive clearance from its boundary. The locally uniform upper bound permits a constant $\epsilon_N \downarrow 0$ such that $v_N = \mathcal{H} + \epsilon_N - u_N \geq 0$ on this neighborhood. It is superharmonic and locally integrable; zeros of $Z_N(2 \cos \eta)$ give only logarithmic singularities, and the anchor ensures this holomorphic function is not identically zero. At the anchor, $v_N = \epsilon_N$.

Choose a finite chain of disks of radius r , with successive centers at distance at most r , leading into the target disk. Take r small enough that, from any point in one disk, the radius- $3r$ disk stays in the chosen neighborhood and covers the next disk. The area mean inequality for a nonnegative superharmonic function implies that if $v_N(z) \leq a_N$ at the chosen point, then its integral over the next radius- r disk is at most $9\pi r^2 a_N$. Some point there therefore has value at most $9a_N$. Starting at the anchor and iterating the fixed finite chain yields $v_N(\eta_N) \leq 9^k \epsilon_N = o(1)$ in the target.

Comparison with positive real fugacities. The disk argument supplies a near-saturating point, not a pointwise lower bound throughout the disk. Positivity of the physical polynomial converts this into the desired lower bound at a positive fugacity. For $\chi = 2$, take target disks about small positive real values of η , sufficiently narrow that $|2 \cos \eta| < 2$ throughout. Positivity of coefficients gives $Z_N(2) \geq |Z_N(2 \cos \eta_N)|$. Thus the lower limit of $\log Z_N(2)/\log N$ is at least the infimum of $\mathcal{H}$ on each such disk. Letting these disks approach zero gives $1/6$; the upper bound at zero matches it.

If $0 < \chi < 2$, put $\eta_\chi = \arccos(\chi/2)$. Choose real disk centers decreasing to η_χ from above, and radii sufficiently small that $|2 \cos \eta| \leq \chi$ throughout. The same positivity argument gives the lower exponent $\mathcal{H}(\eta_\chi)$, matching the upper bound there. If $\chi > 2$, use $\eta_\chi = i \operatorname{arcosh}(\chi/2)$ and disk centers approaching it along the imaginary axis from smaller imaginary parts. Again their radii can be chosen so that $|2 \cos \eta| \leq \chi$. This proves the remaining formula. The argument obtains a lower bound from the polynomial's positivity and the interior anchor; it requires no lower estimate on a single complex contour sector. $\square$

8 A uniform bound for the number of separating polygons

The fixed-fugacity cylinder asymptotic determines the nesting exponent. A different consequence of the same finite sector identity controls the number of separating polygons, including counts that grow with the width. For that purpose the fugacity must be allowed to grow without a fixed upper bound. The real, untruncated sector estimate provides this uniformity directly.

Use the physical balanced two-marker cylinder with even width $N = 2m$, band parameters 2λ on the first half and λ on the second, and $\lambda = \pi/8$. For each integer $a \geq 0$, let $w_N(a)$ be the critical mass of families of exactly a pairwise vertex-disjoint polygons, each separating the two markers.

The weight of a family is $\prod_P \rho^{|P|}$, with $\rho = (2 + \sqrt{2})^{-1/2}$ per polygon vertex. In particular the factor two for a polygon is not included in $w_N(a)$. The physical partition is

$$Z_N(\chi) = \sum_{a \geq 0} w_N(a) \chi^a, \quad w_N(a) \geq 0, \quad w_N(0) = 1. \quad (49)$$

This is the separating-loop contraction defined in Section 2. At fixed N it is a polynomial with finite coefficients; a cut between the two markers bounds the number of disjoint separating polygons.

Theorem 8.1 (Uniform fugacity bound). *There are constants $C < \infty$ and N_0 such that, for every even $N \geq N_0$ and every real $v \geq 0$,*

$$Z_N(2 \cosh v) \leq \exp(C \log N (1 + v^2)). \quad (50)$$

The constants are independent of v .

Proof. Set $L = (\log N)/\gamma$, where $\gamma = 4/3$, and fix the smoothing parameter used in the real-sector bound. It is fixed independently of N and v . We use the exact sector identity (47) at $\eta = iv$, retaining all valid integer sectors and dimensions. The imbalance M remains a real integer throughout this proof. Thus $v = \text{Im } \eta$, whereas $\text{Im } M = 0$.

Use the normalization (46),

$$G_N(\eta) = \Delta_N Z_N(2 \cos \eta).$$

The scalar Δ_N is nonzero and independent of η . We first bound $G_N(iv)$, and then divide by $|\Delta_N|$.

Recall the quantities in that identity. The explicit configuration ζ consists of exterior main particles and all far particles; n_e counts its integration variables, with a fused double counted as one variable, and

$$\ell_e = \sum_{z \in \zeta} (|\text{Re } z| - L)_+.$$

The interior dimensions $m'_\sigma(M)$ are nonnegative integers at most N . Each explicit species retains its factorial divisor; its elementary multiplicity is one for a singleton and two for a fused double. The frozen prefactor has modulus at most $\exp(C_h n_e)$. It is independent of η and M .

At $\eta = 0$, part 1 of Proposition 5.1 gives

$$|F_{m'(M)}(M)| \leq \exp \left(C_h L + C_h n_e - c_h \left(\frac{M^2 + n_e^2}{L} + \ell_e \right) \right). \quad (51)$$

Its dimension range $0 \leq m'_\sigma \leq C e^{\gamma L}$ includes every valid sector because $e^{\gamma L} = N$. In particular, (51) requires no polynomial restriction on M , n_e , or ℓ_e . That assertion uses the coarse real reference-source inequality and compensated quadratic integration; the later sharp source estimate and interpolation of dimensions are not needed here.

The phase of the exact sector identity is $\exp(i(\eta + \pi f)M)$, where the far-particle integer f is fixed by ζ . Replacing $\eta = 0$ by iv therefore multiplies its modulus by $e^{-vM} \leq e^{v|M|}$. Every other factor in (51) is unchanged. Combining the prefactor and this estimate bounds the divided numerator by

$$e^{C_h L} \sum_{M \in \mathbb{Z}} e^{-c_h M^2 / L + v|M|} \sum_{\zeta} e^{C'_h n_e - c_h \ell_e}.$$

We have discarded the nonpositive n_e^2 term and relaxed the valid-count constraints, which only enlarges the positive majorant. All integrals in the last sum retain their species factorials.

Here is the explicit-configuration estimate. For a far species its real parameter ranges over $\mathbb{R}$, and

$$\int_{\mathbb{R}} e^{-c_h(|u|-L)_+} du = 2L + 2/c_h.$$

For an exterior main species it ranges over $|u| > L$, giving $2/c_h$. Fixed contour-measure and activity constants can be absorbed into $e^{C'_h}$. For each of the finitely many species, with parameter domain D_I , the factorial sum is at most

$$\sum_{n=0}^{\infty} \frac{e^{C'_h n}}{n!} \left(\int_{D_I} e^{-c_h(|u|-L)_+} du \right)^n \leq e^{C''_h(L+1)}.$$

Taking the product gives $e^{C_h L}$, for $L \geq 1$. The elementary multiplicity two of a fused double affects the admissibility constraint already dropped; it does not remove its factorial divisor or introduce a second integration variable.

Only the imbalance sum remains. Young's inequality gives

$$-c_h M^2/L + v|M| \leq -\frac{c_h M^2}{2L} + \frac{Lv^2}{2c_h},$$

so

$$\sum_{M \in \mathbb{Z}} e^{-c_h M^2/L + v|M|} \leq C_h \sqrt{L} e^{C_h L v^2}.$$

Absorbing $\sqrt{L}$ into $e^{C_h L}$ proves $|G_N(iv)| \leq e^{C_h L(1+v^2)}$ for the divided numerator G_N in (47), uniformly for all $v \geq 0$.

Finally divide by the η -independent denominator Δ_N . The balanced denominator theorem, Theorem 2.11, gives $\log |\Delta_N| = (5/24) \log N + o(\log N)$. In particular it is nonzero and $|\Delta_N|^{-1} \leq e^{C \log N}$ for all sufficiently large even N . This bound is independent of v . The physical value $Z_N(2 \cosh v)$ is nonnegative by (49). Dividing the numerator estimate and using $L = (\log N)/\gamma$ proves (50). $\square$

Corollary 8.2 (Tail in the polygon count). *There are constants $c, C > 0$ such that for every sufficiently large even N and every integer $a \geq 0$,*

$$w_N(a) \leq \exp \left(C \log N - \frac{ca^2}{\log N} \right). \quad (52)$$

Proof. For every $v \geq 0$, positivity in (49) gives

$$w_N(a)(2 \cosh v)^a \leq Z_N(2 \cosh v).$$

Since $\log(2 \cosh v) \geq v$, Theorem 8.1 yields

$$w_N(a) \leq \exp \left(C \log N (1 + v^2) - av \right).$$

Take $v = a/(2C \log N)$. The exponent becomes $C \log N - a^2/(4C \log N)$, as required. This includes $a = 0$; indices beyond the polynomial degree have zero mass. $\square$

9 Winding pressure and essential polygons

Throughout the pressure and nesting arguments, except during the explicitly indicated variation of the loop fugacity, put

$$\lambda = \frac{\pi}{8}, \quad \beta = 2\lambda = \frac{\pi}{4}, \quad \gamma = \frac{\pi}{3\beta} = \frac{4}{3}, \quad \rho = \frac{1}{\sqrt{2 + \sqrt{2}}}.$$

A geometric honeycomb polygon has weight $\rho^{|P|}$, where $|P|$ counts vertices. Polygons are unrooted and unoriented; when translation classes are used, the translation group will always be specified.

The cylinder exponent concerns polygons separating two marked points. To obtain a nesting law around one planar point, we must remove two kinds of polygon with a bounded multiplicative cost: essential polygons that wind around the cylinder, and contractible polygons too large to fit near either marker. We treat them separately. The first two derivatives of an annular pressure control the total essential-polygon weight and its horizontal-span moment. Section 10 obtains the planar diameter tail needed for contractible polygons from a separate open-strip pressure. Section 11 then compares the surviving configurations with two independent planar nests and derives the strip-length estimates. The present section supplies the essential-polygon removal bound.

9.1 The annular transfer and its interpolation data

Let $N = 2m$, and assign the additive site parameters

$$b_1 = \cdots = b_m = \lambda/2, \quad b_{m+1} = \cdots = b_{2m} = -\lambda/2.$$

The auxiliary argument $z_0 = 3\lambda/2$ in the column of boxes $R(z - b_i)$ gives the physical box arguments λ and 2λ , respectively. In the transfer considered here an essential closed curve receives fugacity χ , whereas a contractible closed curve receives zero. This convention is specific to the winding pressure.

A state records a cyclic noncrossing partial matching together with a marked complementary region, namely the region containing the past end of the cylinder. Equivalently, draw the matching in a disk with a marked puncture and no strand ending at the puncture. There are finitely many states: an ordinary cyclic noncrossing matching has finitely many regions in which to place the puncture. This information determines composition. Indeed two realizations of the same state decompose the disk into disks and one punctured disk; homeomorphisms of these pieces fixing their boundary arcs identify their sewing rules and distinguish essential from contractible cycles.

Let $T_\chi(z)$ carry the auxiliary slot around all sites, joining its final output to its initial input. The local identities of the strip calculation apply, since every cycle internal to one of those identities is contractible. At $(\chi, z) = (0, z_0)$, the empty-coordinate functional is invariant and the restriction to vectors with zero empty coordinate has spectral radius less than one. To see the latter assertion in the refined state space, a nonzero diagram with nonempty initial connectivity either closes a killed loop or contains a path joining its two longitudinal cuts. The periodic-slab spanning estimate in Lemma 2.5 therefore gives decay of its powers, exactly as for the unmarked connectivity space. Consequently the eigenvalue 1 is simple and strictly dominant, with analytic continuation under small changes of χ and of the site parameters.

The matrices $T_\chi(z)$ commute as z varies. Insert an invertible crossing between two auxiliary lines, move it through the sites by the three-box identity, and cancel it at the other end; the resulting diagram reverses the two columns. This proves commutation at generic parameters and hence as a rational identity. The simple eigenspace at z_0 is therefore invariant under every $T_\chi(z)$. Denote its eigenvalue by $\Lambda(\chi; z)$, normalizing an eigenvector to have empty coordinate one. At $\chi = 0$, the empty-coordinate rule gives $\Lambda(0; z) = 1$.

Lemma 9.1 (Annular pinch and analytic data). *The function $\Lambda(\chi; z)$ is rational in e^{2iz} , with denominator dividing $\prod_i \mathcal{D}(z - b_i)$, where $\mathcal{D}$ is the local denominator in (2), and tends to $1 + \chi$ at both ends of this multiplicative coordinate. At the balanced specialization it is invariant under $z \mapsto 3\lambda - z$. For $u = b_i$ and generic remaining sites,*

$$\Lambda(\chi; u + j\lambda)\Lambda(\chi; u) = \begin{cases} \Lambda(\chi; u + \lambda), & j = 2, \\ 1, & j = 3. \end{cases} \quad (53)$$

Set $w = 2i(z - z_0)$ and, with derivatives evaluated at $\chi = 0$, define

$$\begin{aligned} f(w) &= \partial_\chi \Lambda(\chi; z), & h_2(w) &= \partial_\chi^2 \Lambda(\chi; z), & g(w) &= h_2(w) - f(w)^2, \\ (\mathcal{L}v)(w) &= v(w + i\beta) + v(w - i\beta) - v(w). \end{aligned} \quad (54)$$

Then $\mathcal{L}f$ and $\mathcal{L}g$ vanish to orders $2m, m, m$ at $0, i\beta, -i\beta$, respectively. The functions $f - 1$ and h_2 are even rational functions of e^w , vanish at both real infinities, and have poles only at the translates of these three knots by $i\pi$, modulo $2\pi i$, with the same respective order bounds.

Proof. The common eigenvector is independent of z . Computing the empty output of the transfer on that vector proves the rationality and denominator bound. A shift $z \mapsto z + \pi$ inserts occupancy signs that cancel around the closed auxiliary. At either imaginary infinity, $A, U \rightarrow 0$, $B \rightarrow 1$, and all entries remain bounded. Only empty input reaches empty output; it contributes the empty diagram and one essential auxiliary circle, with weights 1 and χ . Reflection reverses the row order and negates all b_i ; the balanced list is thereby preserved, and simplicity identifies the reflected eigenline with the original one.

For the pinch, pass the two auxiliaries in order $[u + j\lambda, u]$. At a host with $x = u - b_k$ their crossing is

$$M_x = R_1(x + j\lambda)R_2(x), \quad (I \otimes S_j^\dagger)M_x = \hat{R}(S_j^\dagger \otimes I).$$

At the marked host, $x = 0$ and $M_0 = (S_j \otimes I)(S_j^\dagger \otimes I)$. Close the paired auxiliary trace and move the transpose splitter backwards through all other hosts. At the marked host the remaining contraction is

$$(I \otimes S_j^\dagger)(S_j \otimes I) = \hat{R}|_{x=0}.$$

For completeness, this identity follows by multiplying on the right by $S_j^\dagger \otimes I$, using the transposed splitting identity at $x = 0$, and cancelling that factor. Cancellation is valid because $S_j^\dagger$ is surjective: insert vacancy, or one through strand into a fixed output and divide by its nonzero coefficient. The fused auxiliary is the one of argument $u + \lambda$ for $j = 2$, and disappears for $j = 3$. These are identities of annular diagrams, so they prove (53) with the essential-loop fugacity retained.

The first two derivatives of $\log \Lambda$ are f and g . Taking their derivatives in (53) makes $\mathcal{L}f$ and $\mathcal{L}g$ vanish at the points corresponding to both $b_i + \lambda$ and $b_i + 2\lambda$. At generic sites the factors giving these zeros are distinct. Analyticity of the normalized eigenvector near the balanced physical sites, and regularity of the boxes at all the evaluation points, permit confluence. The two families give the stated multiplicities. The two sine factors in each $\mathcal{D}(z - b_i)$ give the pole list. Since the denominator is independent of χ , taking two χ -derivatives does not increase those orders. $\square$

9.2 Uniform Hardy-space interpolation

The interpolation conditions just obtained will estimate the pressure at the physical point $w = 0$. Their number grows with the cylinder width, so invertibility alone is insufficient: the inverse must

remain bounded as the knots coalesce and their multiplicities grow. The following Hardy-space realization separates that issue from the particular pressure derivatives.

We establish the analytic estimates with constants independent of N . Let H be the Hardy space on $|\operatorname{Im} w| < h$, where

$$h = \frac{3\beta}{2} = \frac{\pi}{2\gamma}.$$

Its squared norm is the sum of the squared $L^2(dx)$ norms on the two boundary lines. With Fourier convention $\widehat{v}(k) = \int e^{-ikx} v(x) dx$,

$$\|v\|_H^2 = \int_{\mathbb{R}} 2 \cosh(3\beta k) |\widehat{v}(k)|^2 \frac{dk}{2\pi}. \quad (55)$$

The reproducing kernel is a positive constant times $\operatorname{sech}(\gamma(w - \bar{v})/2)$. One way to justify the boundary description is to start with uniformly bounded L^2 norms on interior horizontal lines. The Cauchy–Riemann equation gives their Fourier transforms as $e^{-ky}\widehat{v}(k)$ at height y , initially in distributions on compact frequency intervals. Monotone convergence of the resulting weighted integrals gives (55) and the two boundary traces.

The prescribed jets determine a finite Hardy model space. We use the strip form of the standard model-space construction [20, Section 1], proving its repeated-zero formulation directly below. A jet of order r at a point means the value and the first $r - 1$ derivatives there. Let P_N be the orthogonal projection onto their representing vectors for the three knots in Lemma 9.1, with orders $2m, m, m$. List their imaginary coordinates, with multiplicities, as θ , and put

$$B_N(w) = \prod_{\theta} \frac{e^{\gamma w} - e^{i\gamma\theta}}{e^{\gamma w} + e^{-i\gamma\theta}}, \quad A_N = \gamma^{-1} \log \left(\sum_{\theta} 2 \cos(\gamma\theta) \right) = \gamma^{-1} \log(3N). \quad (56)$$

Each factor has modulus one on the boundary and at most one in the strip. Thus multiplication by B_N is an isometry on H , and

$$(I - P_N)H = B_N H. \quad (57)$$

Indeed division by the finite product cancels precisely the specified zeros, preserves boundary norms, and leaves bounded interior L^2 norms. This also proves that the jet space has the asserted dimension.

For $0 < a < 1$, define the Fourier multiplier

$$D_a(k) = \frac{(2 \cosh(\beta(k + ia)) - 1) \cosh(3\beta k)}{\sin(\pi(a - ik))}. \quad (58)$$

We use the same symbol for the resulting bounded operator on H .

Lemma 9.2 (Coercive jet interpolation). *For every fixed $0 < a < 1$, the operators D_a are bounded and satisfy*

$$\operatorname{Re} D_a(k) \geq c_a > 0 \quad (k \in \mathbb{R}), \quad \|(P_N D_a|_{P_N H})^{-1}\| \leq c_a^{-1}. \quad (59)$$

Moreover, writing $\|\cdot\|_{\partial,p}$ for the sum of the two boundary L^p norms, one has

$$\|e^{aw}(1 - f)\|_{L^p(\operatorname{Im} w = y)} \leq C_a e^{aA_N}, \quad p = 2, 4, \quad |y| \leq h + \beta. \quad (60)$$

For fixed $0 < a < 1/2$,

$$|\mathcal{L}f(s)| + |\mathcal{L}g(s)| \leq C_a |B_N(s)| (1 + e^{2a(A_N - s)}), \quad s \geq 0. \quad (61)$$

For $\mathcal{L}f$ alone, $2a$ can be replaced by any fixed $a \in (0, 1)$.

Proof. We first solve the jet equation in the Hilbert norm. We then obtain the fourth-norm estimate needed for the quadratic term f^2 , and divide out the prescribed zeros to estimate the residual.

The jet equation and coercivity. Partial fractions for the rational functions in Lemma 9.1 use derivatives of $(1 + e^{w-i\theta})^{-1}$, up to the specified pole orders, and a constant determined by the limit at positive infinity. That constant vanishes for $1 - f$ and h_2 . The beta integral, followed by a pole-free contour shift, gives

$$\int_{\mathbb{R}} \frac{e^{(a-ik)x}}{1 + e^{x-i\theta}} dx = \frac{\pi e^{i\theta(a-ik)}}{\sin \pi(a-ik)}.$$

Differentiated terms contribute polynomials in k . Weighting by e^{aw} changes the shift multiplier to $2 \cosh(\beta(k+ia)) - 1$. Kernel jet vectors have transforms equal to polynomials times $e^{-k\theta} / \cosh(3\beta k)$. The knot multiset is symmetric, and consequently

$$e^{aw} \mathcal{L}(1 - f) = D_a x_N, \quad x_N \in P_N H, \quad e^{aw} \mathcal{L} h_2 \in D_a P_N H. \quad (62)$$

All contour shifts used here lie before the first poles of the rational functions.

To check coercivity, symmetry reduces the problem to $k \geq 0$. The real part of $2 \cosh(\beta(k+ia)) - 1$ is positive. If $a \geq 1/2$, its imaginary part and that of $\sin \pi(a-ik)$ have the same sign, so the real part of their quotient is positive. If $a < 1/2$, their imaginary parts have opposite signs, but

$$\frac{\operatorname{Im}(2 \cosh(\beta(k+ia)) - 1)}{\operatorname{Re}(2 \cosh(\beta(k+ia)) - 1)} \leq \frac{2 \sin(\beta a)}{2 \cos(\beta a) - 1} < \tan(\pi a).$$

The last inequality follows, with $x = \beta a \in (0, \pi/8)$, by multiplying positive denominators: the difference of the cross-products is $2 \sin(3x) - \sin(4x) > 0$. This is sufficient since the absolute ratio of the imaginary and real parts of the denominator is at most $\cot(\pi a)$. Finally

$$D_a(+\infty) = e^{i(\pi/2-3\beta a)}, \quad D_a(-\infty) = e^{-i(\pi/2-3\beta a)}.$$

These limits have positive real part. Continuity proves the uniform lower bound and boundedness. Compressing the real-part inequality to the finite-dimensional jet space proves the inverse bound.

The product defining B_N gives, throughout the closed strip,

$$|1 - B_N(w)| \leq C \min\{1, e^{-\gamma(\operatorname{Re} w - A_N)}\}. \quad (63)$$

For large positive real part, sum the deviations of individual factors from one; for the remaining points use $|B_N| \leq 1$. It follows by direct integration, for $p = 2, 4$, that

$$\|e^{aw}(1 - B_N)\|_{\partial, p} \leq C_a e^{aA_N}.$$

The jet conditions give the exact interpolation equation

$$P_N D_a x_N = P_N [e^{aw}(1 - B_N)]. \quad (64)$$

Thus $\|x_N\|_H + \|D_a x_N\|_H \leq C_a e^{aA_N}$.

The product estimate. The second pressure derivative involves f^2 . We therefore need a fourth-norm estimate in addition to the Hilbert-space bound just proved; it will control this product without losing a power of N . Here are the additional bounds needed for products of analytic functions. Embed H in the two-component boundary space, and let Π be the orthogonal projection onto its traces. In each Fourier fiber it projects onto (e^{-hk}, e^{hk}) . Its diagonal entries differ from the appropriate positive- or negative-frequency projections by bounded square-integrable symbols; its off-diagonal entries decay exponentially. Such remainders map L^2 into $L^2 \cap L^\infty$, hence into

L^4 . The half-frequency projections are bounded on L^4 , the $p = 4$ case of the classical M. Riesz conjugate-function boundedness theorem; for a real-line reference see [11, 12]. Only boundedness, not a sharp norm, is used here. An elementary proof for real Schwartz data writes its positive-frequency part as $u + iv$, with u half the original data. Fourier support gives $\int (u + iv)^4 = 0$, whence

$$\int v^4 \leq 6 \left(\int u^4 \int v^4 \right)^{1/2}.$$

The identity is justified first for frequency cutoffs, whose inverse transforms are bounded with $O(|x|^{-1})$ tails; density and complex linearity then prove the assertion. Therefore Π is bounded from boundary $L^2 \cap L^4$ to itself, with the sum norm.

On the full boundary space,

$$P_N = \Pi - B_N \Pi \overline{B_N}.$$

Boundary multiplication by B_N preserves every L^p norm. Hence these projection bounds are uniform in N . Moreover, the commutator of P_N with any fixed constant diagonal matrix maps L^2 to L^4 uniformly: only the off-diagonal entries of Π occur in its commutator, and multiplication by B_N commutes with the diagonal matrix.

Let D_{edge} be the diagonal matrix whose two entries are the limiting values of D_a in the dominant frequency direction of the corresponding boundary trace. On H , $D_a - D_{\text{edge}}$ maps boundary L^2 to L^4 . On each dominant half-axis this follows from exponential convergence of the symbol; on the other half-axis the trace has exponential decay relative to the weighted norm (55). Since $x_N = P_N x_N$, the preceding commutator estimate yields

$$P_N D_a x_N = D_{\text{edge}} x_N + E_N, \quad \|E_N\|_{\partial,4} \leq C_a \|x_N\|_H.$$

The right side of (64) has boundary fourth norm at most $C_a e^{aA_N}$; the diagonal entries of D_{edge} are nonzero. This proves the same fourth-norm bound on x_N and on $D_a x_N$.

Invert the weighted shift multiplier in (62). To move a trace from height h to height $h + \beta$, or from $-h$ to $-h - \beta$, the multiplier is, up to a constant phase,

$$\frac{e^{\pm \beta k}}{2 \cosh(\beta(k + ia)) - 1}.$$

It has finite constant limits on the two half-axes and exponentially decreasing remainders, so the preceding L^2 and L^4 argument applies. Interior lines follow by the two-boundary Fourier representation. This proves (60).

The second derivative and the residual. Now fix $a < 1/2$, and write $Y_2 = e^{2aw} \mathcal{L} h_2 \in D_{2a} P_N H$. Since $\mathcal{L} g$ has the prescribed jets,

$$P_N Y_2 = P_N [e^{2aw} (\mathcal{L}(f^2) - B_N)]. \quad (65)$$

If $v = 1 - f$, then $\mathcal{L}(f^2) = 1 - 2\mathcal{L}v + \mathcal{L}(v^2)$. The second-norm bound at weight $2a$, the fourth-norm bound at weight a on the shifted lines, and (63) show that the interpolation input has H -norm at most $C_a e^{2aA_N}$. Coercivity gives this bound for Y_2 as well.

Subtract the right interpolation input from each left input in (64) and (65). The differences lie in $B_N H$. Division by B_N is isometric, and evaluation at a real point has uniformly bounded norm in H . In the first equation the difference is $e^{aw} (B_N - \mathcal{L} f)$; in the second it is $e^{2aw} (\mathcal{L} g + B_N)$. This proves (61) and its first-derivative variant. $\square$

9.3 The size and first spatial moment of winding pressure

We now turn the uniform interpolation estimate into two bounds at the physical point: the first derivative has order N^{-1} , and the second logarithmic derivative has absolute value $O(N^{-1})$. Their polygon interpretation will then bound the total essential weight and its first horizontal-span moment.

We will use the inverse of $\mathcal{L}$. Its convolution kernel on the real line is

$$\rho_*(s) = \frac{1}{\beta\sqrt{3}} \frac{\sinh(2\pi s/(3\beta))}{\sinh(\pi s/\beta)}, \quad \widehat{\rho}_*(k) = \frac{1}{2 \cosh(\beta k) - 1}, \quad (66)$$

where the quotient at zero is interpreted continuously. This is exactly the mean density ϱ from Section 3: it canceled the particle-site sources there, and here it inverts the shift operator $\mathcal{L}$. The Fourier identity follows by a residue computation, or by integrating after the substitution $u = e^{\beta k}$. In particular

$$\rho_*(s) > 0, \quad \int \rho_* = 1, \quad \rho_*(s) \leq C e^{-\gamma|s|}, \quad \lim_{s \rightarrow +\infty} e^{\gamma s} \rho_*(s) = \frac{1}{\beta\sqrt{3}}. \quad (67)$$

Lemma 9.3 (Winding pressure bounds). *There are constants $0 < c < C < \infty$ such that, for all sufficiently large even N ,*

$$\frac{c}{N} \leq f(0) \leq \frac{C}{N}, \quad |g(0)| \leq \frac{C}{N}. \quad (68)$$

In fact $e^{\gamma A_N} f(0)$ has a finite strictly positive limit.

Proof. The residual estimate already gives both upper bounds. The lower bound requires a separate argument: after shifting to the edge $A_N = \gamma^{-1} \log(3N)$, we identify a limiting interpolation problem and show that its contribution at the center is strictly positive.

Upper bounds at the center. On the positive real axis the product in (56) satisfies

$$|B_N(s)| \leq \exp\{-ce^{\gamma(A_N-s)}\}, \quad s \geq 0. \quad (69)$$

For an individual factor, subtract its squared modulus from one; on $e^{\gamma s} \geq 1$ the result is bounded below by a positive constant times $e^{-\gamma s}$. Multiplying the factors proves (69). The functions f, g are analytic on a neighborhood of the lines required for $\mathcal{L}$ and tend exponentially to constants at both real infinities. Fourier inversion after subtracting those constants therefore gives

$$f = \rho_* * \mathcal{L}f, \quad g = \rho_* * \mathcal{L}g.$$

By evenness, (61), and (67), their values at zero are bounded by a constant times

$$\int_0^\infty e^{-\gamma s} e^{-ce^{\gamma(A_N-s)}} (1 + e^{2a(A_N-s)}) ds \leq C e^{-\gamma A_N} = \frac{C}{3N}.$$

The last inequality follows on substituting $u = e^{\gamma(A_N-s)}$; the remaining integral is bounded by $\gamma^{-1} \int_0^\infty e^{-cu} (1 + u^{2a/\gamma}) du$.

The translated interpolation limit. For the lower bound, choose $a = \gamma/2$ and $p = a/\gamma = 1/2$. Translate $w = A_N + t$ in (64) and multiply its unknown and right side by e^{-aA_N} . Expansion of each Blaschke factor, uniformly on compact subsets of the closed strip, gives

$$B_N(A_N + t) \longrightarrow B_\infty(t) := \exp(-e^{-\gamma t}). \quad (70)$$

Indeed the sum of the first-order terms is $e^{-\gamma t}$, by the definition of A_N , whereas the sum of their squared errors is $O(N^{-1})$ on compact sets. Both products have boundary modulus one. Their multiplication operators and adjoints converge strongly by dominated convergence, so the boundary projection formula gives strong convergence of the translated projections to

$$P_\infty = I - \text{proj}(B_\infty H).$$

Also (63) supplies an integrable boundary majorant, proving strong convergence in H of the translated input to $e^{at}(1 - B_\infty(t))$.

For a projection P , extend its compression to $PD_aP + (I - P)$ on H . Its real part is bounded below by $\min(c_a, 1)$, as is the real part of its adjoint. It is thus invertible with uniformly bounded inverse, including at $P = P_\infty$. Write the extended operators as $\mathcal{A}_N$ and their strong limit as $\mathcal{A}$. The identity $\mathcal{A}_N^{-1} - \mathcal{A}^{-1} = \mathcal{A}_N^{-1}(\mathcal{A} - \mathcal{A}_N)\mathcal{A}^{-1}$ shows strong convergence of their inverses. The normalized unknowns consequently converge to $x \in P_\infty H$ satisfying

$$P_\infty D_a x = P_\infty [e^{at}(1 - B_\infty(t))]. \quad (71)$$

Bounded evaluation on the real line implies

$$\mathcal{L}f(A_N + t) \longrightarrow b_\infty(t) := 1 - e^{-at}(D_a x)(t).$$

The first-derivative version of (61) dominates this expression by $Ce^{-ce^{-\gamma t}}(1 + e^{-at})$. This is integrable after multiplication by $e^{-\gamma t}$. Hence

$$\lim_{N \rightarrow \infty} e^{\gamma A_N} f(0) = \frac{2}{\beta\sqrt{3}} \int_{\mathbb{R}} e^{-\gamma t} b_\infty(t) dt. \quad (72)$$

Here the original convolution integrates over $s \geq 0$; after translating, its lower endpoint is $-A_N$, and the same majorant justifies extending by zero and applying dominated convergence.

Strict positivity of the limiting moment. The upper estimates have not yet ruled out a zero limit in (72). To prove strict positivity, we identify the limiting model space with a half-line space, solve its compressed multiplier equation by one-sided factorization, and evaluate the remaining weighted integral. Define

$$(\mathcal{W}l)(t) = \int_{\mathbb{R}} \exp\left(\frac{v - \gamma t}{2} - e^{v - \gamma t}\right) l(v) dv. \quad (73)$$

It maps $L^2(dv)$ onto H , with

$$\widehat{\mathcal{W}l}(\xi) = \gamma^{-1} \Gamma(1/2 + i\xi/\gamma) \widehat{l}(\xi/\gamma), \quad \|\mathcal{W}l\|_H^2 = \frac{2\pi}{\gamma} \|l\|_2^2.$$

These formulas follow by the substitution $r = e^{v - \gamma t}$ and $|\Gamma(1/2 + iu)|^2 = \pi / \cosh(\pi u)$; alternatively the autocorrelation of the integral kernel is a constant times the reproducing kernel of H . Its nonvanishing Fourier multiplier proves surjectivity.

Writing $r = e^{-\gamma t}$ and $y = e^v$ gives

$$\mathcal{W}l(t) = \sqrt{r} \int_0^\infty e^{-ry} y^{-1/2} l(\log y) dy.$$

The substitution is an isometry from $L^2(dv)$ to the data space $L^2(dy)$. Multiplication by $B_\infty = e^{-r}$ translates these data to the right by one, with zero extension. Thus P_∞ corresponds exactly to restriction to $v < 0$. The operator D_a corresponds to the multiplier $D_a(\gamma k)$ on the v -line; denote it by $\mathcal{D}$. At $p = 1/2$,

$$e^{at}(1 - B_\infty) = \mathcal{W}(e^{pv} \mathbf{1}_{v < 0}), \quad e^{at} = \mathcal{W}(e^{pv}),$$

where the second formula is a convergent pointwise integral. Therefore $l = \mathcal{W}^{-1}x$ is supported on $v < 0$ and

$$\mathcal{D}l = e^{pv} \quad (v < 0), \quad e^{at}b_\infty(t) = \mathcal{W}(e^{pv} - \mathcal{D}l)(t). \quad (74)$$

We solve this half-line equation by scalar Wiener–Hopf factorization [23]. Here the factors can be constructed directly in the convolution algebra of a delta mass plus an integrable kernel. At this value of a , the smooth coercive symbol $D_a(\gamma k)$ tends to the same positive constant 1 at both infinities, exponentially with all derivatives. The continuous logarithm tending to zero has a Schwartz inverse Fourier transform q . Its conjugation symmetry makes q real. Split $q = q_+ + q_-$ by support on the positive and negative half-lines and exponentiate each piece in the convolution algebra of a delta mass plus L^1 . The absolutely convergent series give

$$\mathcal{D} = K_+ * K_-,$$

where $K_\pm$, and their convolution inverses, are supported on their respective half-lines. For $s > 0$,

$$d_+(s) := \int e^{-sv} K_+(dv) = \exp\left(\int_0^\infty e^{-sv} q(v) dv\right) > 0.$$

Since $K_- * l$ is supported on $v \leq 0$, applying the positive-support inverse of K_+ to (74) at $v < 0$ gives

$$(K_- * l)(v) = d_+(p)^{-1} e^{pv} \mathbf{1}_{v < 0}.$$

Consequently the residual on the entire line is

$$e^{pv} - \mathcal{D}l = d_+(p)^{-1} K_+ * (e^{pv} \mathbf{1}_{v > 0}). \quad (75)$$

The half-line identities use ordinary L^2 convolution on the left; on the right they are valid with the displayed exponential weights. Total variation of the L^1 kernels justifies all exchanges below. Integrating (73) against $e^{-(\gamma+a)t} dt$, then substituting $e^{v-\gamma t}$, yields

$$\begin{aligned} \int_{\mathbb{R}} e^{-\gamma t} b_\infty(t) dt &= \frac{\Gamma(3/2 + p)}{\gamma} \int_{\mathbb{R}} e^{-(1+p)v} (e^{pv} - \mathcal{D}l)(v) dv \\ &= \frac{\Gamma(3/2 + p)}{\gamma} \frac{d_+(1+p)}{d_+(p)} > 0. \end{aligned} \quad (76)$$

Combining (72), (76), and $e^{\gamma A_N} = 3N$ completes the proof. $\square$

Proposition 9.4 (Suppression of essential polygons). *On the balanced cylinder with period $V = iN\sqrt{3}/2$, let $[\ell]$ range over simple essential polygons modulo integer horizontal translations, and set $W(\ell) = \rho^{|\ell|}$. If $d_x(\ell)$ is their horizontal span, then*

$$\sum_{[\ell]} W(\ell) \asymp N^{-1}, \quad \sum_{[\ell]} W(\ell) d_x(\ell) \leq C. \quad (77)$$

The total weight of essential polygons separating the two marked points in the balanced two-marker geometry is bounded uniformly in N . At each fixed nonnegative fugacity, deleting all these polygons changes its partition function by at most a bounded multiplicative factor.

Proof. In a slab of M columns with vacant outer cuts, differentiate the logarithm of the loop partition at $\chi = 0$. Its first derivative counts single essential polygons. Its second derivative is minus the weight of ordered incompatible pairs, including a polygon paired with itself. After division by M , strict spectral dominance gives convergence of the two derivatives to those of $\log \Lambda(\chi; z_0)$: the spectral projection has nonzero vacuum overlap, equal to one at zero fugacity, and the remaining spectrum decays exponentially, locally uniformly in the analytic parameter. Combinatorially each finite polygon or intersecting pair occupies a finite consecutive interval of columns; its number of placements divided by M increases to one. Monotone convergence gives

$$\begin{aligned} f(0) &= \sum_{[\ell]} W(\ell), \\ -g(0) &= \sum_{[\ell], [\ell']} W(\ell) W(\ell') \sum_{j \in \mathbb{Z}} \mathbf{1}_{\{\ell \cap (\ell' + j) \neq \emptyset\}}. \end{aligned} \tag{78}$$

The physical tiles encode precisely the honeycomb polygons with their vertex weights, and on this embedded trivalent graph intersection is precisely incompatibility.

Let Γ be the triangular translation lattice modulo V . Its integer horizontal subgroup has index N . Fix vertices P, Q of ℓ of the same bipartite type, with $\Delta_x = x(Q) - x(P) > 0$, and put $\Delta = Q - P \in \Gamma$. For a translating vector $a \in \Gamma$, let $I_P(a)$ indicate that P lies strictly in the right-hand component of the complement of $\ell' + a$. Along each horizontal coset it is one on the negative tail and zero on the positive tail. Moreover $I_Q(a) = I_P(a - \Delta)$. The difference is finitely supported, and

$$\sum_{a \in \Gamma} (I_Q(a) - I_P(a)) = N \Delta_x.$$

For an integer horizontal Δ , this is ordinary telescoping in each of the N cosets. In general multiply Δ by N , which is an integer horizontal translation modulo V , and telescope the sum of the N translates of the original difference. If the indicators disagree, the path in ℓ from P to Q meets $\ell' + a$; a point on its boundary also gives intersection. Thus at least $N \Delta_x$ translating vectors give an intersection. Choosing the vertices near the two horizontal extremes, and changing one to a neighbor when necessary to match bipartite types, gives $\Delta_x \geq d_x(\ell) - C$.

Average the inner intersection count in (78) over representatives of $\Gamma/\mathbb{Z}$. Translation by each representative permutes the horizontal polygon classes and preserves their weights. The resulting inequality is

$$f(0) \sum_{[\ell]} W(\ell) (d_x(\ell) - C) \leq -g(0).$$

Now apply Lemma 9.3; in particular $-g(0) \geq 0$ by (78). This proves (77).

Every separating essential polygon meets a fixed arc joining the two markers. That arc has horizontal span $O(N)$. A polygon class can therefore have at most $CN + d_x(\ell) + C$ translates meeting it. The two estimates in (77) bound their total weight. If this bound is C , summing all sets of removed polygons without any compatibility restrictions costs at most $\sum_{k \geq 0} (\chi C)^k / k! = e^{\chi C}$. Positivity provides the opposite inequality by retaining only configurations with no removed polygon. $\square$

10 Open-strip pressure and the planar polygon tail

Essential polygons are now controlled. For large contractible polygons we need a planar estimate, and the next transfer counts all polygons inside a strip. Let $[P]$ range over unoriented planar polygons modulo all triangular-lattice translations. Index the triangular bands by integers and let $H(P)$ be the difference between the highest and lowest occupied band indices. Thus P occupies $H(P) + 1$ bands between its extremes. The observable to be recovered from the derivative at zero loop fugacity is

$$Q_N = \sum_{[P]} \rho^{|P|} (N - H(P))_+.$$

The decisive precision is $Q_N = c_1 N + c_2 + O(N^{-1})$: positivity and comparison at two widths will convert that error into a N^{-2} height tail. Here translation classes are taken modulo the full triangular lattice; this differs from the horizontal classes in Proposition 9.4.

10.1 The variable-fugacity transfer and its boundary terms

We first derive the transfer identities, including the boundary terms created by varying the fugacity.

For the transfer construction, vary λ near $\lambda_0 = \pi/8$, and put $n = -2 \cos(4\lambda)$, $d = \sin(2\lambda)$. Thus n is a local analytic coordinate, since $dn/d\lambda = 8$ at λ_0 . A closed cycle now receives weight n . Define the raw two-slot box $r(y)$ by

$$\begin{aligned} e_y &= d \sin(3\lambda) + \sin y \sin(3\lambda - y), & A_y &= d \sin(3\lambda - y), \\ U_y &= d \sin y, & B_y &= \sin y \sin(3\lambda - y), \\ E_y &= \sin(2\lambda - y) \sin(3\lambda - y), & F_y &= \sin(y - \lambda) \sin y. \end{aligned} \tag{79}$$

These are the dilute $O(n)$ weights listed in [24, equation (1.2)], with the common factor $\sin(2\lambda) \sin(3\lambda)$ restored. They are, respectively, the coefficients of vacancy, a single vertical strand, a cup or cap alone, a single diagonal strand, two vertical strands, and a cap together with a cup. Put

$$L_y = \sin(2\lambda + y) \sin(3\lambda + y), \quad R_y = r(y)/L_y, \quad P_y = r(y)/e_y.$$

At λ_0 , $e_y = L_y$, so both normalized boxes coincide with the original box. Use the same splitter diagrams, now with cup coefficients $q_2 = (2 \cos \lambda)^{-1}$, $q_3 = 1$, and with occupied-leg coefficient q_2 for S_2 .

Lemma 10.1 (Local identities away from zero fugacity). *With $\hat{P} = P_{x+\lambda}$ for $j = 2$, and the identity for $j = 3$,*

$$\begin{aligned} R_y R_{-y} &= I, & P_{j\lambda} &= S_j S_j^t, \\ (P_x)_2 (P_{x+j\lambda})_1 (I \otimes S_j) &= d_j(x) (S_j \otimes I) \hat{P}, & d_j(x) &= 1 + n q_j \frac{A_{x+j\lambda} U_x}{e_{x+j\lambda} e_x}. \end{aligned} \tag{80}$$

The raw boxes also satisfy the three-box Yang-Baxter identity.

Proof. For unitarity, the one-occupancy sector is the two-by-two matrix with diagonal A and off-diagonal B . Direct substitution in (79) gives its inverse after normalization by $L_y L_{-y}$. In the even-occupancy sectors the required identities are

$$\begin{aligned} e_y e_{-y} + n U_y U_{-y} &= E_y E_{-y} = L_y L_{-y}, \\ e_{-y} U_y + (E_y + n F_y) U_{-y} &= 0, \\ E_y F_{-y} + F_y E_{-y} + n F_y F_{-y} + U_y U_{-y} &= 0. \end{aligned}$$

They prove $R_y R_{-y} = I$. The factorization at $j\lambda$ follows by substituting $y = 2\lambda, 3\lambda$.

We record the coefficient check for splitting to retain the terms arising from the nonzero cycle weight. In the following tables, subscript 1 means argument $x + j\lambda$, subscript 2 means argument x , and every letter has been divided by the corresponding e . Unsubscripted letters belong to $\hat{P}$; for deletion take $B = 1$ and the other nonempty two-slot coefficients zero. Write $q = q_j$ and $k = q_2$ for $j = 2$, $k = 0$ for $j = 3$. Outputs are numbered 1, 2, 3, with 1, 2 the split slots; a parenthesized pair denotes an additional cup. In the last block a parenthesized output pairs the two input strands, whereas an unparenthesized pair denotes through strands. The coefficient equalities are

input	output	two boxes	fused value
00	$\emptyset$	d_j	d_j
	12	$U_1 A_2 + q B_1 B_2$	$d_j q$
	13	$U_1 B_2 + q B_1 A_2$	$d_j k U$
	23	$U_2 + q A_1 (E_2 + n F_2)$	$d_j k U$
10	1	$A_1 + q (F_1 + n E_1) U_2$	$d_j k A$
	2	$B_1 A_2 + q U_1 B_2$	$d_j k A$
	3	$B_1 B_2 + q U_1 A_2$	$d_j B$
	3, (12)	$q F_1 E_2$	$d_j q B$
	1, (23)	$A_1 U_2 + q (E_1 E_2 + F_1 F_2 + n E_1 F_2)$	0

and, for $j = 2$,

01	1	$k(B_1 + U_1 U_2)$	$d_2 k B$
	2	$k(A_1 A_2 + B_2)$	$d_2 k B$
	3	$k(A_1 B_2 + A_2)$	$d_2 A$
	1, (23)	$k(B_1 U_2 + U_1 F_2)$	0
	3, (12)	$k U_1 E_2$	$d_2 q A$
11	$\emptyset$	$k(U_1 + B_1 U_2)$	$d_2 U$
	12	$k(E_1 A_2 + A_1 B_2)$	0
	(12)	$k F_1 A_2$	$d_2 q U$
	13	$k(E_1 B_2 + A_1 A_2)$	$d_2 k E$
	(13)	$k F_1 B_2$	$d_2 k F$
	23	$k B_1 E_2$	$d_2 k E$
	(23)	$k(U_1 U_2 + B_1 F_2)$	$d_2 k F$

The terms proportional to n are exactly the closed cups in the contraction. These tables can be verified without diagrammatic simplifications: put $[s] = T^s - T^{-s}$, $T = e^{i\lambda}$, and $p = x/\lambda$. At argument $x + l\lambda$, the normalized tuple (A, B, U, E, F) is

$$\frac{([2][3 - p - l], [p + l][3 - p - l], [2][p + l], [3 - p - l][2 - p - l], [p + l - 1][p + l])}{[2][3] + [p + l][3 - p - l]}.$$

Substitute $q_2 = [1]/[2]$, $n = -(T^4 + T^{-4})$, and clear denominators. Multiplication gives the displayed right columns, proving all occupancy sectors of (80).

For the three-box identity, specialize its second argument to $y = j\lambda$, $j = 2, 3$. Splitting, transposition, and reflection reduce both sides to the same fused diagram with scalar $d_j(x)$. Unitarity gives the further specializations $y = j\lambda - x$; $y = 0$ follows from $R_0 = I$. After clearing scalar denominators, each occupancy component has exponents between -4 and 4 in e^{iy} , all of fixed parity. The five values modulo π just listed therefore determine the polynomial at generic x , and rational continuation proves the identity everywhere it is defined. $\square$

Let x_i be host arguments near $\lambda/2$, and use the notation

$$\bar{z} = 3\lambda - z, \quad \mathbf{m}(z) = \frac{\sin(3\lambda/2 - z)}{\sin(3\lambda/2 + z)}, \quad K(a) = \text{vacancy} + a \text{ cup}.$$

The bar here denotes this affine reflection, not complex conjugation. Define an open transfer $t(z)$ on ordinary disk states by passing auxiliaries $[\bar{z}, z]$, the rightmost crossing each host first, with normalized box $P_{\text{auxiliary}-\text{host}}$. Insert $K(\mathbf{m}(\bar{z}))$ at the bottom and contract with $K(\mathbf{m}(z))^t$ at the top. Figure 2 records the auxiliary order and the boundary tensors. The caps are part of the transfer: they contribute the two scalar factors when auxiliaries are fused.

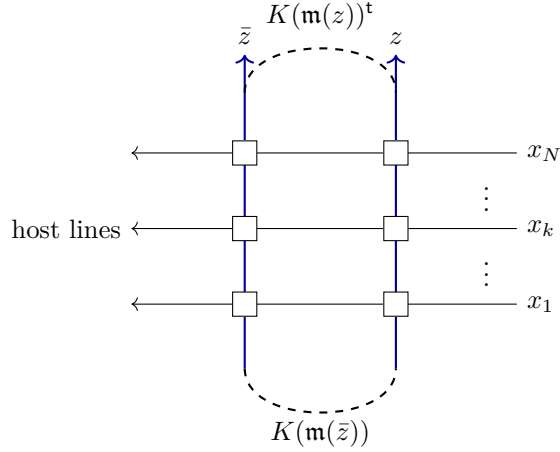


Figure 2: The open transfer, drawn as an operator diagram rather than a physical embedding. The ordered auxiliaries are $[\bar{z}, z]$, so the rightmost line crosses each host first; each box has argument auxiliary minus host. A dashed boundary arc denotes the tensor $K(a) = \text{vacancy} + a \text{ cup}$, not a compulsory occupied arc. At $z = \bar{z} = 3\lambda/2$ both cup coefficients vanish, leaving the two physical columns with vacant auxiliary boundaries.

Lemma 10.2 (Open transfer and its pinches). *The matrices $t(z)$ commute and are invariant under $z \leftrightarrow \bar{z}$. At $\lambda = \lambda_0$, $x_i = \lambda/2$, and $z = 3\lambda/2$, this is the transfer of two physical columns with vacant auxiliary boundaries. Its simple eigenvalue continuing 1 will be denoted by $\Lambda_o(n; z)$. It is rational and π -periodic in z , equals one at $n = 0$, and tends to $1 + n$ at both imaginary infinities. For $u = x_i$, $v = u + j\lambda$, its pinch rule is*

$$\Lambda_o(n; v)\Lambda_o(n; u) = c_b c_t \prod_k d_j(u - x_k) d_j(\bar{v} - x_k) \begin{cases} \Lambda_o(n; u + \lambda), & j = 2, \\ 1, & j = 3, \end{cases} \quad (81)$$

where $w_0 = u + v - 3\lambda$ and

$$c_b = \frac{e_{w_0} + nq_j \mathbf{m}(\bar{u}) A_{w_0}}{L_{w_0}}, \quad c_t = \frac{e_{-w_0} + nq_j \mathbf{m}(v) A_{-w_0}}{L_{-w_0}}. \quad (82)$$

Proof. Exchanging adjacent auxiliaries v, u uses the crossing R_{v-u} at the bottom, moved to the top by the three-box identity. For exchanging the two lines of one transfer, the boundary identity is

$$R_{\bar{z}-z} K(\mathbf{m}(\bar{z})) \text{ is proportional to } K(\mathbf{m}(z)).$$

The proportionality follows from, at $y = \bar{z} - z$,

$$U_y + \mathbf{m}(\bar{z})(E_y + nF_y) = \mathbf{m}(z)(e_y + n\mathbf{m}(\bar{z})U_y).$$

The transposed inverse crossing contributes the reciprocal scalar, proving reflection symmetry.

For two pairs, the exchanges taking $[\bar{v}, v, \bar{u}, u]$ to $[\bar{u}, u, \bar{v}, v]$ have operator product

$$(R_{-w_0})_2(R_h)_3(R_{-h})_1(R_{w_0})_2, \quad h = v - u, \quad w_0 = v + u - 3\lambda,$$

with the rightmost factor acting first. They take the product of bottom caps exactly to its reversal. By unitarity, this reduces to left-right symmetry of $(r(h))_3(r(w_0))_2[K(a) \otimes K(b)]$, where $a = \mathbf{m}(\bar{v})$, $b = \mathbf{m}(\bar{u})$. Only the paired output types 12, 34 and 13, 24 need comparison; their equalities are

$$\begin{aligned} aA_{w_0}e_h + ab(nE_{w_0} + F_{w_0})U_h &= e_{w_0}U_h + bA_{w_0}(E_h + nF_h), \\ (a - b)B_{w_0}A_h &= (1 - ab)U_{w_0}B_h. \end{aligned} \tag{83}$$

Substitution from (79) and the sine formula for $\mathbf{m}$ proves them. Replacing v, u by $\bar{u}, \bar{v}$ and transposing gives the top exchange, hence commutation.

At the physical center both cap coefficients vanish. Spectral simplicity of the open-strip transfer gives an analytic simple eigenspace nearby, invariant under all $t(z)$ by commutation. The empty-coordinate rule at $n = 0$ makes its eigenvalue identically one. At imaginary infinity, $A_y/e_y, U_y/e_y \rightarrow 0$, $B_y/e_y \rightarrow 1$, and $\mathbf{m}(z)\mathbf{m}(\bar{z}) \rightarrow 1$. Thus the empty-output computation is $1 + n$. Occupancy signs prove periodicity, and computing on the fixed common eigenvector proves rationality.

For the pinch, exchange the middle lines using R_{w_0} on the bottom caps and R_{-w_0} on the top, reaching order $[\bar{v}, \bar{u}, v, u]$. The descending pair v, u crosses first. At host $x_i = u$, the factorization of $P_{j\lambda}$, together with $P_0 = I$, inserts a transpose splitter at input. Move it backwards through the other hosts by the transposed version of (80). Its contraction against the exchanged bottom caps splits the first pair by S_j , with scalar c_b . In raw coefficients the necessary identities are

$$\begin{aligned} aA_{w_0} + q_j ab(nE_{w_0} + F_{w_0}) &= q_j(e_{w_0} + nq_j bA_{w_0}), \\ aB_{w_0} + abU_{w_0} &= U_{w_0} + bB_{w_0} \\ &= \mathbf{m}(\bar{u} - \lambda)(e_{w_0} + nq_j bA_{w_0}) \quad (j = 2). \end{aligned} \tag{84}$$

Here $a = \sin(v - 3\lambda/2)/\sin(9\lambda/2 - v)$, and likewise for b ; multiplying out proves (84), just as for (83). Move the first pair's splitter through the hosts by the reflected splitting identity. The reflected top calculation, with $\bar{u}, \bar{v}$, gives c_t . The remaining descending pair has both an input and output transpose splitter; transposed intertwining reduces it to the other fused transfer. Cancelling the surjective input splitter is valid as in the annular proof. Every crossed host contributes the two factors displayed in (81). This proves that identity for generic other sites, hence by continuation. $\square$

10.2 The forced interpolation problem

We have obtained the open transfer and its exact boundary scalars. We next differentiate its pinches to identify the forced interpolation problem whose value at the physical center is the planar polygon pressure. Define $F_o(z) = \partial_n \Lambda_o(n; z)|_{n=0}$. In all formulas that follow, raw coefficients and auxiliary arguments are evaluated at λ_0 . Movement of evaluation arguments while differentiating costs nothing, because $\Lambda_o(0; z)$ is constant. Introduce

$$\begin{aligned} \phi_j(x) &= q_j \frac{U_x A_{x+j\lambda}}{e_x e_{x+j\lambda}}, \\ b_j(u) &= -\frac{U_{w_0} U_{-w_0}}{e_{w_0} e_{-w_0}} + q_j \left[\mathbf{m}(\bar{u}) \frac{A_{w_0}}{e_{w_0}} + \mathbf{m}(u + j\lambda) \frac{A_{-w_0}}{e_{-w_0}} \right], \quad w_0 = 2u + (j - 3)\lambda. \end{aligned} \tag{85}$$

The first term of b_j comes from differentiating scalar unitarity on the vacuum sector. Formula (81) gives the remaining terms.

There is one additional evaluation. At $z = 0$, the top cap has $\mathbf{m}(0) = 1$ and folds the two columns into one another. A quarter-turn of a raw box changes its argument to $3\lambda - y$, and $e_y = e_{3\lambda-y}$. Each folded pair therefore cancels by unitarity at $x_k, -x_k$, contributing $1 + nU_{x_k}U_{-x_k}/(e_{x_k}e_{-x_k})$; the remaining bottom factor is $1 + n\mathbf{m}(3\lambda)$. Hence

$$F_o(0) = \mathbf{m}(3\lambda) + \sum_k \frac{U_{x_k}U_{-x_k}}{e_{x_k}e_{-x_k}}. \quad (86)$$

At $8\lambda = \pi$, direct sine identities give

$$\begin{aligned} \phi_3(x) &= \phi_2(x) + \phi_2(x + \lambda), & b_3(u) &= b_2(u) + b_2(u + \lambda), \\ b_2(\lambda) &= \mathbf{m}(3\lambda), & \phi_2(\lambda - x) + \phi_2(-x) &= \frac{U_x U_{-x}}{e_x e_{-x}}. \end{aligned} \quad (87)$$

Combining the two differentiated pinches using (87) shows that the even residual

$$F_o(z + \lambda) + F_o(z - \lambda) - F_o(z) - \sum_k [\phi_2(z - \lambda - x_k) + \phi_2(2\lambda - z - x_k)] - b_2(z - \lambda)$$

vanishes at $z = x_i + \lambda, x_i + 2\lambda$ and their reflections about $3\lambda/2$. Formula (86) gives zeros at $z = \lambda, 2\lambda$ as well.

Now set every $x_i = \lambda/2$ and use $s = 2i(z - 3\lambda/2)$. To distinguish the physical center from the folded evaluation (86), write

$$\mathcal{F}_o(s) = F_o(3\lambda/2 + s/(2i)).$$

Thus $\mathcal{F}_o(0)$ is the pressure derivative at the physical center $z = 3\lambda/2$, while $F_o(0)$ in (86) is the folded value at $z = 0$. Set

$$Y(s) = \mathcal{L}\mathcal{F}_o(s) - N\phi(s) - b(s), \quad \phi(s) = \phi_2(y) + \phi_2(-y), \quad b(s) = b_2(\lambda/2 + y), \quad y = s/(2i). \quad (88)$$

Confluence yields zeros of orders $2N, N, N$ at $0, i\beta, -i\beta$, and simple zeros at $\pm i\beta/2$. The function $\mathcal{F}_o$ is even, rational in e^s , tends to one at both real infinities, and has poles of orders at most $N + 1$ at $\pm 3i\beta$, and $2N$ at $4i\beta$, modulo $2\pi i$. These bounds count the two box denominators per host and the two boundary denominators. Differentiation does not increase the cleared denominator at zero fugacity: if $A(n, s)/D(n, s)$ equals one at $n = 0$, its derivative is $(A_n - D_n)/D$ there.

Both forcing terms are rational in e^s and are $O(e^{-|\operatorname{Re} s|})$ at real infinity. The function ϕ is analytic in a neighborhood of $|\operatorname{Im} s| \leq h$, and b has at most simple poles at the two boundary points $\pm ih$. To check the possibly closer poles of b , use the raw coefficients at $y = \lambda$:

$$\frac{U_{-2\lambda}}{A_{-2\lambda}} = \frac{\sin(-2\lambda)}{\sin(5\lambda)} = \mathbf{m}(7\lambda/2), \quad \frac{U_{2\lambda}}{e_{2\lambda}} = q_2.$$

The two singular terms at $y = \lambda$ cancel; reflection gives the cancellation at $-\lambda$. This cancellation is independent of the hosts. Thus all preceding evaluations and coalescences, including the shifted $\mathcal{F}_o$ terms, occur at regular points.

Lemma 10.3 (Open-strip interpolation estimate). *For some fixed $a \in (\gamma/2, 1)$, and uniformly in N ,*

$$|Y(s)| \leq C e^{-cNe^{-\gamma s}} (1 + N^{a/\gamma} e^{-as}), \quad s \geq 0. \quad (89)$$

Consequently

$$\mathcal{F}_o(0) = (\rho_* * (N\phi + b))(0) + O(N^{-1}). \quad (90)$$

Proof. The poles of $\mathcal{F}_o - 1$ determine the space of unknowns, whereas zeros of Y determine the interpolation equations. These spaces have the same dimension but different two-dimensional additions to their common base. We first show that the resulting interpolation remains uniformly invertible. We then remove the bulk forcing $N\phi$, treat the two boundary poles of b , and estimate what remains.

The unknown and equation spaces. Use the Hardy space H on $|\operatorname{Im} s| < 3\beta/2$, with norm (55), and the multiplier D_a in (58). Let P project onto the base jets of orders $2N, N, N$ at $0, i\beta, -i\beta$. Let P' increase the two outer orders by one, and let P'' instead add the simple knots $i\beta/2, -i\beta/2$. Their complementary inner functions are B, B', B'' , respectively. The extra orders of P' come from the two boundary denominators of the transfer; the extra knots of P'' come from the folded evaluation (86). Partial fractions, as in (62), give

$$e^{as}\mathcal{L}(\mathcal{F}_o - 1) = D_ax, \quad x \in P'H. \quad (91)$$

We first prove that

$$P''D_a : P'H \longrightarrow P''H \quad \text{has inverse norm bounded independently of } N \quad (92)$$

for all sufficiently large N and all a in a fixed neighborhood of $\gamma/2$.

At $a = \gamma/2$, the compression to PH is uniformly invertible by (59), and $D_a(k) \rightarrow d_0 = 1$ at both frequency infinities. The orthogonal decompositions of the two enlarged spaces are

$$P'H = PH \oplus BE', \quad P''H = PH \oplus BE'',$$

where E' and E'' are the fixed two-dimensional jet spaces at $\pm i\beta$ and $\pm i\beta/2$, respectively. For each fixed vector e in either space,

$$\|(D_a - d_0I)Be\|_H + \|(D_a^* - d_0I)Be\|_H \longrightarrow 0.$$

Indeed $B(s) \rightarrow 0$ boundedly at every real point, so $Be \rightarrow 0$ in real-line L^2 by domination. Its weighted Fourier mass on a fixed compact interval consequently tends to zero. Outside that interval $D_a - d_0$ is uniformly small, while $\|Be\|_H = \|e\|_H$. The same argument applies to the adjoint; finite dimensionality makes convergence uniform on unit balls. The off-diagonal blocks of $P''D_a|_{P'H}$ therefore tend to zero, and its extra block tends to d_0 times the overlap projection from E' to E'' . The cross-Gram matrix of the kernel-vector bases is, up to the fixed positive kernel constant,

$$\begin{pmatrix} \sec(\pi/4) & \sec(\pi/12) \\ \sec(\pi/12) & \sec(\pi/4) \end{pmatrix}, \quad \det = 2 - \sec^2(\pi/12) > 0.$$

Thus the limiting extra block is invertible. The block inverse proves (92) at $a = \gamma/2$. Operator-norm continuity of D_a , uniform for a in compact subsets of $(0, 1)$, extends the bound to a fixed neighborhood. Choose a slightly larger than $\gamma/2$ within it.

Removing the bulk forcing. Set $\mu = \gamma - a$ and

$$X = D_a^{-1}(e^{as}\phi(s)).$$

The unrestricted vector NX solves the bulk part of the equation exactly. Its projection $NP'X$ is an admissible unknown; the following estimate ensures that projecting it introduces an error of order $N^{a/\gamma}$, rather than order N . We claim

$$\|(I - P')X\|_H \leq CN^{-\mu/\gamma}. \quad (93)$$

First, X extends to a slightly wider spatial strip and obeys

$$|X(s)| \leq C e^{-\mu |\operatorname{Re} s|} \quad \text{on that wider strip.} \quad (94)$$

Here is the required Fourier-contour justification. Since ϕ is analytic on a wider strip, the transform of $e^{as}\phi$ decays exponentially in $|\operatorname{Re} k|$ at a rate greater than h . Choose $\epsilon > 0$ such that

$$\mu + \epsilon < \min(\gamma/2, a).$$

Its continuation between the horizontal frequency lines at heights $\pm(\mu + \epsilon)$ has only a possible simple pole at $i(1 - a)$, originating in its leading positive-real tail. Explicitly, subtract a suitable multiple of $e^{as}/(1 + e^s)$. This removes that tail; the remaining negative-real tail is $O(e^{as})$, sufficient because $a > \mu + \epsilon$. Shifting the spatial contour inside the wider analytic strip proves the same large-frequency decay on all these continued frequency lines. Multiplication by D_a^{-1} cancels the pole at $i(1 - a)$ through its sine numerator. Its shift denominator introduces at most a simple pole at $i\mu$; the poles arising from $1/\cosh(3\beta k)$ begin at $\pm i\gamma/2$, outside the chosen region. On horizontal frequency tails the bounds remain unchanged up to constants. Fourier inversion on the displaced contours, including the residue at $i\mu$, proves (94).

To estimate the complementary projection, pair X with $B'G$, $\|G\|_H = 1$, using the two boundary integrals. On each boundary $\overline{B'} = 1/B'$. Move the upper integral a small fixed distance upward and the lower integral downward, extending the conjugate of G by reflection across the original boundary. Choose the distance so that the new lines remain in the wider strip of (94) and before the nearest poles of the finite Blaschke product. The reflected points are interior to H , so the test functions have uniformly bounded L^2 norms there. The contour move is justified first for finite linear combinations of reproducing kernels, which are dense and have exponential tails, and then by norm convergence. On the displaced lines, the coordinate $e^{\gamma s}$ has negative real part. The explicit factors give

$$|1/B'(s)| \leq C \exp(-cN e^{-\gamma |\operatorname{Re} s|}).$$

Cauchy–Schwarz and (94) now bound the pairing by the square root of

$$C \int_{\mathbb{R}} e^{-2\mu|u|} e^{-cN e^{-\gamma|u|}} du \leq C N^{-2\mu/\gamma}.$$

This proves (93).

Removing the boundary forcing. It remains to interpolate $e^{as}(b - 1)$, whose boundary poles prevent it from belonging directly to H . Choose b_p with simple poles at $\pm ih$, having the residues of b divided by the values of B'' there. These values have modulus one. One explicit choice is a sum of the corresponding polar parts multiplied by Gaussians equal to one at their own pole; this gives uniform decay on every fixed horizontal line and a uniform bound on the real line. Put

$$U(s) = e^{as}[b(s) - 1 + B''(s) - B''(s)b_p(s)]. \quad (95)$$

The poles cancel, and U has the same prescribed jets as $e^{as}(b - 1)$. Furthermore

$$\|U\|_H \leq C N^{a/\gamma}, \quad |b(s) - 1 - e^{-as}U(s)| \leq C |B''(s)| \quad (s \in \mathbb{R}). \quad (96)$$

For the first assertion, $e^{as}(B'' - 1)$ has norm $O(N^{a/\gamma})$ by (63). Near either boundary pole, the canceled polar parts are bounded by $C \min(N, |u|^{-1})$, since the tangential derivative of B'' is $O(N)$ on the boundary. Their squared integral is $O(N)$. Away from the poles, the rational tails and the Gaussian terms have bounded weighted norm. Since $a/\gamma > 1/2$, the pole contribution is absorbed. Removability of the poles and the boundary bounds imply $U \in H$. The second assertion follows directly from (95) and the real-line bound on b_p .

The residual and the center value. The exact jet equation for (88) is

$$P''(D_a x - N D_a X - U) = 0.$$

Compare x with $N P' X$, using (92). Since $N \|(I - P')X\|_H \leq C N^{a/\gamma}$, we obtain

$$\|D_a x - N D_a X - U\|_H \leq C N^{a/\gamma}.$$

This difference is in $B''H$, so divide by B'' and use bounded real-line evaluation. Equations (91) and (96) give

$$|Y(s)| \leq C |B''(s)| (1 + N^{a/\gamma} e^{-as}).$$

The product estimate (69) proves (89). Evenness gives the corresponding estimate on the negative half-line.

Finally, the real-line functions are regular, and $\mathcal{F}_o$ tends exponentially to one. Thus the inverse shift formula gives $\mathcal{F}_o = \rho_* * (N\phi + b + Y)$. The error at zero is at most

$$C \int_0^\infty e^{-\gamma s} e^{-c N e^{-\gamma s}} (1 + N^{a/\gamma} e^{-as}) ds = \frac{C}{\gamma N} \int_0^N e^{-cu} (1 + u^{a/\gamma}) du = O(N^{-1}),$$

proving (90). □

10.3 Polygon density and the diameter tail

We now identify the analytic derivative with a positive polygon sum. There are two normalization factors to keep track of: one transfer step advances two columns, and a polygon shape has several vertical placements in the strip.

Proposition 10.4 (Linear open-strip pressure). *For the polygon sum Q_N defined at the start of this section, there are constants c_1, c_2 such that*

$$Q_N := \sum_{[P]} \rho^{|P|} (N - H(P))_+ = c_1 N + c_2 + O(N^{-1}). \quad (97)$$

Proof. At the moving center $z = 3\lambda/2$, and with the hosts also taken at $\lambda/2$, both auxiliary caps are vacancies for all nearby λ . The local vacuum weights for P_y are identically one. Differentiating the finite-strip partition at $n = 0$ therefore counts exactly one polygon, with its original critical honeycomb weight: derivatives of local occupied weights have an additional factor n and disappear at zero. Changing the arguments with λ contributes nothing to the eigenvalue derivative, since the base eigenvalue is constant.

The transfer t advances two horizontal columns. Spectral dominance, as in (78), identifies $\mathcal{F}_o(0)$ with twice the polygon density per single column. Normalize a representative of a full translation class so that its lowest occupied band is zero. Its lowest band can then be placed at precisely $0, 1, \dots, N - H(P) - 1$ in the strip. Horizontal half-unit offsets caused by a one-band translation do not affect density: for each of these $(N - H(P))_+$ placements, integer horizontal translates have density one per column. Monotone counting of finite longitudinal placements thus gives $Q_N = \mathcal{F}_o(0)/2$. All passages from finite length use nonnegative polygon sums and the analytic finite-dimensional transfer; they require no interchange of an uncontrolled infinite derivative. Equation (90) proves (97), with

$$c_1 = \frac{1}{2}(\rho_* * \phi)(0), \quad c_2 = \frac{1}{2}(\rho_* * b)(0).$$

□

Theorem 10.5 (Planar polygon tail). *Uniformly for $R \geq 0$,*

$$\sum_{[P]: \text{diam}(P) > R} \rho^{|P|} \leq \frac{C}{(1+R)^2}. \quad (98)$$

The sum is over unrooted, unoriented polygons modulo triangular-lattice translations.

Proof. Divide (97) by N . Since $(1 - H(P)/N)_+$ increases to one, monotone convergence gives $\sum_{[P]} \rho^{|P|} = c_1 < \infty$. Define

$$D_N = c_1 N - Q_N = \sum_{[P]} \rho^{|P|} \min(N, H(P)).$$

This sequence is increasing, and $D_N = -c_2 + O(N^{-1})$. Choose an admissible integer $M \asymp N/2$. Every polygon with $H(P) > N$ contributes $N - M$ to $\min(N, H(P)) - \min(M, H(P))$, so

$$(N - M) \sum_{[P]: H(P) > N} \rho^{|P|} \leq D_N - D_M = O(N^{-1}).$$

Thus the height tail is $O(N^{-2})$. This argument also works if the pressure formula is used only along even widths, by taking M, N even. Rotate the lattice to obtain the same bound in the three band directions. The Euclidean diameter is bounded by a constant times the largest of these three band spans, plus a fixed lattice constant. The union bound proves (98) for large R ; total summability covers bounded R . $\square$

11 Planar nests and strip-length consequences

The two pressure calculations have supplied the needed geometric inputs: essential polygons have bounded total weight in the two-marker geometry, and planar polygon shapes have a quadratic diameter tail. We first turn the latter into a bounded removal cost for large contractible polygons. The cylinder exponent then gives the planar and strip nesting exponents. The final step converts these nest sums into length-weighted path sums.

11.1 Removing large polygons and comparing two nests

Lemma 11.1 (Macroscopic removal). *Fix $c_0 > 0$. In an N -band strip, the total weight of polygons enclosing a specified face center and having diameter greater than $c_0 N$ is bounded by a constant depending only on c_0 . The same is true of contractible polygons enclosing a specified marker on the balanced cylinder, with diameter measured on a planar lift. At each fixed nonnegative fugacity, deleting these polygons changes a nesting partition function by at most a bounded multiplicative factor.*

Proof. For a planar shape of diameter comparable to R , the number of translations fitting in the strip and enclosing the specified point is at most $CN(1+R)$: there are $O(N)$ possible vertical placements, and enclosure forces its horizontal range to contain the point. For dyadic $R \geq c_0 N$, Theorem 10.5 bounds the weight of all such shapes by CR^{-2} . Thus their total weight is bounded by

$$C \sum_{j \geq 0} N(1 + 2^j c_0 N)(1 + 2^j c_0 N)^{-2} \leq C_{c_0}.$$

A simple contractible cylinder polygon lifts to translates of a simple planar polygon with disjoint interior disks. The boundary lifts are disjoint; if two of their bounded disks overlapped, planar separation would force nesting, which is impossible for distinct translates of equal-area bounded disks. Thus its disk projects injectively to the cylinder. For a fixed planar translation class, the number of placements modulo the period containing the marked point is again at most $CN(1 + R)$: there are N vertical translation classes modulo period and its horizontal range must straddle the marker. Counting all these placements, including those not valid under projection, is an upper bound. The same dyadic estimate applies.

For a fixed remaining nest, sum the removed polygons while ignoring disjointness both among themselves and with the nest. If their total single-polygon weight is at most C_{c_0} , this contributes at most $e^{\chi C_{c_0}}$. Positivity gives the reverse inequality by taking the empty removed set. $\square$

Let $P_\chi(r)$ be the planar partition over finite disjoint nests around one face center, with every polygon of diameter at most r , each weighted by $\chi \rho^{|P|}$; include the empty nest. Recall that $Z_N(\chi)$ is the balanced-cylinder partition in which every polygon separates the two prescribed markers.

Theorem 11.2 (Normalization by planar nests). *For every fixed $\chi \geq 0$, and every sufficiently small fixed $c_0 > 0$,*

$$P_\chi(c_0 N)^2 \leq Z_N(\chi) \leq C_{\chi, c_0} P_\chi(c_0 N)^2 \quad (99)$$

for all sufficiently large even N . Consequently, for every fixed $\chi > 0$,

$$P_\chi(r) = r^{\tau(\chi)/2 + o(1)} \quad (r \rightarrow \infty), \quad (100)$$

where τ is the exponent in Theorem 1.1. In particular,

$$P_2(r) = r^{1/12 + o(1)} \quad (r \rightarrow \infty). \quad (101)$$

Here $P_\chi(r)$ includes the empty nest, and the diameter cutoff applies separately to every polygon in a nest.

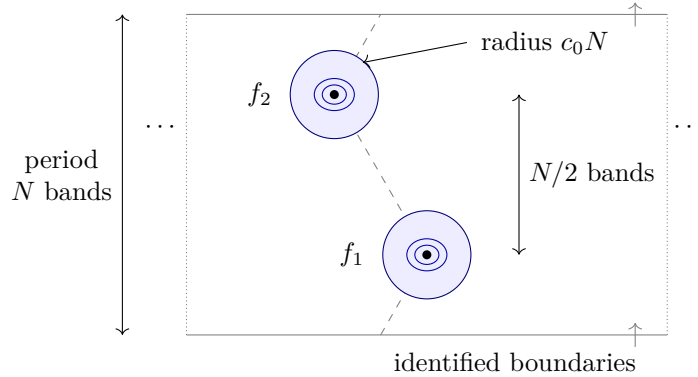


Figure 3: A schematic fundamental band of the cylinder, with the lattice suppressed. The dashed column cut follows the two balanced row blocks; its marked seams have displacement $-N/4 + iN\sqrt{3}/4$ and are separated vertically by half the period. For small fixed c_0 , their radius- c_0N neighborhoods are disjoint and embed isometrically in the plane. After essential and larger polygons are deleted at bounded cost, each remaining polygon belongs to a nest in one of these two neighborhoods. The independent choices give the square of the planar partition in (99).

Proof. The two cylinder markers have disjoint neighborhoods of radius c_0N , each isometric to its planar neighborhood, if c_0 is small enough: along the standard balanced column cut their displacement is $-N/4 + iN\sqrt{3}/4$, and their vertical separation is half the balanced period. Figure 3 shows the two embedded neighborhoods. Delete all separating essential polygons, using Proposition 9.4, and all contractible polygons of lifted diameter greater than c_0N , using Lemma 11.1 at both markers. The product of the deletion costs is uniformly bounded.

Every remaining polygon is contractible and encloses exactly one marker in its disk. A planar simple curve of diameter at most r enclosing a point lies in the radius- r neighborhood of that point: the point is in the curve's convex hull, so its distance to each curve point is at most the curve's diameter. Hence each remaining polygon lies in its marker's embedded neighborhood. Conversely, a planar nest counted by $P_\chi(c_0N)$ projects into that neighborhood and separates the cylinder markers. The neighborhoods are disjoint, so the two nests can be chosen independently. Retaining just these configurations gives the lower bound, and the preceding deletion gives the upper bound in (99).

For each fixed $\chi > 0$, Theorem 1.1 gives $Z_N(\chi) = N^{\tau(\chi)+o(1)}$ along even N . Taking logarithms in the comparison yields $P_\chi(c_0N) = N^{\tau(\chi)/2+o(1)}$. Monotonicity in the radius and bracketing an arbitrary radius between consecutive admissible values prove (100). At $\chi = 2$, $\tau(2) = 1/6$, giving (101). $\square$

11.2 Nests around the middle of a strip

Let S_k be the partition sum of nests around a face center f on the middle line of the strip $\mathcal{S}_{2k}$ of $2k$ triangular bands. Thus S_k sums the finite vertex-disjoint nests enclosing f , with weight $\prod_P 2\rho^{|P|}$, including the empty nest.

Corollary 11.3 (Middle-face nesting). *As $k \rightarrow \infty$,*

$$S_k = k^{1/12+o(1)}. \quad (102)$$

Proof. Choose a fixed $c_0 > 0$ so small that every nest enclosing f , with each polygon of diameter at most c_0k , fits inside the strip. By Lemma 11.1, deleting all larger polygons has a bounded cost. Therefore

$$P_2(c_0k) \leq S_k \leq CP_2(c_0k).$$

Apply Theorem 11.2. $\square$

11.3 A positive comparison between length and nests

We use the boundary winding identity to count visits to one internal edge by summing over all ordered boundary sources and endpoints. This local comparison will convert the nesting exponent into a length exponent. Recall $t = 3/8$, $c = \cos(3\pi/8)$, and $d = 1/\sqrt{2}$.

For a finite simply connected union D of triangular tiles with simple polygonal boundary and a boundary port a , the boundary identity of [19, Lemma 2.1] is

$$\sum_{b \in \partial D} \sum_{\gamma: a \rightarrow b} \rho^{n(\gamma)} e^{itW(\gamma)} = 1, \quad (103)$$

where W measures signed turning from the inward initial direction to the outward final direction. Its local rule sends the current weight into the two new continuations with multipliers $\rho e^{i\lambda}$ and $\rho e^{-i\lambda}$; their sum is one. At a first self-collision, reversing the closing cycle gives the canceling winding increments $\pm 4\pi/3$. In a convex domain every boundary-to-boundary path satisfies $|W| \leq \pi$, and

therefore $c \leq \cos(tW) \leq 1$. These conventions also specify the modified interior-source calculation below.

For a finite convex union D of triangular tiles and an internal honeycomb mid-edge p , define

$$L_p(D) = \sum_{a,b \in \partial D} \sum_{\gamma: a \rightarrow b} \rho^{n(\gamma)} \cos(tW(\gamma)) \mathbf{1}_{\{p \in \gamma\}}.$$

The boundary ports are ordered, and $W(\gamma)$ is the total signed turning from the initial inward direction to the final outward direction. Convexity gives $c \leq \cos(tW(\gamma)) \leq 1$. For a point r not on a polygon, let $\mathcal{Z}_r(D)$ be the sum over finite sets of pairwise vertex-disjoint simple polygons, all strictly enclosing r , with weight $\prod_{\ell} 2\rho^{|\ell|}$; the empty set has weight one. The same definition applies when r is a mid-edge, excluding polygons through that mid-edge.

Lemma 11.4 (Length and polygon nests). *Let f_1, f_2 be the two honeycomb face centers adjacent to p . Then*

$$d(\mathcal{Z}_{f_1}(D) + \mathcal{Z}_{f_2}(D)) \leq L_p(D) \leq \mathcal{Z}_{f_1}(D) + \mathcal{Z}_{f_2}(D).$$

The inequalities also hold, with finite quantities, in every homogeneous infinite strip of fixed positive height.

Proof. Cut the edge at p into two stopping outlets. For a boundary source a , the winding splitting identity still has total one: every collision stem approaches its closing cycle from outside, so the two orientations cancel. Subtract this identity from the one without the cut. The total winding weight of paths visiting p equals that of paths from a stopping at the cut. Reversing these paths and taking real parts gives $L_p(D) = F_+(D) + F_-(D)$, where F_σ is the real winding-weighted sum of paths from the specified side p^σ to the exterior. The initial direction points into the neighboring vertex on that side.

Apply the same splitting procedure from this interior source, retaining all stopping possibilities. A return to the other side of the cut traces a polygon ℓ' through p . For a fixed initial side it is counted once and has turning $\pm 2\pi$, hence real factor $\cos(2\pi t) = -d$. A collision with a polygon not enclosing p cancels with its reversed orientation. If its polygon ℓ strictly encloses p , the stem approaches from inside. The closing turn of the counterclockwise polygon at the collision vertex is $-\pi/3$, whereas the turn from the stem into that orientation is $+\pi/3$. The added winding is therefore $8\pi/3$, or $-8\pi/3$ for the other orientation. Both have factor $\exp(\pm it 8\pi/3) = -1$.

For a fixed enclosing ℓ , sum the stems in its interior, excluding its vertices. This is exactly the corresponding F_σ for the smaller domain: the stem stops at the mid-edge before its aimed-at vertex, whose weight belongs to ℓ . If its initial direction already points at an excluded vertex, this stem sum is one. No further inner polygon is then possible. Write D_ℓ for the domain strictly inside an enclosing polygon ℓ , with its vertices excluded and with stopping outlets on the incident half-edges. Conservation of the initial mass, followed by the collision classification above, gives the recurrence

$$F_\sigma(D) = 1 + d \sum_{\ell' \ni p} \rho^{|\ell'|} + 2 \sum_{\ell \text{ enclosing } p} \rho^{|\ell|} F_\sigma(D_\ell).$$

The factor two in the last sum comes from the two orientations of the enclosing polygon, each with real factor -1 on the stopped side of the conservation identity. The initial-source convention just specified gives $F_\sigma(D_\ell) = 1$ if its first vertex is excluded.

Iterate this recurrence, choosing an outermost enclosing polygon first and then polygons strictly inside it. Every nested family is obtained once in this order. The iteration terminates either with the constant term one or with a polygon through p ; such a polygon must be the innermost selected

cycle: p lies on its boundary, so no disjoint cycle strictly inside its bounded disk can enclose p . Every enclosing cycle removes vertices, so the iteration is finite. It gives, for either sign,

$$F_\sigma(D) = \mathcal{Z}_p(D) + d \sum_{\ell' \ni p} \rho^{|\ell'|} \mathcal{Z}_{\text{around}}(\ell'),$$

where $\mathcal{Z}_{\text{around}}(\ell')$ sums the disjoint outer polygons surrounding ℓ' , with fugacity two per polygon. This recursion remains valid if the opposite neighbor of p is excluded; in that case there is no return to the cut.

A polygon enclosing exactly one of f_1, f_2 uses their separating edge. There can be at most one such polygon in a disjoint nest, and it is the innermost polygon of that nest: any smaller polygon around the same face would also have to use that edge. Consequently

$$\mathcal{Z}_{f_1}(D) + \mathcal{Z}_{f_2}(D) = 2\mathcal{Z}_p(D) + 2 \sum_{\ell' \ni p} \rho^{|\ell'|} \mathcal{Z}_{\text{around}}(\ell').$$

Together with the preceding expression for $F_+ + F_-$ and $0 < d < 1$, this proves the finite-domain claim.

Exhaust an infinite strip by parallelograms. The top/bottom path sums and the nesting sums increase. The terms with a distant side-wall endpoint tend to zero. To justify the latter assertion, split a path at p and discard avoidance between its arms. Keep a fixed group of columns containing p . On either side, the vacuum transfer supplies a uniformly bounded half-state. A top/bottom endpoint is bounded by the sum over its source column followed by one-defect propagation to the fixed columns; this sum converges by the fixed-height transfer decay. A side-wall endpoint contributes a decaying one-defect power, with only finitely many wall ports. Thus each arm is uniformly summable and its side-wall part tends to zero. The upper bound on L_p obtained in this way and the lower finite-domain inequality also prove finiteness of the limiting nesting sums. $\square$

11.4 Raw and conditioned first-length estimates

We now keep a single bottom boundary port a fixed. Let $\mathcal{P}_N(a)$ be the self-avoiding port paths in the open N -band strip from a to either boundary, with no other boundary visit. These are the bottom arches and the bridges. Their terminal ports are summed over their actual lattice positions; there is no additional translation sum. Write

$$\mathcal{M}_N(a) = \sum_{\gamma \in \mathcal{P}_N(a)} \rho^{n(\gamma)} n(\gamma),$$

where $n(\gamma)$ counts visited honeycomb vertices. This is an unnormalized first-length mass.

Proposition 11.5 (Raw first-length mass). *There is a constant C , independent of the boundary source and $N \geq 1$, such that*

$$\mathcal{M}_N(a) \leq C(1 + N S_{N+1}) \leq N^{13/12+o(1)} \quad (N \rightarrow \infty). \quad (104)$$

Proof. Sum $L_p(\mathcal{S}_N)$ over one representative of each internal mid-edge orbit under integer horizontal translation. For each summand, translate its source port to the fixed representative of its boundary orbit. The marked internal edge then runs through all visits of the translated path. This is a bijection between the two sets of path-edge pairs, preserving both length and winding; hence translation invariance turns the sum into the winding-weighted number of internal edge visits for paths from the chosen source representatives. There is one bottom-port orbit and one top-port

orbit, whose source totals agree by a half-turn. Thus, if $\mathcal{E}_N$ is the chosen set of internal mid-edge representatives, the orbit-count identity is

$$\sum_{p \in \mathcal{E}_N} L_p(\mathcal{S}_N) = 2 \sum_{\gamma \in \mathcal{P}_N(a)} \rho^{n(\gamma)} (n(\gamma) - 1) \cos(tW(\gamma)).$$

A path with n vertices has $n - 1$ internal mid-edge visits, and its winding cosine is between c and one. There are $O(N)$ internal edge orbits. These positive sums may be rearranged by monotone convergence.

Every face-center nesting mass in the strip is at most S_{N+1} : embed the strip in a $2(N+1)$ -band strip whose middle line contains the chosen face center. Lemma 11.4 now bounds the sum of internal edge visits by CNS_{N+1} . To replace $n - 1$ by n , add the boundary partition mass $\mathcal{A}_N + \mathcal{B}_N \leq 1/c$, using the strip identity $c\mathcal{A}_N + \mathcal{B}_N = 1$. This proves the first inequality; Corollary 11.3 proves its asymptotic form. $\square$

Proposition 11.6 (Length conditional on reaching macroscopic height). *Give $\mathcal{P}_N(a)$ its normalized critical weights. Conditional on the path reaching the top boundary of the bottom $\lfloor N/4 \rfloor$ bands, its expected number of visited vertices is $N^{4/3+o(1)}$.*

Proof. Put $k = \lfloor N/4 \rfloor$, and take N large enough that $k \geq 1$. A path is confined to the bottom k -band strip exactly when it stays below its top line: crossing that line uses a port between the two adjacent bands. Thus the conditioning event consists of all bridges and precisely the bottom arches not counted by $\mathcal{A}_k$. Its unnormalized mass is

$$q_N = \mathcal{B}_N + \mathcal{A}_N - \mathcal{A}_k = \mathcal{B}_k/c - (1/c - 1)\mathcal{B}_N \asymp N^{-1/4}.$$

Indeed monotonicity gives $\mathcal{B}_k \leq q_N \leq \mathcal{B}_k/c$, and the strip theorem gives $\mathcal{B}_k \asymp N^{-1/4}$.

Restrict the orbit sum in the preceding proof to internal mid-edges whose heights lie in the central third. For sufficiently large N , each adjacent face center is at distance at least $N/3 - O(1) > k$ bands from either boundary. Its centered $2k$ -band strip therefore fits inside $\mathcal{S}_N$, so its nesting mass is at least S_k . There are at least $c_1 N$ such internal edge orbits, and Lemma 11.4 gives a lower bound $c_2 NS_k$ for their total winding-weighted visit mass. Any bottom-source path counted there reaches the top boundary of the bottom k bands; the rotated statement holds for a top source. Since the winding cosine is at most one and the number of counted visits is at most the path length, the unnormalized first-length mass of the conditioning event is at least $c_3 NS_k$. Its upper bound is $\mathcal{M}_N(a)$. By Corollary 11.3 and Proposition 11.5, these bounds both have exponent $13/12$. Division by q_N gives $N^{13/12+1/4+o(1)} = N^{4/3+o(1)}$. $\square$

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